Beyond Conventional Mathematics

An Experimental Journey into AI-Discovered Mathematical Systems

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Preface

This book is written by a research laboratory rather than by a single author. The laboratory — a small team of cooperating AI research roles (an Inventor, an Explorer, a Formal Verifier, a Skeptic, an Application Researcher, a Historian and an Author) — set out to explore mathematical systems built from deliberately minimal assumptions: distinguishable objects, equality, and explicitly stated transformation rules. Numbers, positional notation, multiplication and even "value" were not assumed; where they appear in these pages, an earlier chapter shows how they *emerge*.

The book has one rule that overrides every other consideration:

> **Nothing is presented as established unless it has been proved, and nothing is presented as new unless we looked for it and failed to find it.**

Every theorem, conjecture and refuted hypothesis carries a label:

Label	Meaning
PROVEN (Lean)	machine-checked in Lean 4 with Mathlib, standard axioms only
PROVEN (computer-assisted)	a finite certificate checked by a program; the trust base is stated
PROVEN (written)	a complete written proof, reviewed by the lab's independent Skeptic
COMPUTATIONALLY SUPPORTED	experiments support the claim; no proof
CONJECTURE	a precise claim awaiting proof
REFUTED	a counterexample is given — and kept, because failures teach
REDISCOVERY	the result is already in the literature (references given)
POTENTIALLY NOVEL	no equivalent found in the sources searched; *not* a priority claim

Most of what the laboratory found is, honestly, rediscovery: the commutative monoid of counts, chip-firing, rational base numeration, the Zeckendorf game, Knuth's Fibonacci multiplication. We regard this as a success of the method — independent rediscovery from minimal assumptions is evidence that the structures are natural — and we document the literature carefully. Where something appears to be new (for example, the "glider" of borderline exchange rules, or the density of irreducibles for Knuth's circle product), we say exactly how far our search went. We also report where a deeper, independent audit found our "novelty" already in the literature: Chapter 2 tells how a formula we first believed new turned out to follow, in most cases, from 2022 and 2026 papers on the homology of one-relator monoids.

How the book is organised. The ten volumes follow the plan with which the laboratory started, adjusted to what the research actually produced. Volumes 1–5 and 7–8 contain the research chapters, one per research cycle: foundations (quantity, exchange, termination, derivations), invented operations, emergent structures, alternative representations of quantities, transformations and relationships, computational

experiments, and applications. Volumes 6, 9 and 10 are *catalogues generated from the laboratory's database*: every verified theorem and open conjecture, every refuted hypothesis with its counterexample, and every open problem. They cannot drift from the evidence, because they are rebuilt from it each time the book is built. Volumes are unequal in size; no chapter was written to fill a volume.

Each chapter follows the same plan: motivation, definitions, examples, results with proofs or proof attempts, the computational experiments behind them (with experiment identifiers that can be re-run from the repository), counterexamples, applications, comparison with existing mathematics, limitations, and open questions.

How to verify anything in this book. Every experiment identifier (EXP-C01-05, …) names a deterministic script in the repository; `python -m lab run EXP-C01-05` re-runs it and checks the recorded result hash. Every claim identifier (CLM-C08-01, …) has a record in `db/claims/` with its full status history. Lean theorem names refer to `proofs/lean/Infinity/`.

1 How the laboratory works

1.1 Minimal assumptions, stated twice

Each research branch begins from a *foundational system* whose object-level axioms are written down explicitly (repository file `research/systems/SYSTEMS.md`):

- **System A** — words over distinguishable symbols, juxtaposition, rewriting rules; no numbers.
- **System B** — a carrier with one invented binary operation; no algebraic law is assumed, every law is a hypothesis to be tested.
- **System C** — representations of quantities (digit strings with a valuation), admissible only after System A has *derived* counting.
- **System D** — structures defined by relations only (game options, graphs).
- **System E** — local-rule dynamics over invented algebras.

The *meta level* — the mathematics we use to prove things about these systems — is ordinary mathematics (formally, Lean's type theory with Mathlib). When we say that a concept "emerges", we mean that it is definable from the system's primitives and forced by them up to isomorphism, as shown by a meta-level proof.

1.2 A research cycle

Every cycle runs ten steps, and the cycle report in the repository is organised by them:

1. generate hypotheses — including at least one *bold* hypothesis that might fail;
2. state them precisely;
3. design experiments that could falsify them;
4. implement the experiments (deterministic, seeded scripts);
5. execute them (results are hashed; re-running checks the hash);
6. analyse the results;
7. attempt proofs or find counterexamples;
8. compare with the literature (OEIS and web searches; references recorded);
9. look for applications, with baselines and honest negatives;
10. document everything in the journal and the database.

1.3 Kinds of proof

Lean proofs. Statements are formalised in Lean 4 with Mathlib; a file counts as verified only if it compiles with no `sorry` and `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound`.

Computer-assisted proofs. Some results are proved by finite certificates that a program checks: a completed rewriting system whose convergence is re-verified independently, an automaton whose count sequence obeys a recurrence of bounded order, a recurring window of a monotone map. For these the book states the *trust base* — the program code — and the cross-checks used to guard against implementation errors.

Written proofs are complete arguments reviewed by the lab's Skeptic, who works independently from the authors and tries to break every claim.

1.4 Novelty

The Historian searches the OEIS (all integer sequences we produce are looked up, and the cached answers are kept in the repository) and the literature. Matches are *read*, not counted: in Chapter 9 a sequence agreed with an OEIS entry for 24 terms and then diverged; the agreement had a simple explanation and was recorded as a coincidence.

Volume 1 — Foundations and Alternative Mathematical Objects

The first volume builds quantity from nothing but symbols and rules. Chapter 2 shows that counting is the complete invariant of free rearrangement and that modular arithmetic is what a single relation does to counting. Chapter 3 lets place value emerge from local exchanges of tokens. Chapters 4 and 5 study the transformation systems themselves: when they stop, how long they take, and which games they define.

2 Quantity from transformation

Volume 1 — Foundations and alternative mathematical objects. Research cycle C01.

2.1 Motivation

Can numbers be *derived* rather than assumed? We start with nothing but distinguishable symbols, the empty word, juxtaposition, and rules that declare two words interchangeable. If quantity is real, it should appear as whatever survives the transformations — an *invariant* — and the invariant should be forced, not chosen.

2.2 Definitions

Let Σ be a finite alphabet of q symbols. A *relation* $u \leftrightarrow v$ allows replacing a factor u by v (or back) anywhere in a word. The *congruence* generated by a set of relations identifies two words when one can be transformed into the other by finitely many such replacements. An *invariant* is a map that is constant on congruence classes; it is *complete* if it separates different classes.

A *border* of a word w is a nonempty proper prefix of w that is also a suffix; w is *unbordered* if it has none. The *autocorrelation polynomial* of w ($|w| = k$) is $c_w(x) = 1 + \sum_b x^{k-|b|}$, the sum over the borders b of w.

2.3 Counting emerges

Theorem 2.1 (counting is the complete invariant of commutation) *[PROVEN — Lean: `commEq_iff_count`; REDISCOVERY]*. Modulo the relations $xy \leftrightarrow yx$ ($x, y \in \Sigma$), two words are congruent if and only if every symbol occurs equally often in both. The projection of a word onto a symbol s — delete all other symbols — is the unary word $s \cdots s$.

So the only information that survives free rearrangement is a family of unary words, one per symbol. This is the laboratory's justification for introducing the natural numbers: they are the classes of the one-symbol free monoid, and they *are* the invariant.

Theorem 2.2 (one symbol: a clock with a tail) *[PROVEN — Lean: `gen_iff, exactly_m_add_p_classes`; REDISCOVERY]*. On words over one symbol a, the relation $a^m = a^{m+p}$ ($p \geq 1$) produces exactly $m + p$ classes: $a^x \sim a^y$ iff $x = y$, or $x, y \geq m$ and $x \equiv y \pmod{p}$.

Theorem 2.3 *[PROVEN — written; REDISCOVERY]*. Several relations $a^{m_i} = a^{n_i}$ generate index $M = \min m_i$ and period $P = \gcd(n_i - m_i)$. (Two independent algorithms agree with this on 6159 relation sets, EXP-C01-02.)

Modular arithmetic is therefore also emergent: it is what one relation does to counting.

Theorem 2.4 (multiplication is composition) *[PROVEN — written; REDISCOVERY]*. The substitutions $a \mapsto a^k$ of the one-symbol free monoid, with pointwise juxtaposition as addition and composition as multiplication, form a commutative semiring isomorphic to $(\mathbb{N}, +, \times)$. With two symbols modulo commutation, composition is matrix multiplication and is no longer commutative.

2.4 A bold conjecture fails

Conjecture (H-C01-4) [REFUTED]. Any single relation that only rearranges symbols — like $aab \leftrightarrow aba$ — eventually identifies all words with the same symbol counts.

Counterexample. Under $aab \leftrightarrow aba$, the word baa contains neither side, so its class is $\{baa\}$, while aab, aba, baa have the same counts. All 33 rearranging relations of length 3 and 4 fail in the same way (EXP-C01-04).

2.5 An unexpected Fibonacci pattern

The failed experiment produced something better than a confirmation. For each relation, count the classes of words of length n over $\{a, b\}$:

relation	$n = 1, 2, 3, \ldots$
all six length-3 relations	2, 4, 7, 12, 20, 33, 54, 88, 143, 232, 376
25 of 27 length-4 relations	2, 4, 8, 15, 28, 52, 96, 177, 326, 600, 1104
$aaba \leftrightarrow abaa$ (and complement)	2, 4, 8, 15, 28, 52, **97**, 181, 338, 631, 1178

The first row is $F_{n+3} - 1$ (partial sums of Fibonacci numbers), the second the partial sums of tribonacci numbers. The explanation is the self-overlap structure of the words involved.

Theorem 2.5 (border containment) *[PROVEN — written, Skeptic-reviewed; the list identities of the critical-pair join are checked in Lean (`common_border_joinable`); REDISCOVERY: Kurth 1987].* Let $u \neq v$ have length k and suppose every border of u is a border of v. Then the rewriting system $\{u \to v\}$ terminates and is confluent; the classes of length-n words correspond to the words avoiding u; and

$$\sum_{n \geq 0} \#\{\text{classes of length } n\}\, x^n = \frac{c_u(x)}{x^k + (1 - qx)\, c_u(x)}.$$

For unbordered u this is $1/(1 - qx + x^k)$; for $q = 2$ the coefficients are partial sums of $(k-1)$-bonacci numbers.

Proof sketch. Every step changes the word in the same lexicographic direction inside a finite set, so derivations terminate. Two overlapping occurrences of u overlap along a border b of u; since b is also a border of v, writing $u = pb = bs$ and $v = p'b = bs'$ one checks $vs = p'u \to p'v$ and $pv = us' \to vs' = p'v$, so the critical pair joins in one step on each side. Newman's lemma gives confluence, and the count of u-avoiding words is the Guibas–Odlyzko formula. ∎

The exceptional relations are those whose two sides *share* a border. Testing that observation led to a precise conjecture.

Conjecture 2.6 (common borders) *[COMPUTATIONALLY SUPPORTED; partly settled by the literature; POTENTIALLY NOVEL only in the case "common border without containment"].* For all $u \neq v$ of length k, the class-count series is $c(x)/(x^k + (1 - qx)c(x))$ with $c(x) = 1 + \sum_b x^{k-|b|}$, the sum over the words b that are borders of *both* u and v.

Evidence. No counterexample among all 2159 relations with binary $k \leq 6$ and ternary $k \leq 4$ (up to $n = 12$, resp. 7), including 1326 relations not covered by Theorem 2.5 (EXP-C01-06).

Theorem 2.7 *[PROVEN — computer-assisted; REDISCOVERY for 1620 of the 1632 relations].* Conjecture 2.6 holds for all n for 1632 of these relations. *Method:* Knuth–Bendix completion finds a finite convergent system (independently re-verified: rules

decrease, all critical pairs join, u and v have equal normal forms); its irreducible words are counted by an automaton with s states, so the count sequence obeys a recurrence of order $\leq s$; the conjectured series obeys one of order k; equality of the first $s + k + 5$ terms therefore implies equality forever (EXP-C01-07). Every certificate is stored with the experiment, every completed rule is re-derived from $u = v$ by search, and the Skeptic re-verified all 1632 certificates with independently written code. The remaining 527 relations have infinite completions, e.g. $aba \leftrightarrow bab$ completes to $bab \to aba$ together with $ba^j ba \to abaab^{j-1}$ for all $j \geq 2$.

2.5.1 What the literature already knew

Our Historian's independent audit found that much of this territory had been mapped by a different route — homological algebra of one-relator monoids:

- Theorem 2.5 is the length-preserving case of Kurth's (1987) one-rule confluence criterion, and Kurth's criterion is an *iff*: $u \to v$ is confluent exactly when every border of u is a border of v (we confirmed this on 28,850 pairs, EXP-C01-08).

- When u and v share **no** border, Conjecture 2.6 follows from a 2026 theorem of Gray and Steinberg together with Anick's theory of strongly free relations:

  Theorem 2.8 *[PROVEN — corollary of published theorems, every step checked by the Skeptic; REDISCOVERY].* If $u \neq v$ of length k have no common border, the class-count series is $1/(1 - qx + x^k)$.

  Proof idea. No common border means the monoid is "incompressible", so it has Hochschild cohomological dimension at most 2 (Gray–Steinberg 2026, Thm 8.1); the trivial module of $K\langle\Sigma\rangle/(u - v)$ then has the resolution $0 \to A(-k) \to A(-1)^q \to A \to K \to 0$, whose Euler characteristic is the formula. ∎ Every one of the 527 relations that our completion method could not settle lies in this case, so all 2159 relations of EXP-C01-06 are now proved.

- With exactly **one** common border, growth identities read off from Gray–Steinberg (2022) imply the conjectured formula *provided* a certain multiplication map on a left ideal is injective. The Skeptic found that this lemma was asserted but not proved; it holds in all 2591 cases tested, so the case is "proved modulo one lemma".

What remains open is the case of two or more common borders without containment. The formula holds in every case tested: the Skeptic's search covered 295,358 relation orbits and 1,491 targeted relations with one to four common borders, up to $n = 27$, without a single failure. The episode is recorded as a lesson: two quick web searches were not enough to establish novelty.

2.6 Experiments

Experiment	Question	Result
EXP-C01-01	Is counting the complete invariant?	25 exact slices, no exception
EXP-C01-02	Index/period of one-symbol relations	6159 relation sets, 0 disagreements
EXP-C01-03	Do substitutions form $(\mathbb{N}, +, \times)$?	all laws on $a^0 .. a^{12}$; non-commutative with 2 symbols

Experiment	Question	Result
EXP-C01-04	Does one rearrangement give counting?	refuted 33/33; Fibonacci pattern found
EXP-C01-05	Unbordered-side hypothesis	756 relations, 0 failures
EXP-C01-06	Common-border conjecture	2159 relations, 0 failures
EXP-C01-07	Proofs for all n	1632 proved, 527 open

2.7 Comparison with existing mathematics

Theorems 2.1–2.4 are classical: Parikh vectors (1966) and trace monoids (Cartier–Foata 1969), monogenic semigroups, the endomorphism semiring of $\mathbb{N}$. Theorem 2.5 is Kurth (1987) with Guibas–Odlyzko (1981). For Conjecture 2.6 see the previous section: the no-common-border case is a corollary of Gray–Steinberg (2026) and Anick (1982), and the homological counterpart of the whole common-border dichotomy is Gray–Steinberg (2022, Theorem B). The explicit growth formula with common borders was not found.

2.8 Applications and limitations

Complete invariants turn equivalence testing from a search into a computation (counting decides commutation-equivalence in linear time). Theorem 2.5 also measures how much information a single “invisible” local rewrite destroys: modulo $aab = aba$, binary words carry $\log_2 \varphi \approx 0.694$ bits per symbol instead of 1. These are observations, not benchmarked algorithms.

2.9 Open questions

1. Prove the injectivity lemma that completes the one-common-border case, then the case of two or more common borders, perhaps by induction over Adjan–Oganesyan compression.
2. Is there a common-border formula for two or more relations?
3. Formalise Theorem 2.5 completely in Lean.

3 Numeration from exchange

Volume 1 — Foundations. Research cycle C02.

3.1 Motivation

Chapter 2 showed that a multiset of tokens *is* a number. Positional notation is usually introduced by fiat ("the digit in the second place is worth two"). Here we ask whether place value can be *discovered*: put tokens on levels $0, 1, 2, \ldots$ and allow only local exchanges.

3.2 Definitions

A *configuration* is a finitely supported vector $c = (c_0, c_1, \ldots)$ of token counts. A **carry rule** $m[k] \to \sum_{j=1}^{J} b_j [k+j]$ removes m tokens from a level holding at least m and adds b_j tokens j levels higher. A configuration is *stable* if every $c_k < m$. The *carry polynomial* is $P(x) = \sum_j b_j x^j - m$ and the *emergent base* r is its unique positive root. Nothing about value is assumed — but solving $m w_k = \sum_j b_j w_{k+j}$ for exchange-invariant weights forces $w_k = r^k$: **value is the invariant of the exchange rule**, $V(c) = \sum_k c_k r^k$.

3.3 Examples

- $2[k] \to [k+1]$: from n tokens at level 0 one obtains the binary digits of n. Base b arises from $b[k] \to [k+1]$.
- $3[k] \to 2[k+1]$: base $3/2$ (the rational base of Akiyama, Frougny and Sakarovitch): $1, 2, 20, 21, 22, 210, \ldots$ (OEIS A024629).
- $3[k] \to [k+1] + [k+2]$: base $r = (\sqrt{13} - 1)/2 \approx 1.3028$, digits 0..2: $1, 2, 110, 111, 112, 220, 221, 222, 12100, \ldots$ (not in the OEIS).
- Fibonacci exchanges $[k] + [k+1] \to [k+2]$, $2[k] \to [k+1] + [k-2]$ (with boundary versions) produce Zeckendorf representations.

3.4 Results

Theorem 3.1 (abelian property) *[PROVEN — written; REDISCOVERY: chip-firing].* For a carry rule, firings at different levels commute. Hence from any configuration either all maximal derivations are infinite, or all are finite, end at the same stable configuration, have the same length, and fire every level the same number of times.

Theorem 3.2 (integer bases) *[PROVEN — written].* Under $b[k] \to [k+1]$ every derivation from n tokens ends at the base-b expansion after exactly $(n - s_b(n))/(b-1)$ exchanges (s_b = digit sum).

Theorem 3.3 (termination dichotomy) *[PROVEN — written; REDISCOVERY (folklore)].* If $\sum b_j < m$, every derivation terminates. If $\sum b_j > m$, then $r < 1$ and no derivation from $n > (m-1)/(1-r)$ tokens terminates, because every stable configuration has value at most $(m-1)/(1-r)$.

The borderline $\sum b_j = m$ is the interesting case: the rule $2[k] \to [k+1] + [k+2]$ started from 6 tokens leaves a pattern $(2, 1)$ that *glides* upward forever. In the upward sweep the

“window” of the next J levels evolves by a monotone map whose fixed point is the tail-sum vector $T = (\sum_{j \geq 1} b_j, \sum_{j \geq 2} b_j, \ldots)$: every level fires exactly once.

Theorem 3.4 (critical rules) *[PROVEN — two written proofs; POTENTIALLY NOVEL — Propp (2017) proposed this rule family without results].* If $\sum b_j = m$, every derivation from n tokens is infinite for all sufficiently large n. Hence a carry rule terminates from every unary start **if and only if** $\sum b_j < m$.

Proof idea. Write the window map as $F(x) = Ax - \varphi(x)\,b$ with $\varphi \in [0, 1)$ the floor error. The eigenvalues of A are $1/\mu$ for the roots of $\sum_j b_j \mu^j = m$; when the support of b has gcd 1, $\mu = 1$ is a simple root and every other root satisfies $|\mu| > 1$. So A has the simple eigenvalue 1 (eigenvector T, left eigenvector $(1, \ldots, 1)$ — token conservation) and all other eigenvalues inside the unit disk. The floor errors stay bounded off the T-direction, so after $O(\log n)$ levels the window is a large multiple of T plus a bounded error, hence dominates T, and the glider runs forever. ∎ For all 98 critical rules with $m \leq 5$, $J \leq 4$ every start beyond 124 is certified non-terminating (EXP-C02-07).

The lab’s Skeptic, asked to break this theorem, instead found a second proof that uses only positivity: the linearised window map is a *primitive* non-negative matrix (its cycles have lengths j with $b_j > 0$, of gcd 1), so some power of it is strictly positive and a large enough start dominates T after a bounded number of levels. This second proof was checked line by line by the laboratory’s Formal Verifier, so the theorem rests on a reviewed proof; the Skeptic also simulated 668,793 critical rules, all of which glide.

Theorem 3.5 (radix order) *[PROVEN — written; REDISCOVERY in substance: Akiyama–Frougny–Sakarovitch 2008, Prop. 11].* For terminating carry rules, $n < n'$ implies that the representation of n precedes that of n' in radix order. (Add one token and stabilise. If H is the highest level that fires, the levels $H + j$ with $b_j > 0$ gain tokens and never fire, so the highest level whose count changes lies above H: it only gained tokens. An earlier version of this argument had a gap, found by the Skeptic.)

Theorem 3.6 (uniqueness) *[PROVEN — Lean core: `digits_unique`; REDISCOVERY in substance: Akiyama–Frougny–Sakarovitch 2008, Lemma 5].* Let g be the gcd of the levels reachable in one exchange and $\tilde{P}$ the carry polynomial in the variable $y = x^g$. If $\tilde{P}$ is primitive and irreducible over $\mathbb{Q}$, distinct stable configurations on the reachable levels have distinct values; in particular, when $\sum b_j < m$, each integer has exactly one representation. If $\tilde{P}$ is not primitive and $\sum b_j \leq m$, uniqueness fails. (Both side conditions are needed: the Skeptic’s counterexamples are $2[k] \to 4[k+1]$, non-primitive yet without collisions, and $2[k] \to 3[k+1]$, which has no stable configuration of value 2.)

Proof idea. A difference Δ of two digit strings with equal value vanishes at the root; an irreducible $\tilde{P}$ must then divide Δ over $\mathbb{Q}$ (a Bézout identity evaluated at the root rules out coprimality), hence over $\mathbb{Z}$ by Gauss’s lemma; but a nonzero multiple of $\tilde{P}$ has a trailing coefficient divisible by m, while digit differences are smaller than m. ∎

A refuted hypothesis. We first conjectured uniqueness for *every* terminating rule. The rule $4[k] \to 2[k+1]$ (base 2 with digits 0..3) refutes it: “2” and “10” both have value 2. All failures among the 31 terminating small rules have a non-primitive reduced carry polynomial.

3.5 The Fibonacci exchanges are the Zeckendorf game

Theorem 3.7 *[PROVEN — written; REDISCOVERY].* The Fibonacci exchanges terminate from every configuration at the Zeckendorf representation of its value, and the shortest derivation from n tokens has $n - z(n)$ steps ($z(n)$ = number of Zeckendorf terms). Termination uses the potential (token count, $\sum_k c_k\, g(k)$) with $g(0) = -\frac{1}{2}$, $g(k) = k$; the shortest derivation adds tokens one at a time, which needs only merges.

This system is exactly the *Zeckendorf game* of Baird-Smith, Epstein, Flint and Miller (2018), who proved termination and the shortest game $n - Z(n)$, and whose Figure 4 already lists the *longest* games for $n \le 49$ — identical to our exhaustive values for $n \le 40$ ($0, 1, 2, 3, 5, 6, 8, 10, \ldots, 78$). Cusenza et al. (2022) proved the greedy strategy optimal; a closed form is known only for $n = F_k - 1$. The sequence is not yet in the OEIS.

3.6 Limitations and open questions

1. An explicit bound n_0 for Theorem 3.4, and its Lean formalisation.
2. Which carry rules give *regular* languages of representations?
3. A closed form for the longest Zeckendorf game.

4 When does rewriting stop?

Volume 1 — Foundations. Research cycle C03.

4.1 Motivation

A transformation system is only a *computation* if it stops. For the simplest non-trivial systems — a single rule moving a block of 0s past a block of 1s while multiplying them,

$$0^p 1^q \to 1^r 0^s \qquad (p,q,r,s \geq 1),$$

we ask: from which rules does every sequence of transformations terminate?

4.2 Easy cases

Theorem 4.1 *[PROVEN — Lean: `terminating_of_count_lt`].* If a rule decreases the number of some letter ($s < p$ or $r < q$), it terminates.

Theorem 4.2 *[PROVEN — written; swap case in Lean].* Rules that conserve a letter count ($s = p$ or $r = q$) terminate.

Theorem 4.3 *[PROVEN — Lean: `not_terminating_of_loop`, `not_terminating_double`].* A derivation $u \to^+ xuy$ (a *loop*) proves non-termination, and every rule in the *doubling region* $r \geq 2q$, $s \geq 2p$ has a two-step loop.

4.3 Two hypotheses that failed

H-C03-1 [REFUTED]: "a rule terminates iff some letter count never grows." Zantema's problem $0011 \to 111000$ grows both counts and yet terminates — a famous published result (Geser 1993), which our own search corroborates on all 3747 words of length ≤ 12 containing 0011, but which we did not prove independently.

H-C03-2 [REFUTED, Lean-checked]: "a rule terminates iff $r < 2q$ or $s < 2p$." The rule $011 \to 11100$ has $r = 3 < 4 = 2q$ yet loops: $001111 \to^4 111\,001111\,00100$ (Lean: `not_terminating_0_11`). There are 40 counterexamples with parameters ≤ 5.

4.4 Divisibility

The decisive surprise was a pair of almost identical rules: $0011 \to 11100000$ ($s = 5$) appears to terminate, while $0011 \to 111000000$ ($s = 6$) loops. The difference is whether the produced block of zeros is a whole number of consumed blocks.

Lemma 4.4 (lifting) *[PROVEN — Lean: `step_map`, `loop_map`].* The morphism $0 \mapsto 0^p$ carries every loop of $0\,1^q \to 1^r 0^k$ onto a loop of $0^p 1^q \to 1^r 0^{kp}$.

One base loop thus certifies infinitely many rules. Together with exhaustive search this classifies the family almost completely:

Theorem 4.5 *[PROVEN — computer-assisted; REDISCOVERY].* Of the 625 rules with $p,q,r,s \leq 5$, 525 terminate (Theorems 4.1–4.2) and 76 do not (re-checked loop certificates); 24 remained open in our hands, all predicted to terminate, including Zantema's problem.

Theorem 4.6 (Zantema–Geser) *[PROVEN in the literature; REDISCOVERY — our conjecture is equivalent to it].* $0^p 1^q \to 1^r 0^s$ is non-terminating iff ($r \geq 2q$ and $s \geq 2p$), or (

$s \geq 2p$, $q < r < 2q$ and $p \mid s$), or ($r \geq 2q$, $p < s < 2p$ and $q \mid r$).

We first stated this as a conjecture (zero contradictions on 625 rules). The laboratory's Historian later located the 1994 technical report of Zantema and Geser, whose Theorem 1.1 says exactly this — rule terminates iff $p \geq s$, or $q \geq r$, or $p < s < 2p$ and $q \nmid r$, or $q < r < 2q$ and $p \nmid s$ — and whose page 2 lists the three non-terminating cases in the same words. So the 24 open rules all terminate, and our intermediate hypothesis H-C03-3 ("non-terminating iff both counts grow and $r \geq 2q$ or $s \geq 2p$") is refuted by the published theorem: $0011 \to 11100000$ terminates.

Loops can be long: the rule $0\,1^{6} \to 1^{7}\,0^{6}$ needs 9,332 steps and grows a 56,035-letter word before its starting word reappears inside it.

4.5 Comparison with existing mathematics

The family was completely characterised by Zantema and Geser (tech. report UU-CS-1994-44; RTA 1995; AAECC 2000). Our first attempts to retrieve the theorem failed (the usual repository hosts refused the download); the Historian found the scanned report on an older archive host, and Conjecture 4.6 turned out to be equivalent to their theorem (Theorem 4.6 above): their Theorem 1.1 characterises termination, our conjecture listed the non-terminating cases, and the lists match case by case. Their Lemma 4.1 is our doubling loop, and their Lemma 4.3 reduces the divisible cases to $q = 1$ exactly as our lifting lemma does. Hard terminating cases such as Zantema's problem admit no simple matrix interpretations (none exist in the tiers we searched) and need the Zantema–Geser method. Hard terminating cases such as Zantema's problem admit no simple matrix interpretations (none exist in the tiers we searched) and need match-bounds or the Zantema–Geser method.

4.6 Open questions

1. ~~Compare Conjecture 4.6 with the Zantema–Geser characterisation.~~ Done: identical (Theorem 4.6).
2. An independent, machine-checkable termination proof for the 24 rules (e.g. RFC match-bounds) — they terminate by the published theorem.
3. Explain the k^{q} growth of loop lengths in the base family.

5 How long does rewriting take?

Volume 1 — Foundations: transformation systems. Research cycle C04.

5.1 Motivation

Chapter 4 asked whether a transformation system stops. Once it does, the *length* of a derivation is itself a quantity produced by the system, without any arithmetic having been put in. This chapter asks three questions about the smallest systems — a single rule $l \to r$ over the letters a, b:

1. how long can derivations from words of length n be?
2. when does the *order* in which rewrites are applied not matter?
3. what games does a rule define if two players take turns rewriting?

5.2 Definitions

A step rewrites one occurrence: $xly \to xry$. A *normal form* contains no occurrence of l. $D(w)$ is the maximal number of steps from w to a normal form, and the *derivational complexity* is $D(n) = \max_{|w|=n} D(w)$. A rule has **random descent** (RD; van Oostrom) if from every word all maximal derivations have the same length *and* end in the same normal form. A *border* of l is a nonempty proper prefix of l that is also a suffix. In the *rewriting game* two players alternately make one step; a player who cannot move loses (normal play), and positions get Sprague–Grundy values.

5.3 Which rules terminate?

Theorem 5.1 *[PROVEN — computer-assisted].* Of the 196 rules with $1 \le |l| \le 3$, $0 \le |r| \le 3$, exactly the 34 in which l is a factor of r are non-terminating. Each of the other 162 has a checked certificate: length decrease (70 rules), lexicographic monotonicity of length-preserving steps (70), a linear interpretation over $\mathbb{N}$ (20; e.g. $ab \to bba$ with $a \mapsto 3x$, $b \mapsto x+1$), or a factor count that drops at every step (2).

The bold generalisation fails. Hypothesis H-C04-0 — "a rule with $|l| \le 3$ terminates iff l is not a factor of r" — is **refuted** as soon as $|r| = 4$ is allowed: $ab \to bbaa$ does not terminate although ab does not occur in $bbaa$:

$$aab \to a\,bbaa = abbaa \to bbaa\,baa = bb \cdot aab \cdot aa.$$

The extended census ($|r| \le 5$, 868 rules) has 18 such rules; two symmetry classes ($aab \to ababa$, $aab \to babaa$) were left undecided here (EXP-C04-02) and are settled — both terminate — in Chapter 21.

5.4 One rule, one invariant

The following elementary fact organises the rest of the chapter.

Theorem 5.2 (constant-step potential) *[PROVEN — written; consequences in Lean: `steps_length_unique, length_changing_unique`].* For every rule $l \to r$ with $l \ne r$ there are $\Phi\colon \Sigma^* \to \mathbb{Z}$ and $\delta \ne 0$ such that every step $u \to v$ satisfies $\Phi(v) = \Phi(u) + \delta$. Consequently all derivations between two given words have the same length, and a terminating one-rule system is confluent if and only if it has random descent.

Proof idea. If $|l| \neq |r|$ take $\Phi(w) = |w|$. Otherwise pick a letter c whose position sets P in l and Q in r differ; the power sums $\sum_{j \in P} j^m$ and $\sum_{j \in Q} j^m$ differ for some $m < |l|$ (Vandermonde), and for the least such K the potential $\Phi(w) = \sum_{i : w_i = c} i^K$ changes by the same amount at every offset by the binomial theorem. ∎ Checked on 470,927 steps of 5,365 rules (EXP-C04-06), where $K \leq 2$ always sufficed.

Theorem 5.3 (local criterion) *[PROVEN — Lean: `random_descent`; REDISCOVERY: van Oostrom 2007].* If every local peak $b \leftarrow a \rightarrow c$ can be joined by two derivations of equal length, the system has random descent (from any word that has a normal form).

Theorem 5.4 *[PROVEN — Lean: `unbordered_diamond`, `unbordered_random_descent`; REDISCOVERY].* If l is unbordered, two different one-step reducts of a word have a common one-step reduct; hence $l \rightarrow r$ has random descent. (Two overlapping occurrences of l would exhibit a border.)

A refuted converse. Hypothesis H-C04-1 claimed RD *iff* l is unbordered. The rule $aa \rightarrow a$ has a bordered left side but random descent: every step shortens one run of a's, so every derivation from w has exactly $\sum_{\text{runs}} (\text{length} - 1)$ steps and ends in the same word. Eighteen bordered rules with $|l|, |r| \leq 3$ have RD (EXP-C04-05).

5.5 Derivational complexity

Theorem 5.5 *[PROVEN — Lean: `inv_step`, `normal_iff_inv_zero`, `derivation_length_eq_inv`, `p_positions`; REDISCOVERY].* Under $ab \rightarrow ba$ each step lowers the number of inversions by one; every complete derivation from w has exactly $\mathrm{inv}(w)$ steps, so $D(n) = \lfloor n^2/4 \rfloor$; and in the rewriting game the losing positions for the player to move are the words with an even number of inversions.

This is the extremal case of Métivier's conjecture, proved by Bertrand (1994): length-preserving one-rule systems never need more than $n^2/4$ steps.

Theorem 5.6 *[PROVEN — written].* For $ab \rightarrow bba$ and $ab \rightarrow baa$, $D(n) = 2^{n-1} - 1$. (Scanning w from the right, an a that meets B letters b crosses them in B steps and leaves $2B$ behind it.)

Observation 5.7 *[COMPUTATIONALLY SUPPORTED].* The 162 terminating rules fall into 55 symmetry classes with complexity linear (144 rules), quadratic (14: $\lfloor n^2/4 \rfloor$, $\lfloor n(n-1)/6 \rfloor$, $\lfloor n^2/8 \rfloor$) or exponential (4: $2^{n-1} - 1$); 54 classes have a simple closed form on all computed terms (EXP-C04-03).

One class does not: the rule $aab \rightarrow ba$ refutes the hypothesis (H-C04-G) that every complexity sequence satisfies a linear recurrence.

Theorem 5.8 *[PROVEN — written; POTENTIALLY NOVEL (small)].* For $aab \rightarrow ba$,

$$D(n) = n - 1 - \kappa(n), \qquad \kappa(n) = \min\{k \geq 0 : 2^k + k \geq n\},$$

so $D(n) = n - \log_2 n + O(1)$; D stalls exactly at $n = 2^k + k + 1$ $(4, 7, 12, 21, 38, \ldots)$, and $\sum D(n) x^n$ is not a rational function.

Proof idea. Write $w = a^{x_0} b\, a^{x_1} b \cdots b\, a^{x_m}$. Rewriting aab moves two a's before the $(k+1)$-th b to one a after it: this is the exchange $2[k] \rightarrow [k+1]$ of Chapter 3 — a *binary counter* — with the last level inert. The value $\sum_k x_k 2^k$ is invariant, each step removes one token, and aab is unbordered, so every derivation has the same length, determined by the binary expansion of the value. Optimising over w gives $n - D(n) = \min_m (m + \lceil (n-m)/2^m \rceil) = 1 + \kappa(n)$.

Since the stall positions have unbounded gaps, no linear recurrence can produce them. ∎ (Checked for all words of length ≤ 21 and as an identity to $n = 20000$.)

The same numbers turned up somewhere unexpected. OEIS A247983 counts how many terms of the series $\log 2 = \sum_{h \ge 1} 1/(h\,2^h)$ are needed for n bits of accuracy; the entry gives no formula.

Theorem 5.9 *[PROVEN — written, Skeptic-reviewed; POTENTIALLY NOVEL (small)].* For every $n \ge 1$, $\mathrm{A247983}(n) = n - \max\{j \ge 0 : 2^j + j \le n+1\} = D(n+2)$ for $aab \to ba$. (The Skeptic noted that the identity fails at $n = 0$, outside the OEIS offset.)

Proof idea. The tail $\log 2 - \sum_{h \le k} 1/(h2^h)$ lies between $2^{-k}/(k+2)$ (Jensen) and $2^{-k}/(k+1)$, which pins down the first k with tail below 2^{-n}. ∎ (Checked with exact arithmetic for $n \le 1500$, EXP-C04-10.) The coincidence has a simple reason: both quantities measure how far the counter $2^k + k$ has advanced.

5.6 Games from rewriting

Theorem 5.10 *[PROVEN — written; REDISCOVERY].* If a rule has random descent, the Grundy value of its game at w is $D(w) \bmod 2$: such games are decided by parity alone.

Theorem 5.11 *[PROVEN — written; REDISCOVERY: octal games as rows of pins].* For a terminating rule $a^k \to u$ the game on w is the disjoint sum of games on the maximal runs of a's, and each run game is a take-and-break game. In particular

rule	game
$aa \to b$	Dawson's Kayles (0.07) — period 34 from $n = 53$
$aa \to ab$	Officers (0.6) — periodicity **unknown**
$aaa \to b$	Treblecross (0.007) — periodicity unknown
$aaa \to ab$	0.06 — no period ≤ 300 up to $n = 3000$
$aaa \to aba$	0.4 — period 34

So the two-letter, one-rule system $aa \to ab$ already encodes a famous open problem: whether the Grundy values of Officers are ultimately periodic (they have been computed to beyond 10^{14} without a period appearing).

A bold hypothesis refuted. H-C04-6 guessed that Grundy values of these games never exceed 3 (true on all words of length ≤ 10). The single word a^{14} under $aa \to b$ has Grundy value 4.

5.7 Experiments

Experiment	Question	Result
EXP-C04-01	termination, $\lvert l\rvert, \lvert r\rvert \le 3$	162 terminating, 34 looping, 0 unknown
EXP-C04-02	termination, $\lvert r\rvert \le 5$	472 / 388 / 8 undecided; $ab \to bbaa$ loops
EXP-C04-03	complexity census	144 linear, 14 quadratic, 4 exponential
EXP-C04-04	$aab \to ba$	formula confirmed to $n = 20000$

Experiment	Question	Result
EXP-C04-05	random descent vs. borders	84 unbordered all RD; 18 bordered RD
EXP-C04-06	constant-step potential	5,365 rules, 470,927 steps, 0 violations
EXP-C04-07	game parity and run decomposition	0 violations
EXP-C04-08	octal games	agree with octal recursions to $n = 3000$
EXP-C04-09	OEIS	0.06 Grundy values not in the OEIS
EXP-C04-10	A247983 identity	0 mismatches, $n \le 1500$

5.8 Comparison with existing mathematics

Random descent and its local criterion are due to van Oostrom (RTA 2007) and have been formalised in Isabelle; our Lean development is an independent re-formalisation. Quadratic and exponential one-rule complexities are standard examples (Kobayashi; Métivier's conjecture, settled by Bertrand). The octal games are classical (Guy–Smith 1956; *Winning Ways*). A brief search found no prior statement of Theorem 5.2 or of the analysis of $aab \to ba$; Theorem 5.2 is elementary enough to be folklore and is labelled accordingly in the database ("not assessed").

5.9 Applications and limitations

For one-rule systems the cost of normalisation is predictable without simulation (Theorem 5.2): any one derivation tells the length of all of them. Closed forms such as $\lfloor n^2/4 \rfloor$, $2^{n-1} - 1$ and $n - \log_2 n + O(1)$ are exact worst-case costs of bubble-sort-like swaps, doubling processes and binary-counter carries. These are observations; no practical improvement over existing tools is claimed.

5.10 Open questions

1. ~~Do $aab \to ababa$ and $aab \to babaa$ terminate?~~ Yes — see Chapter 21, which also gives their exact (exponential) complexities.
2. For $|l| = 4$, do complexities other than polynomial, exponential and $n - \Theta(\log n)$ appear?
3. Is the Grundy sequence of $aaa \to ab$ (octal game 0.06) ultimately periodic? (For $aa \to ab$ this is the open Officers problem.)

Volume 2 — Invented Operations and Their Properties

Here a single binary operation is invented and nothing about it is assumed: every law is a hypothesis. Chapter 6 maps which laws have finite models of which sizes; Chapter 7 anchors an operation to a successor and finds that the sizes admitting a model form arithmetic sets — powers of two for left-distributivity (Laver’s tables), the odd numbers, and the moduli in which the golden ratio exists.

6 Which sizes can a single law allow? Spectra of magma identities

Volume 2 — Invented operations and their properties. Research cycle C05.

6.1 Motivation

The barest invented operation is a *magma*: a set and a multiplication table, with no laws at all. Impose a single equation, say $x = y(x(xy))$, and most tables are cut away. This chapter asks the simplest question one can put to such a law: for which sizes n does *any* table satisfy it? There are no numbers in the definition, yet the answers are pieces of arithmetic — congruences modulo 3 and 4, single forbidden sizes, long gaps, as in triple systems and Euler's 36 officers. We found these patterns before learning that the Equational Theories Project (ETP) had studied such laws in 2024–25, so the census is a rediscovery; we then pushed into the sizes its notes leave open.

6.2 Definitions

A *magma* $(M, *)$ of *order* $n = |M|$ is given by its Cayley table; we write xy for $x * y$. A *law* is an identity $x = t$, where t is a term in the variables x, y; its *order* is the number of multiplications in t. The *spectrum* of a law is the set of $n \geq 1$ for which some magma of order n satisfies it for all values of the variables. Numbers such as "law 167" refer to the ETP's list of equations.

A law is *nontrivial* if x occurs in t but is neither its leftmost nor its rightmost leaf. Otherwise only $n = 1$ works (x absent: every element equals $t(y)$), or a projection $ab = a$ or $ab = b$ is a model of every order. A law and its *mirror* (t read right to left) have the same spectrum, via the opposite table $a *' b = b * a$. A *quasigroup* is a magma whose table is a Latin square. An *affine* model over $\mathbb{Z}/n$ is a table $xy = ax + by + c$. We write spectra on $n = 1, 2, \ldots$ as strings: 1 = a model exists, 0 = none, ? = undecided within the solver budget.

Model existence is decided by SAT: one variable per table entry and value, plus one per subterm, assignment of x, y and value. The encoding is exact (`proofs/paper/C05.md`) and matches brute force on all magmas of orders 2, 3.

6.3 The census

Observation 6.1 *[COMPUTATIONALLY SUPPORTED; REDISCOVERY: Equational Theories Project]*. There are 115 nontrivial laws $x = t(x, y)$ of order 2–4 (2, 15 and 98 of orders 2, 3 and 4). Of these, 52 have models of every order $n \leq 6$; 2 have none of any order $2 \leq n \leq 6$; and 61 have a restricted spectrum. Mirror pairs always agree. Extended to $n \leq 8$, the 63 non-full laws fall into 15 patterns:

pattern, $n = 1..8$	**laws**	**example (ETP number)**	**what is known**
10111011	16	$x = y(x(xy))$ (63), $x = y((xx)y)$ (115)	115: conjectured all $n \neq 2, 6$

pattern, $n = 1..8$	laws	example (ETP number)	what is known
10110010	4	$x = y(x(yy))$ (66)	$n \equiv 0, 1 \pmod 3$, $n \neq 6$ (Mendelsohn)
10011??1	1	$x = (yx)(xy)$ (167)	$n \equiv 0, 1 \pmod 4$ (Theorem 6.2, which also settles the two ? as absent)
11011011	8	$x = y((xy)(yy))$ (883)	contains $n \equiv 1, 2 \pmod 3$
10001011	8	$x = y(x(x(yy)))$ (467)	open
1000101?	4	$x = y(x((yx)y))$ (677), $x = y(((yx)x)y)$ (1313)	open; order 16 occurs (Theorem 6.10), so Conjecture 6.6 is refuted
10001000	2	$x = y((x(xy))y)$ (1076)	open
1010001?	2	$x = y((yx)(xy))$ (907)	open; known sizes 1, 3, 7, 9, 13
10011001	2	$x = y(y(x(xy)))$ (501)	open
10011011	2	$x = y((y(xx))y)$ (1110)	open
100110??	2	$x = y(x((xy)y))$ (670)	open
10110011	2	$x = y((x(yx))y)$ (1083)	open
10101111	4	$x = y(x(y(xy)))$ (474)	all $n \neq 2, 4$
10111111	4	$x = (yx)((xy)y)$ (1685)	all $n \neq 2$
1000001?	2	$x = y(((xy)x)y)$ (1286)	open; smallest nontrivial model has order 7

The last column comes from the ETP sources under *Comparison*; our data agree with every size stated there. The 16 Dupont-type laws have one model up to isomorphism at each of $n = 3, 4, 5$, always a medial quasigroup; the order-3 model of law 907 is the Steiner quasigroup; law 167 has 58 models of order 4 up to isomorphism, not all quasigroups. The census's non-existence answers are SAT results that we did not certify further.

6.4 A law that counts modulo 4

Theorem 6.2 *[PROVEN — Lean: `law167_card_mod_four`; REDISCOVERY: Equational Theories Project].* Every finite magma satisfying $x = (yx)(xy)$ has order $n \equiv 0$ or $1 \pmod 4$.

Proof. Send an unordered pair $\{a, b\}$ to $F\{a, b\} = \{ab, ba\}$. The law gives $(ab)(ba) = b$ and $(ba)(ab) = a$, so F is an involution. If $a \neq b$ then $ab \neq ba$, since otherwise $F(F\{a, b\})$ would be a single element. So F permutes the $\binom{n}{2}$ pairs of distinct elements. It fixes none of them: $\{ab, ba\} = \{a, b\}$ forces $a = b$ through the law in both possible cases. A fixed-point-free involution on a finite set has an even number of points, so $4 \mid n(n-1)$. ∎

Lean follows this argument via Mathlib's `Sym2` and its fixed-point congruence for maps of prime-power order, and checks models of orders 4, 5 (`T4_law167`, `T5_law167`) and the absence of order 2 (`no_law167_order2`). The converse is the ETP write-up's construction:

pair up the two-element subsets as $\{a,b\},\{c,d\}$ and set $ab=c$, $ba=d$, $cd=b$, $dc=a$, $xx=x$. So the spectrum is exactly $\{n \equiv 0,1 \pmod 4\}$.

6.5 Pushing the open cases

Theorem 6.3 *[PROVEN — Lean: `left_mul_bijective_of_law; REDISCOVERY (standard)`]*. If a finite magma satisfies $x = y\,s(x,y)$ for some term s, every left translation $a \mapsto ya$ is a bijection: each row of the table is a permutation.

Proof. $y \cdot s(x,y) = x$ shows that $a \mapsto ya$ is onto, and an onto map from a finite set to itself is one-to-one. ∎

Theorem 6.3, and fixing $0*0 \in \{0,1\}$ (relabelling always allows it), are *sound* extra constraints for the solver. With them we decided the 15 laws of unknown spectrum in the ETP notes for all $n \le 10$ but a few UNKNOWN sizes, with no disagreement with any size stated in the notes. What is new relative to the notes:

Theorem 6.4 *[PROVEN — Lean `decide` certificates; POTENTIALLY NOVEL relative to the ETP notes]*. Law 677, $x = y(x((yx)y))$, has models of orders 5, 7 and 9. Laws 1286, $x = y(((xy)x)y)$; 1313, $x = y(((yx)x)y)$; and 1516, $x = (yy)(x(xy))$, each have a model of order 9.

Proof. We give explicit Cayley tables found by SAT. Lean's kernel checks each law on each table (`E677_order5_law`, `E677_order7_law`, `E677_order9_law`, `E1286_order9_law`, `E1313_order9_law`, `E1516_order9_law`). ∎

The same pipeline certified an order-9 model of law 467. That size is *not* new: the notes construct models of 467 on $\mathbb{Z}[i]/(k+il)$ whenever k^2+l^2 is odd, and $9 = 3^2+0^2$. We first missed this and recorded it as new; a second reading corrected it.

Theorem 6.5 *[PROVEN — computer-assisted, drat-trim-verified; POTENTIALLY NOVEL for $n \ge 5$]*. The following laws have no model of the given orders: law 501, $x = y(y(x(xy)))$, orders 6 and 7; law 677, order 6; law 883, $x = y((xy)(yy))$, order 6; law 1076, $x = y((x(xy))y)$, order 8; law 1083, $x = y((x(yx))y)$, orders 5 and 6. The same holds for 501 at order 3 and 677 at orders 3 and 4. These are new relative to the notes but small enough that ETP searches probably settle them.

Proof. For each case Glucose 4 logs a DRUP refutation of the CNF formula, and the independent checker drat-trim verifies it (all 48 frontier refutations passed). What remains trusted is drat-trim, the CNF generator and the soundness of the two extra constraints. ∎

Conjecture 6.6 *[REFUTED — by the Skeptic's review; see Theorem 6.10]*. "Laws 677 and 1313 both have spectrum $\{1\} \cup \{n \text{ odd} : n \ge 5\}$."

Both laws have models of orders 5, 7, 9 and none of orders 2, 3, 4, 6, all certified except orders 5 and 7 of law 1313 (stated in the notes, found again by SAT, tables re-checked). Orders 8 and 10 were undecided; the order-5 models are idempotent quasigroups. The first version of this book stated the conjecture on this evidence. It is false: both laws have models of order 16 (Theorem 6.10). The evidence was too thin — four odd sizes and four small gaps say little about larger even sizes — and the search space had a blind spot: the lab looked for affine models only over $\mathbb{Z}_n$, whose additive group is cyclic, never over the fields $\mathrm{GF}(2^k)$.

6.6 Three bold hypotheses fail

Hypothesis 6.7 (H-C05-1) *[REFUTED].* "A single law $x = t$ has spectrum $\{1\}$ or all of $\mathbb{N}$: one identity cannot restrict sizes." Law 167 is a counterexample (Theorem 6.2), and 61 of the 115 census laws are others.

Hypothesis 6.8 (H-C05-Q) *[REFUTED].* "Every nontrivial law forces its finite models to be quasigroups." Counterexample: this model of law 167, whose row 1 repeats the value 2 (`T4_law167`, `T4_not_latin`):

*	**0**	**1**	**2**	**3**
0	0	3	1	0
1	0	2	2	2
2	2	0	1	3
3	1	3	1	3

Eighteen census laws already have non-quasigroup models of order at most 3.

Hypothesis 6.9 (H-C05-A) *[REFUTED].* "A law has models of every order $n \le 6$ iff it has an affine model over every $\mathbb{Z}/n$, $n \le 6$." Counterexample: $x = (yx)(x(yy))$. It has models of every order up to 6, and of order 2 it is satisfied by NAND, $xy = \neg(x \wedge y)$: $yy = \neg y$, $x(yy) = \neg x \vee y$, $yx = \neg x \vee \neg y$, so $(yx)(x(yy)) = \neg((\neg x \vee \neg y) \wedge (\neg x \vee y)) = x$ (`nand2_law`). Yet no affine operation over $\mathbb{Z}/2$ satisfies it (`no_affine_model_mod2`); for $n \le 6$ its affine models exist only at $n = 1$ and $n = 5$. Three more laws (two mirror pairs in all) behave the same way, and in 39 restricted laws some sizes of the spectrum have no affine model.

6.7 After the review: linear models over finite fields

The Skeptic found the counterexample to Conjecture 6.6 by looking at linear operations over a field with 16 elements. We then searched every prime power $q \le 128$ systematically (EXP-C05-07). For $x * y = ax + by$ a term $t(x, y)$ evaluates to $\alpha x + \beta y$, where α and β are polynomials in a and b; the law $x = t$ holds exactly when $\alpha = 1$ and $\beta = 0$ in GF(q).

Theorem 6.10 *[PROVEN — computer-assisted, three tables Lean-checked; NOT ASSESSED: linear models are a standard tool of the ETP].* For the 15 open laws of this chapter, the prime powers $q \le 128$ admitting a linear model over GF(q) are those listed in EXP-C05-07. In particular:

- laws 677 and 1313 have models of order 16 — e.g. $x * y = (1 + t^3)x + t^3 y$ for 677 and $x * y = t^3 x + (1 + t^3)y$ for 1313, in GF(16) = GF(2)[t]/($t^4 + t + 1$) (the Skeptic's examples); our search finds other pairs (a, b) as well;
- laws 670, 1076 and 1286 also have models of order 16, and 1286 one of order 32;
- law 670 has a model of order 9, a size the SAT search had left UNKNOWN;
- no linear model exists at any order that the SAT search, the DRUP certificates or the ETP notes exclude (0 conflicts).

Proof. The search is exhaustive over GF(q)^2, and GF(q) is unique up to isomorphism, so "no linear model of order q" does not depend on the representation. Every model found was re-checked on its full table, and Lean's kernel checks the tables `E677_order16_law`, `E1313_order16_law` and `E670_order9_law`. ∎

Because direct products of models are models, the spectra of 677 and 1313 now contain every product of the orders 5, 7, 9, 16, … — including all $16 \cdot m$ with m in the spectrum. Which even orders below 16 occur (8, 10, 12, 14) is open.

6.8 Experiments

Experiment	Question	Result
EXP-C05-01	spectra ($n \le 6$) of the 115 nontrivial laws	52 full, 2 only $n = 1$, 61 restricted; 1 UNKNOWN (law 167 at $n = 6$, settled by Theorem 6.2)
EXP-C05-02	the 63 non-full laws to $n \le 8$; models up to isomorphism for $n \le 5$	15 patterns; 52 of 63 have only quasigroup models ($n \le 5$)
EXP-C05-03	the 15 open ETP laws, $n \le 10$	0 disagreements with the notes; Theorems 6.4–6.5
EXP-C05-04	Lean certificates for the models found	9 of 9 tables checked by the kernel
EXP-C05-05	affine models over $\mathbb{Z}/n$, $n \le 6$	H-C05-A refuted by 4 laws
EXP-C05-06	DRUP proofs for the gaps	48 of 48 verified by drat-trim
EXP-C05-07	linear models over GF(q), $q \le 128$ (after the review)	0 conflicts; order 16 for laws 677 and 1313 (Theorem 6.10)

6.9 Comparison with existing mathematics

Classical spectra. Steiner triple systems exist exactly for $n \equiv 1, 3 \pmod 6$ (Kirkman, 1847). Mendelsohn triple systems exist for $n \equiv 0, 1 \pmod 3$, $n \neq 6$ (N. S. Mendelsohn, 1971). Pairs of orthogonal Latin squares exist for all $n \neq 2, 6$ (Bose, Shrikhande and Parker, 1959–60; Euler's 36 officers). Spectra of quasigroup identities have a long literature, for example Bennett and Mendelsohn on Stein quasigroups.

The Equational Theories Project (T. Tao et al., https://teorth.github.io/equational_theories/; paper arXiv:2512.07087, https://arxiv.org/abs/2512.07087) settled the implications between the 4694 laws of order at most 4. Two notes on the project's Zulip treat spectra:

- "Spectra of order-3 equations" (https://leanprover.zulipchat.com/user_uploads/3121/C8gCuSGHJXw9lUPe23ouRkVM/Order_3_sp proves that law 66 has the Mendelsohn spectrum and law 167 the spectrum $\{0, 1 \pmod 4\}$. B. Le Floch's notes cite "Ben Gunby-Mann's notes" for law 167, which we take to be this write-up.
- B. Le Floch, "Spectra of equational laws" (July 10, 2025, https://leanprover.zulipchat.com/user_uploads/3121/apvdoKH3jwd1mrnPxZcvlXL_/spectrum.pdf) lists the 48 laws of order at most 4 that may have a non-full spectrum. Of these, 31 have no model of order 2 and 14 have one of order 2 but none of order 3. For many of them it gives only "unknown spectrum starting with …".

Not new here: Observation 6.1 as a whole; Theorem 6.2 (our addition is the Lean proof); Theorem 6.3; the Mendelsohn spectrum of law 66; all gaps at order 2 and law 883's gap at order 3, which the notes list; law 467 at order 9. *New relative to the notes:* the memberships in Theorem 6.4 and the gaps in Theorem 6.5, of which those at $n \ge 5$ are the plausible additions. We retrieved the ETP paper only partly, and its model searches may already contain these facts, so the novelty is relative to the sources we read. The three refuted hypotheses were our own.

6.10 Applications and limitations

The reusable output is a certification pipeline for finite algebra: a model becomes a table checked by Lean's `decide`, a non-existence result a DRUP proof checked by drat-trim, so neither direction rests on trusting the SAT solver. Laws such as 66 or 907 also turn a solver into a generator of triple-system-like designs, a well-known use of quasigroups; we ran no comparison with dedicated design algorithms.

Limitations. Only two-variable laws $x = t$ of order at most 4 were studied; the census stops at $n \le 6$ ($n \le 8$ and 10 for subsets); solver budgets left UNKNOWN entries. **Most non-existence results of the census are trusted from the solver** — only the 48 frontier instances are independently checked. An isomorphism-grouping step of EXP-C05-02 compared non-canonical tables, so its count is not used.

6.11 Open questions

1. Decide laws 677 and 1313 at orders 8, 10, 12 and 14 (Conjecture 6.6 is refuted at 16), and law 1286 at orders 8 and 10, with stronger symmetry breaking.
2. Is the spectrum of law 115 all $n \neq 2, 6$, as the notes conjecture? Our data agree up to $n = 8$.
3. Direct products of models are models, so every spectrum is closed under multiplication. Which multiplicatively closed sets arise from a single law of small order?
4. Which laws admit a finite-order "pair map" like F in Theorem 6.2, and which congruences do such maps force?
5. Extend the census to laws of order 5 and to three variables, where the central groupoid law (spectrum: the squares, by Knuth) already appears.

7 Anchored operations and arithmetic spectra

Volume 2 — Invented operations and their properties. Research cycle C06.

7.1 Motivation

An invented operation with no laws is just a table. A law such as associativity cuts the tables down, and Chapter 6 asks which sizes admit *any* model. Here we add one more ingredient, the simplest link between an operation and counting: we *anchor* the operation to the successor,

$$x \star 0 = x + 1 \ (\mathrm{mod}\ N),$$

and ask, for a single composition law, for which sizes N a model exists and how many there are. The answers turn out to be pieces of arithmetic.

7.2 Definitions

A *successor-anchored magma* of size N is a binary operation $\star$ on $\mathbb{Z}/N$ with $x \star 0 = x + 1$. The *spectrum* of a law is the set of N admitting an anchored model. An isomorphism of anchored structures must fix 0 and commute with $x \mapsto x \star 0$, so it is the identity; we therefore count tables. A table is *affine* if $x \star y = x + ay + 1$ for some $a \in \mathbb{Z}/N$.

7.3 The classical case: left-distributivity

For the left-distributive law $x \star (y \star z) = (x \star y) \star (x \star z)$ the anchor produces the famous **Laver tables**.

Theorem 7.1 *[REDISCOVERY: Laver; Drápal — COMPUTATIONALLY SUPPORTED in the lab].* An anchored left-distributive magma of size N exists iff N is a power of 2, and it is then unique.

Our evidence is independent of Laver's recursion: a complete SAT enumeration finds exactly one model for $N = 1, 2, 4, 8$ and none for the other $N \le 10$ (EXP-C06-02); Laver's recursion gives left-distributive tables exactly at the powers of two up to 64 (EXP-C06-01). Lean checks the table A_3 and the non-existence for $N = 3$. The general theorem is published (Laver; elementary proof by Drápal; Dehornoy's book) and we do not re-prove it.

7.4 When the anchor pins everything

Theorem 7.2 *[PROVEN — Lean: `anchored_assoc_unique`].* An anchored associative operation is $x \star y = x + y + 1$, for every N.

Theorem 7.3 *[PROVEN — Lean: `anchored_rd_unique`].* An anchored right-distributive operation, $(x \star y) \star z = (x \star z) \star (y \star z)$, is $x \star y = x + 1$, for every N.

In both proofs, putting $z = 0$ moves the anchor *along* the law and pins the whole table. Left-distributivity instead sends the anchor through *other rows*, $p \star (q + 1) = (p \star q) \star (p + 1)$, and the consistency of that recursion is a genuine arithmetic constraint on N.

7.5 A bold hypothesis fails

H-C06-2 [REFUTED]. "Restricted spectra are special to left-distributivity: every other law gives either every N or only $N = 1$."

We tested all 382 laws $x(yz) = t$ and $(xy)z = t$ whose right side has three or four leaves using exactly x, y, z (EXP-C06-03): 77 have models for every $N \le 6$, 283 only for $N = 1$, and **22** have a restricted spectrum — only one of them left-distributivity:

spectrum (N ≤ 8)	number of laws	example
$\{1, 2\}$	10	$x(yz) = ((xx)y)z$
$\{1, 3\}$	7	$x(yz) = ((zx)x)y$
$\{1, 5\}$	2	$(xy)z = x(y(zz))$
odd N	2	$(xy)z = x((yz)z)$
powers of 2	1	$x(yz) = (xy)(xz)$

7.6 The odd law and the golden law

Theorem 7.4 *[PROVEN — Lean: `odd_law_even_impossible, odd_law_formula, odd_law_model; POTENTIALLY NOVEL (small)].* The law $(x \star y) \star z = x \star ((y \star z) \star z)$ has an anchored model iff N is odd, and then exactly one: $x \star y = x + 1 + y/2$.

Proof. Put $z = 0$: $(x \star y) + 1 = x \star (y + 2)$. Going $N/2$ times around an even cycle would give $x \star y = x \star y + N/2$; for odd N every y is $2k$ and $x \star y = x + 1 + k$. ∎

Theorem 7.5 *[PROVEN — written, Skeptic-reviewed; Lean (Chapter 8): `golden_law_models; POTENTIALLY NOVEL (small)].* The anchored models of $(x \star y) \star z = x \star (y \star (z \star z))$ are exactly the tables $x \star y = x + 1 + ay$ with

$$a^2 + a \equiv 1 \pmod{N}.$$

So the spectrum is the set of moduli in which the golden ratio exists: $N = 5^e m$ with $e \le 1$ and every prime factor of m congruent to $\pm 1 \bmod 5$ — $1, 5, 11, 19, 29, 31, 41, 55, \ldots$ (OEIS A089270) — and the number of models is the number of roots (OEIS A087781).

Proof idea. Putting $z = 0$ gives $x \star (y \star 1) = (x \star y) + 1$. Iterating $\sigma(y) = y \star 1$ from 0 visits every element (the values $0 \star \sigma^k(0) = 1 + k$ are distinct), so the table is $x \star y = x + 1 + \kappa(y)$ for a bijection κ; the law then forces $\kappa(y + c) = \kappa(y) + 1$ with c a unit, so κ is linear, $\kappa(y) = ay$, and the law reduces to $a^2 + a = 1$. Quadratic reciprocity and Hensel's lemma count the roots. ∎

The predictions were tested before they were proved: the SAT solver finds exactly 1, 1 and **2** models at $N = 1, 5, 11$ and none at the other $N \le 12$ (EXP-C06-04).

A second hypothesis fails. H-C06-3 guessed that *every* spectrum other than Laver's is explained by affine tables. Six laws refute it — e.g. $x(yz) = y(x(yz))$ has models for every N, none of them affine — but for 20 of the 22 restricted laws every model found ($N \le 9$) is affine.

7.7 Two counting theorems

Theorem 7.6 *[PROVEN — written].* The anchored magmas satisfying $(xy)z = (xz)y$ are exactly $x \star y = x + c(y)$ with $c(0) = 1$; there are N^{N-1} of them. (The law says that right translations commute; they all commute with the successor, an N-cycle, whose centraliser consists of its powers.)

Theorem 7.7 *[PROVEN — Lean core: `centralizer_eq`, `comp_eq`, `id_eq`, `leftPerm_assoc`, `leftPerm_comm`; REDISCOVERY in substance — a two-line lemma on left-permutable groupoids with a right identity]*. Anchored magmas satisfying $x(yz) = y(xz)$ correspond bijectively to commutative monoid structures on $\mathbb{Z}/N$ with identity 0, via $a \cdot b = (a - 1) \star b$. Their numbers $1, 2, 9, 94, 1486, \ldots$ are OEIS A058156.

Proof idea. The maps $f_m = L_{m-1}$ commute and satisfy $f_m(0) = m$. Any map g commuting with all of them satisfies $g(m) = g(f_m(0)) = f_m(g(0)) = f_{g(0)}(m)$: the family is its own centraliser. So it is closed under composition and contains the identity — it is the regular representation of a commutative monoid. ∎

The OEIS match came first; the explanation followed from asking *why* a count of magmas should equal a count of monoids.

7.8 Which sets can be spectra?

Theorem 7.8 *[PROVEN — written]*. If a set of laws has anchored models of coprime sizes N and M, it has one of size NM.

Proof. Take the direct product of the two models and transport it to $\mathbb{Z}/NM$ along the Chinese remainder isomorphism, which sends $x + 1$ to $(x + 1, x + 1)$ and therefore preserves the anchor. ∎

Every spectrum contains $N = 1$; Theorem 7.8 adds that a spectrum containing two primes contains their product, so it can never be $\{1\}$ together with an infinite set of primes. Every spectrum we have seen is also closed under *divisors*; whether that is always so is the first question of the next cycle.

7.9 Experiments

Experiment	Question	Result
EXP-C06-01	Laver’s recursion	LD exactly at powers of 2 (N ≤ 64); periods A098820
EXP-C06-02	model counts for 12 laws	LD: 1 at 1, 2, 4, 8; N^{N-1}; A058156
EXP-C06-03	spectra of 382 laws	22 restricted; H-C06-2 refuted
EXP-C06-04	affine models	golden and odd predictions exact; H-C06-3 refuted

7.10 Comparison with existing mathematics

Theorem 7.1 is Laver’s; the number theory in Theorem 7.5 is classical and the moduli are catalogued in the OEIS. Left-permutable groupoids are studied in the literature on groupoid identities, and the centraliser argument is the regular representation. We found no prior statement of the odd law, the golden law or the left-permutable correspondence for anchored operations; each is small.

7.11 Applications and limitations

No application is claimed. The data cover laws with at most four leaves on the right and sizes up to 8–12; spectra of larger laws are unexplored.

7.12 Open questions

1. ~~Prove that the laws with spectra~~ $\{1,2\}$ ~~and~~ $\{1,3\}$ ~~have no other models.~~ Chapter 8 proves it for 16 of the 17 laws (two of them in Lean); only $x(yz) = y((yx)z)$ remains.
2. Which sets of integers are anchored spectra of a single law? They are closed under coprime products (Theorem 7.8); are they closed under divisors?
3. Are there laws other than left-distributivity with non-affine, Laver-like spectra?

Volume 3 — Emergent Mathematical Structures

Laws imply other laws. This volume studies the structure that emerges among the invented laws themselves: which identities follow from which, and what the resulting order looks like.

8 Which sets of integers are spectra?

Volume 3 — Emergent mathematical structures. Research cycle C13.

8.1 Motivation

Chapter 7 anchored an invented operation to the successor, $x \star 0 = x + 1$ on $\mathbb{Z}/N$, and found that single laws cut out arithmetic sets of sizes: the powers of two (Laver), the odd numbers, the moduli in which the golden ratio exists. Every such *spectrum* is closed under products of coprime sizes (Theorem 7.8). This chapter asks the converse question: **which sets of integers are spectra?** Two answers emerge. Through the *affine* models a single law can encode a remarkable range of classical number theory — quadratic, cubic and cyclotomic conditions modulo N. But spectra in general are wilder than the examples suggested: they need not even be closed under divisors.

8.2 Definitions

An anchored table is *affine* if $x \star y = x + ay + 1$ for some $a \in \mathbb{Z}/N$ (under the anchor these are the only tables of the form $\alpha x + \beta y + \gamma$). Evaluating both sides of a law in the affine table turns it into a finite set of polynomials in $\mathbb{Z}[a]$, its *affine system*; the *affine spectrum* $\mathrm{Aff}(E)$ is the set of N for which the system has a common root modulo N. Always $\mathrm{Aff}(E) \subseteq \mathrm{Spec}(E)$; a law is *affinely rigid* if every anchored model is affine, so that the two coincide.

8.3 Affine spectra are arithmetic

Theorem 8.1 *[PROVEN — written].* Affine spectra are closed under divisors and under coprime products, so membership of N depends only on the prime powers dividing N. If the polynomials of the affine system have no common factor over $\mathbb{Q}$, the affine spectrum is the set of divisors of a single integer L; otherwise it is infinite (by Schur's theorem).

The laboratory computed the affine spectra of all 18,581 laws with at most four leaves per side over x, y, z, and of all 75,625 laws over x, y with at most five leaves per side (EXP-C13-01). They fall into 27 and 125 distinct spectra. Every finite one is a divisor set $\mathrm{Div}(L)$, with L ranging over 80 values up to 328. The infinite ones read like a table of contents of elementary number theory:

affine condition	spectrum	OEIS	example law
$2a + 1 = 0$	odd numbers	A005408	$x = x((xx)x)$
$3a + 1 = 0$	$3 \nmid N$	A001651	$((xx)x)x = ((yy)y)y$
$a^2 + 1 = 0$	-1 is a square mod N	A008784	$x(xx) = y(xy)$
$a^2 - 2a - 1 = 0$	2 is a square mod N	A057126	$(xx)(xy) = (yy)(yx)$
$a^2 + 2a - 2 = 0$	3 is a square mod N	A057125	(5 leaves)
$a^2 + a + 1 = 0$	$x^2 + xy + y^2$ represents N primitively	A034017	$x = x(x(xx))$
$a^2 - a - 1 = 0$	golden ratio exists mod N	A089270	$x(xy) = y(yx)$

affine condition	spectrum	OEIS	example law
$2a^2 + 2a + 1 = 0$	all prime factors $\equiv 1 \mod 4$	A004613	(5 leaves)
$2a^2 - 1 = 0$	all prime factors $\equiv \pm 1 \mod 8$	A058529	$x((xy)y) = y((xx)x)$
$a^4 + 1 = 0$	-1 is a fourth power mod N	A192453	(5 leaves)
$a^4 - a^2 + 1 = 0$	all prime factors $\equiv 1 \mod 12$	A354467	(5 leaves)

Of the 45 infinite affine spectra found, 29 have **no match in the OEIS** at all; most come from cubic and quartic conditions — the plastic number ($a^3 - a - 1$), the tribonacci and tetranacci polynomials, the fifth cyclotomic polynomial. Spot checks against the direct definitions up to $N = 1000$ (for example "-1 is a square modulo N") found no discrepancy (EXP-C13-09).

8.4 …but most laws are not rigid

H-C13-2 [REFUTED]. "Every law with an infinite, non-trivial affine spectrum is affinely rigid." The laboratory blocked all affine tables and asked a SAT solver for any other model (EXP-C13-02). Of 199 laws — the three shortest of every class — **198 have non-affine models**, most of them at every size from 3 to 8. The shortest laws in each class are weak (for example $x = x(x(xx))$ only constrains each element's own diagonal), and weak laws have many models: the affine spectrum of $x = x(x(xx))$ is the Eisenstein set A034017, but its true spectrum turns out to be every N except 2 (Theorem 8.6 below).

So affine spectra are lower bounds. To realise an arithmetic set as a *true* spectrum, a law must be shown to be rigid, and the laboratory found a large family where this can be proved.

Theorem 8.2 *[PROVEN — Lean: `right_normed_translation`].* For any term $V(y, z)$, every anchored model of $(x \star y) \star z = x \star V(y, z)$ is a *translation table* $x \star y = x + 1 + \kappa(y)$, with κ a permutation fixing 0.

Proof idea. Putting $z = 0$ gives $x \star \sigma(y) = (x \star y) + 1$ with $\sigma(y) = V(y, 0)$. Iterating σ from 0 therefore adds one each time, so it visits every element exactly once, and κ counts the steps. ∎

Theorem 8.3 (the W-family) *[PROVEN — Lean: `w_family_models`, `w_family_model_of_root`].* For every term $W(z)$ in one variable, the anchored models of

$$(x \star y) \star z = x \star (y \star W(z))$$

are exactly the affine tables with $a \cdot c_W(a) \equiv 1 \pmod N$, where $c_W(a) = \sum_{\text{leaves}} a^{(\text{right depth})}$. The polynomials c_W that occur are exactly the polynomials with non-negative integer coefficients, constant term 1 and no gaps *[PROVEN — written].*

For the right comb $W = z(z(\cdots z))$ with n leaves the condition is that the n-bonacci polynomial $t^n - t^{n-1} - \cdots - 1$ has a root modulo N: the golden law of Chapter 7 is $n = 2$, and $n = 3$ gives a "tribonacci law" whose spectrum $1, 2, 7, 11, 13, 14, 17, \ldots$ is not in the OEIS. For the left comb with $k + 1$ leaves the condition is $ka^2 + a - 1 \equiv 0$, of discriminant $1 + 4k$. The theorem was tested before it was proved: for all 23 terms with at most five leaves and all $N \le 9$, the number of models equals the number of roots (EXP-C13-03).

8.5 Finite spectra, proved by machine

Chapter 7 conjectured that seventeen laws have spectrum exactly $\{1,3\}$ or $\{1,2\}$ (CLM-C06-10). All seventeen agree with the conjecture for every $N \le 12$ (EXP-C13-04).

Theorem 8.4 *[PROVEN — Lean: `law13_size_dvd_three, law12_size_dvd_two`].* The laws $x(yz) = ((zx)x)y$ and $x(yz) = ((xy)z)x$ have spectra exactly $\{1,3\}$ and $\{1,2\}$.

Proof idea for the first. With $x = 0$ the law says $0 \star (y \star z) = (z + 2) \star y$, and with $y = 0$ as well, $0 \star (0 \star z) = z + 3$. Together: $w \star y = (w - 2) \star (y - 2) + 3$. Walking $N/2$ steps around an even cycle gives $N/2 \equiv 0$, impossible; for odd N the walk shows that the table is affine with $a = 1/2$, and then the law forces $3 \equiv 0$. ∎

Such arguments have a common shape — instances of the law at small elements, the anchor, and the fact that the table is a function — and they can be automated.

Theorem 8.5 *[PROVEN — written; computer-assisted applications].* The *offset closure* procedure of the laboratory, which records equalities of the form "entry = element + integer" and merges them, is sound: whenever it derives $N \mid g$, every anchored model has size dividing g.

Combined with the affine models, it proves the exact spectrum of **16 of the 17 laws** of the conjecture in milliseconds each (EXP-C13-08); the seventeenth, $x(yz) = y((yx)z)$, is the one law with a non-affine model (at $N = 2$), and it remains supported only up to $N = 12$. Run over all 18,581 laws with at most four leaves per side, the procedure proves an exact finite spectrum $\mathrm{Div}(g)$ for 6,912 of them (6,045 have only the trivial model). Its trust base is the closure code; checked against the SAT solver on 1,245 sizes of 235 laws, it was never contradicted.

8.6 A bold hypothesis fails: spectra are not closed under divisors

Every spectrum seen in Chapter 7 was closed under divisors, and so are all affine spectra (Theorem 8.1). *H-C13-1* conjectured that this always holds.

Theorem 8.6 *[PROVEN — Lean: `cube_law_no_two, cube0p_model`].* The one-variable law $x = x \star (x \star (x \star x))$ has anchored models of every size except 2. In particular it has a model of size 4 but none of size 2.

Proof. At $N = 2$ the anchor forces $0 \star 0 = 1$, and neither choice of $0 \star 1$ returns 0 after three steps. For $N \ge 3$ let $0 \star 1 = 2$, $0 \star 2 = 0$ and $x \star x = x$ for $x \ne 0$. ∎

Divisor-closure fails often. Among all 253 one-variable laws with at most five leaves per side, 24 violate it for $N \le 12$, with spectra missing exactly $\{2\}$, $\{3\}$ or $\{2,3\}$ (EXP-C13-07). In random samples the proportion is higher: 81 of 300 two-variable laws (sizes up to 9), 29 of 150 three-variable laws and 17 of 80 pairs of laws (sizes up to 8) violate it (EXP-C13-05, EXP-C13-06). The law $((xy)x)y = (y(xx))x$, for instance, has models of sizes 6 and 8 but none of sizes 2, 3, 4 or 5. The same survey refutes a second guess, that one-variable laws have either only the trivial model or models of all large sizes: $x = (xx)x$ has spectrum exactly $\{1,2\}$.

What *is* true is the weaker structure: spectra are closed under coprime products (Theorem 7.8), and divisor-closed whenever the law is affinely rigid.

8.7 Experiments

Experiment	Question	Result
EXP-C13-01	affine spectra of 94,206 laws	27 + 125 spectra; finite ones are Div(L); classical sets
EXP-C13-02	are infinite affine classes rigid?	198 of 199 laws have non-affine models (H-C13-2 refuted)
EXP-C13-03	the W-family	23 terms, $N \le 9$: model counts = root counts
EXP-C13-04	the 17 laws of CLM-C06-10	spectra agree for $N \le 12$
EXP-C13-05	divisor-closure, 300 random 2-variable laws	81 violations
EXP-C13-06	divisor-closure, 3-variable laws and pairs	29 of 150 laws, 17 of 80 pairs violate
EXP-C13-07	one-variable laws	24 of 253 violate divisor-closure
EXP-C13-08	offset-closure proofs	16 of 17 conjectured spectra; 6,912 exact finite spectra; 0 contradictions in 1,245 checks
EXP-C13-09	Skeptic cross-checks	cube-law models to $N = 60$, six-leaf W-family, identifications to $N = 1000$

8.8 Comparison with existing mathematics

The arithmetic sets of the atlas are classical: they are the sets of moduli where a polynomial has a root, or equivalently (for quadratics) the numbers primitively represented by a binary quadratic form, and their OEIS entries say so. What we did not find in the literature is their appearance as spectra of single magma laws with a successor anchor, nor the W-family theorem. The reduction of identities to polynomial conditions on the coefficients of affine (linear) models is the classical method of linear representations of quasigroups and medial magmas; here the anchor removes all freedom except one parameter. Laws whose finite models are counted or constrained by sizes are studied in universal algebra as *spectra of varieties*; our anchored spectra are a variant with a distinguished successor, and we make no claim about the general spectrum problem.

8.9 Applications and limitations

No application is claimed. The atlas covers laws with at most four leaves (three variables) or five leaves (two variables); the rigidity and divisor-closure searches cover sizes up to 8–12 only; the offset-closure prover is sound but incomplete.

8.10 Open questions

1. Prove the spectrum $\{1, 2\}$ of $x(yz) = y((yx)z)$, the last case of CLM-C06-10.
2. Which arithmetic sets are *true* spectra of rigid laws? Can "-1 is a square modulo N" be realised by a rigid law?
3. Is every spectrum of a one-variable law finite or cofinite?

4. Are there laws besides left-distributivity whose spectrum is infinite but of density zero and comes from non-affine models?

Volume 4 — Alternative Representations of Quantities

Once numbers exist, they can be written in many ways. This volume studies operations that live on representations — multiplying digit positions (Knuth’s circle product and its relatives), the arithmetic of the resulting monoids, the cost of counting in different numeration systems, and integer codes.

9 Multiplying digit positions

Volume 4 — Alternative representations of quantities. Research cycle C08.

9.1 Motivation

In base 10, multiplying 10^i by 10^j gives 10^{i+j}: digit positions add. What happens if we *define* a multiplication by adding digit positions in a numeration system where that is not automatically consistent? Let G be an increasing base sequence ($G_0 = 1$), let $n = \sum_i d_i(n) G_i$ be the greedy expansion, and set

$$a \circ_s b = \sum_{i,j} d_i(a)\, d_j(b)\, G_{i+j+s}.$$

For Fibonacci weights $G = 1, 2, 3, 5, 8, \ldots$ and $s = 2$ this is Knuth's *circle multiplication* (1988), famous for being associative for no visible reason. Our implementation reproduces its OEIS table A101330 (EXP-C08-05).

9.2 The associativity atlas

We tested 46 base sequences × shifts $s = 0..5$ exhaustively on $[1, 22]^3$, then every candidate on random triples up to 10^6 (EXP-C08-01..03). The pattern is governed by the roots of the sequence's *minimal* recurrence:

class of G	**sequences**	**associative shifts**
geometric ($G_k = b^k$)	5	all s
Pisot (other roots $\lvert\beta\rvert < 1$)	31	all s beyond a threshold (most cases; see the correction below)
non-Pisot (some $\lvert\beta\rvert \geq 1$)	10	**none** ($s \leq 8$)

Thresholds grow as $\lvert\beta\rvert \to 1$: Pell ($\lvert\beta\rvert = 0.41$) from $s = 1$, Fibonacci (0.62) from 2, tribonacci (0.74) not before 4, 4-bonacci not before 8, $G_{k+1} = G_k + G_{k-2}$ (0.83) not before 11, $G_{k+1} = 2G_k + 2G_{k-1}$ not before 7 and $G_{k+1} = 3G_k + 3G_{k-1}$ (with $G_1 = 5$) not before 12; $G_{k+1} = G_k + G_{k-3}$ ($\lvert\beta\rvert \approx 0.94$) fails at every shift tested ($s \leq 16$ in EXP-C08-07, $s \leq 24$ in the Skeptic's search).

> **Correction (2026-10-18).** The first version of this chapter gave the thresholds 3 (tribonacci), 7 (4-bonacci), 10 ($G_k + G_{k-2}$) and 5 ($G_{k+1} = 2G_k + 2G_{k-1}$). All four are wrong. The laboratory's Historian found the tribonacci counterexample in the literature — $(10016 \circ_3 10016) \circ_3 1 = 3414681926 \neq 3414681924 = 10016 \circ_3 (10016 \circ_3 1)$, the example by which Ambrož, Frougny, Masáková and Pelantová (2003) refuted a conjecture of Arnoux — and one more by computation. A certificate-producing re-test (EXP-C08-06) found the other two. The original tests sampled random triples up to 10^6, and such sampling misses rare failures. Every failure quoted above is now certified by an explicit triple; every "from s" is a search result, proved only where the literature proves it (Fibonacci: Lean; Pell: Messaoudi 2000; tribonacci $s \geq 6$ and 4-bonacci $s \geq 10$: Messaoudi).
>
> **Second correction (2026-10-18).** The re-test was still not good enough. The Skeptic's review found certified failures at shifts the re-test had passed: $G_{k+1} = 2G_k + 2G_{k-1}$ fails at $s = 6$ and $G_{k+1} = 3G_k + 3G_{k-1}$ ($G_1 = 5$) at $s = 10$ and 11,

> e.g. $(173938 \circ 173938) \circ 1 \neq 173938 \circ (173938 \circ 1)$. Both the random triples and the small pairs of the re-test avoid the inputs where failures live: numbers whose conjugate coordinates are extreme, so that the conjugate of a product leaves the window. A third search aimed there (EXP-C08-07) reproduces every failure the Skeptic found; the values in the paragraph above are its results, with the same caveat: lower bounds are certified, "not before" is all we can say.

9.3 Why: a twisted embedding into an algebraic number field

Theorem 9.1 *[PROVEN — Lean: `circ_assoc, circ_comm, key, window; REDISCOVERY (Knuth, Arnoux)].* For every $t \geq 0$ the shifted Knuth product $a \circ_t b = \sum_{k \in Z(a),\, l \in Z(b)} F_{k+l+t}$ (Zeckendorf indices ≥ 2) is associative and commutative. For the two smaller shifts the product is not associative.

Proof. Put $E(n) = \sum_{k \in Z(n)} \varphi^k$ and $E'(n) = \sum_{k \in Z(n)} \psi^k$ ($\psi = -1/\varphi$). Since $\varphi^k = F_k\varphi + F_{k-1}$, $E(n) = n\varphi + n^*$ and $E'(n) = n\psi + n^*$ for an integer n^*. The conjugates of Zeckendorf sums lie in a window of length 1: $|E'(n)| < -\psi$ (the Lean proof) or, sharper, $-\psi^2 < E'(n) < -\psi$. For a product, $\varphi^t E(a)E(b) = P\varphi + Q$ and $\psi^t E'(a)E'(b) = P\psi + Q$ with $P = a \circ_t b$; the second number has absolute value $< \psi^2$, so $|Q - P^*| < \psi^2 - \psi = 1$, forcing $Q = P^*$. Hence $E(a \circ_t b) = \varphi^t E(a)E(b)$ — the product becomes multiplication in $\mathbb{Z}[\varphi]$ twisted by φ^t — and associativity follows because n is recovered from $(E(n), E'(n))$. ∎

The same picture explains the atlas. The conjugate coordinates of a product are multiplied by β^s; if $|\beta| < 1$ large shifts push them into the window of greedy expansions, if $|\beta| \geq 1$ nothing can.

9.4 A surprise: non-monotone thresholds

For $G = 1, 3, 4, 7, 11, \ldots$ (a Lucas-type sequence) no failure is found for $s = 2$, $s = 3$ fails (certified), and no failure is found again for $s \geq 4$ — even under the targeted search of EXP-C08-07. Our explanation: β^s alternates in sign when $\beta < 0$, and the window is asymmetric about 0 (for Fibonacci $(-0.382, 0.618)$), so an odd shift can push a product's conjugate out.

Conjecture 9.2 *[COMPUTATIONALLY SUPPORTED; POTENTIALLY NOVEL (low)].* For quadratic Pisot sequences with positive conjugate root the associative shifts form an upward-closed set; non-monotone sets occur only for negative conjugates — and, as the Historian pointed out, only for *non-canonical* initial values: for canonical sequences Messaoudi's criterion implies upward-closure whatever the sign of β. *Evidence (after the targeted search EXP-C08-07, shifts* $0 \ldots 12$*):* 0 of 30 positive-β sequences non-monotone; 8 non-monotone sequences, all non-canonical with $\beta < 0$. Four of the original ten examples were artefacts of random testing. The Skeptic's independent search agrees: 0 of 231 positive-β sequences non-monotone ($s \leq 16$), 60 of 396 negative-β ones, none canonical.

Conjecture 9.3 *[COMPUTATIONALLY SUPPORTED].* For non-Pisot sequences whose dominant root exceeds 1 the circle product is non-associative at every shift. (The first version omitted "dominant root exceeds 1"; the Skeptic noted that $G_k = k + 1$, with minimal polynomial $(x-1)^2$, gives $a \circ_s b = a + b + s - 1$, associative at every shift.)

9.5 Comparison with existing mathematics

Knuth (1988) proved associativity for Fibonacci; Arnoux (1989) found the $\mathbb{Z}[\varphi]$ embedding; Grabner, Pethő, Tichy and Woeginger (1994) proved eventual associativity for recurrences with non-increasing coefficients and canonical initial values — our Pisot observations are a **rediscovery** of their theorem in that range (the first version of this chapter overstated it as “all Pisot sequences”). A. Messaoudi (Math. Slovaca 2000) characterises associativity by the stability of the conjugate window — exactly our mechanism — and proves the Fibonacci case for all $s \geq 2$; Ambrož, Frougny, Masáková and Pelantová (2003) bound the fractional digits of products of tribonacci integers. We did not find the non-Pisot statement or the sign phenomenon.

9.6 Open questions

1. Prove Conjectures 9.2 and 9.3 by computing the greedy windows (Rauzy-type fractals for higher degree).
2. Determine the exact threshold $s_0(G)$ from the geometry of the window.

10 Primes of an invented multiplication

Volume 4 — Alternative representations. Research cycle C09.

10.1 Motivation

Chapter 9 produced a new associative, commutative operation on the positive integers. Every such operation has an arithmetic: which numbers are *irreducible* (not $a \circ b$), how many are there, and is factorisation unique?

10.2 Exact formulas

From the embedding $v(n) = n\varphi + n^*$ with $n^* = \lfloor (n+1)/\varphi \rfloor$:

$$a \circ b = ab + ab^* + a^* b, \qquad (a \circ b)^* = ab + a^* b^*,$$

validated against the digit definition on 89,401 pairs.

10.3 Structure

Theorem 10.1 *[PROVEN — written, resting on Theorem 9.1].* $(\mathbb{N}_{\geq 1}, \circ)$ is commutative and cancellative, has no identity, and embeds into the multiplicative monoid of $\mathbb{Z}[\varphi]$. The Fibonacci numbers are exactly the elements mapped to units ($v(F_k) = \varphi^k$), and $F_j \circ F_k = F_{j+k}$. The norm $N(n) = n^{*2} + nn^* - n^2$ is multiplicative.

Refuted hypothesis. Factorisation into irreducibles is *not* unique: $8 = 1 \circ 1 \circ 1 = 2 \circ 2$ (EXP-C09-02). The reason is pretty: 1 and 2 are F_2 and F_3, so factorisations of F_k are factorisations of k in the numerical semigroup $\langle 2, 3 \rangle$.

10.4 Circle primes are common

We conjectured (H-C09-1) that circle irreducibles thin out like primes. The data (EXP-C09-01) disagree: up to 10^6 more than half of all integers are irreducible (density 0.600, 0.577, 0.562, 0.549, 0.539 at $10^2 .. 10^6$). A proof explains why.

Theorem 10.2 (window obstruction) *[PROVEN — written; POTENTIALLY NOVEL].* If $n = a \circ b$ then the conjugate $v'(n) = v'(a)v'(b)$ lies in (ψ^3, ψ^2), i.e. the fractional part of $(n+1)/\varphi$ lies in $(0.2360\ldots, 0.8541\ldots)$, an interval of length $1/\varphi$. Hence every n outside it is irreducible, and by Weyl equidistribution these n have density exactly $1/\varphi^2 \approx 0.381966$.

Numerically: not a single reducible number up to 10^6 lies outside the zone, and the forced-irreducible density at 10^6 is 0.381966 (EXP-C09-04).

Theorem 10.3 *[PROVEN — written].* Every factorisation of n into r irreducibles satisfies $|v'(n)| < |\psi|^r$; factorisation lengths are bounded.

Conjecture 10.4 *[CONJECTURE].* The circle irreducibles have density exactly $1/\varphi^2$: almost every n in the product zone is reducible. The irreducible fraction inside the zone falls slowly — 0.316, 0.291, 0.270, 0.255 at $10^3 .. 10^6$.

10.5 A lesson in reading the OEIS

The list of irreducibles matched OEIS A026273 for 24 terms. Reading the entry term by term showed the sequences diverge at 40 ($= 4 \circ 4$): for $n < 40$ every product has the form $1 \circ b$, a Beatty-like set related to the Wythoff sequences, which explains the coincidence. The irreducibles of Knuth's product appear not to be catalogued.

10.6 Open questions

1. Prove Conjecture 10.4.
2. Describe the irreducibles through prime factorisation in $\mathbb{Z}[\varphi]$ plus a unit-window condition.
3. The same questions for other associative circle products.

11 The cost of counting

Volume 4 — Alternative representations. Research cycle C10.

11.1 Motivation

The successor $n \mapsto n+1$ is the most basic operation on quantities. Its *cost* in a representation — the number of digit positions that change — is a natural measure of how well the representation suits counting, and it matters in practice (counters in data structures and hardware).

Which measure. Throughout this chapter the cost is the Hamming distance between the representations of n and $n+1$. Berthé, Frougny, Rigo and Sakarovitch (Adv. Appl. Math., 2020) study the related *carry propagation* measure systematically; under theirs the Zeckendorf value is φ^2, not the 2 we obtain below. The two measures agree for ordinary bases.

11.2 Classical constants

Theorem 11.1 *[PROVEN — written; REDISCOVERY].* The average cost per increment is $b/(b-1)$ in base b (2 in binary, $10/9$ in decimal), the same in negabinary and balanced ternary, $e-1 \approx 1.71828$ in the factorial base, and exactly 1 for the Gray code. *(Digit k changes with frequency b^{-k}, respectively $1/k!$.)*

Measured over $n < 10^6$ (EXP-C10-01): binary 1.999988, factorial 1.7182767, Gray 1.0, skew binary 1.4999955 with worst case 2.

11.3 Zeckendorf counts like binary

The Inventor guessed that the sparse Zeckendorf system (no two adjacent ones) would be cheaper to count in than binary. It is not.

Theorem 11.2 *[PROVEN — written; REDISCOVERY in substance: OEIS A215005].* Let T_m be the total cost of counting from 0 to $G_m - 1$ in the Zeckendorf system ($G = 1, 2, 3, 5, \ldots$). Then

$$T_{m+1} = T_m + T_{m-1} + \lceil m/2 \rceil + 1, \qquad T_m = 2G_m - \lceil m/2 \rceil - 2,$$

so the average cost tends to exactly 2, the same as binary.

Proof idea. The numbers below G_{m+1} split into $[0, G_m)$ and a block of G_{m-1} numbers beginning "10"; the step from $G_m - 1 = 1010\cdots$ to $G_m = 10\cdots0$ costs $\lceil m/2 \rceil + 1$. ∎

The sequence T_m is OEIS A215005; the per-increment cost sequence is not catalogued.

11.4 Exchange systems

For the exchange numerations of Chapter 3 two clean laws hold.

Theorem 11.3 *[PROVEN — written].* A terminating carry rule $m[k] \to \sum b_j [k+j]$ fires on average $1/(m - \sum b_j)$ times per increment (the abelian property makes one-at-a-time counting fire exactly as often as stabilising n tokens at once).

Theorem 11.4 *[PROVEN — written]*. If only one $b_g = q$ is nonzero, every increment changes exactly (firings $+1$) digits, so the average cost is $1 + 1/(m - q)$: 2 for base $3/2$, $3/2$ for base 3.

With two or more nonzero b_j a level can receive several tokens in one increment, and the cost carries a rule-specific excess — 0.500 for $3[k] \to [k+1] + [k+2]$, 0.333 for $4[k] \to [k+1] + [k+2]$, 0.831 for $4[k] \to [k+1] + [k+2] + [k+3]$ (EXP-C10-03). *Conjecture:* these are expectations under the invariant measure of the digit process; we have no formula.

11.5 Application note

As counters, none of the invented exchange representations beats the classical choices: Gray code (cost 1 at every step) or skew binary (average 1.5, worst case 2, used in purely functional random-access lists). We record this as a negative result.

12 Codes from forbidden patterns

Volume 4 — Alternative representations of quantities. Research cycle C11.

12.1 Motivation

A *universal code* writes every positive integer as a self-delimiting bit string, so that a stream of integers of unknown size can be decoded without separators. The Fibonacci code writes the Zeckendorf digits of n (weights $1, 2, 3, 5, \ldots$, least significant first) and appends a 1; as Zeckendorf digits never contain 11, the codeword ends where 11 first appears. The code is defined by a *forbidden pattern*, and 11 is one choice among many. Replacing it by an arbitrary terminator p, we ask which terminator is best, and whether any of the resulting codes is useful against honest baselines, including an improved Fibonacci code from the literature and Golomb–Rice codes tuned to the data. The counting is governed by the autocorrelation polynomial of Chapter 2.

12.2 Definitions

Let p be a binary pattern of length k. The code T_p has as codewords the binary words in which p occurs *for the first time at the very end*, assigned to $n = 1, 2, 3, \ldots$ in order of length and then lexicographically (enumerative coding with a Knuth–Morris–Pratt automaton):

$$T_{11} = \{11,\ 011,\ 0011,\ 1011,\ \ldots\}, \qquad T_{011} = \{011,\ 0011,\ 1011,\ 00011,\ 01011,\ 10011,\ 11011,\ \ldots\}.$$

$f_p(L)$ is the number of codewords of length L, $\mathrm{cum}_p(L) = \sum_{l \le L} f_p(l)$, and the codeword of n has length $L_p(n) = \min\{L : \mathrm{cum}_p(L) \ge n\}$. As in Chapter 2, $c_p(x) = 1 + \sum_b x^{k-|b|}$, summed over the borders b of p, is the *autocorrelation polynomial*; p is *unbordered* if $c_p = 1$. The *growth rate* is $\lambda_p = 1/\rho_p$, with ρ_p the smallest positive root of

$$D_p(x) = x^k + (1 - 2x)\,c_p(x),$$

and $L_p(n)/\log_2 n \to 1/\log_2 \lambda_p$. The *redundancy* of a code on a source is its expected codeword length minus the entropy.

12.3 The pattern-terminated family

Theorem 12.1 *[PROVEN — Lean: `prefix_free, unique_parse_head; REDISCOVERY]`.* For every pattern p of length k: (a) T_p is a prefix code; (b) its Kraft sum $\sum_L f_p(L)\,2^{-L}$ equals 1; (c) its counts have the generating function

$$F_p(x) = \sum_L f_p(L)\,x^L = \frac{x^k}{x^k + (1 - 2x)\,c_p(x)};$$

(d) T_{11} has exactly the codewords of the Fibonacci code, and T_{1^k} those of its order-k generalisation.

Proof idea. (a) A codeword that is a proper prefix of another would put p before the other's end (Lean also checks unique parsing). (b) The sum is the probability that a random infinite bit sequence contains p. (c) If $A(x)$ counts the words avoiding p, appending a letter gives $1 + 2xA = A + F_p$, and appending p (whose first occurrence then ends after a period of p) gives $x^k A = c_p F_p$. (d) A word whose first 11 is at its end is a Zeckendorf digit string followed by the extra 1. ∎ By (c), T_p depends on p only through k and c_p. The 61 implemented pattern codes passed round trips, a prefix check and Kraft partial sums ≥ 0.999999999993 (EXP-C11-01).

Theorem 12.2 *[PROVEN — written; REDISCOVERY: Guibas–Odlyzko correlation ordering].* Among the binary patterns of length k, 1^k and 0^k have the largest growth rate, and every other pattern has a strictly smaller one.

Proof idea. $D_p(1/2) = 2^{-k}$ for every p. Since 1^k has every period, c_{1^k} dominates c_p coefficientwise, and for $x > 1/2$

$$D_{1^k}(x) - D_p(x) = (1-2x)(c_{1^k}(x) - c_p(x)) \le 0,$$

strictly unless $c_p = c_{1^k}$. So D_{1^k} vanishes no later than D_p. ∎ All 2044 patterns of length 2 to 10 agree (EXP-C11-02); for example $\lambda_{11} = 1.618$, $\lambda_{101} = 1.755$, $\lambda_{111} = 1.839$, $\lambda_{0101} = 1.883$ and $\lambda_{1111} = 1.928$.

The bold hypothesis fails. H-C11-2 *[REFUTED]*: "hence 1^k is the best length-k terminator beyond small n." At $n = 1000$ the terminators 1000, 1001, 1010 give codewords one bit shorter than 1111. Among the sampled $n \ge 1000$ (all $n \le 2048$ and quarter-powers of 2 up to 2^{64}), rivals beat 1^k at 762, 972 and 1127 values for $k = 4, 5, 6$, the last as late as $n \approx 1.3 \cdot 10^{19}$ (EXP-C11-02). Theorem 12.2 is about the limit; at practical sizes the constants decide.

12.4 Unbordered terminators

Theorem 12.3 *[PROVEN — written; REDISCOVERY].* If u is unbordered and $|u| = k + 1$, then

$$F_u(x) = \frac{x}{1-x} F_{1^k}(x), \qquad \text{i.e.} \qquad f_u(L) = \mathrm{cum}_{1^k}(L-1).$$

Hence $\lambda_u = \lambda_{1^k}$ is the k-bonacci constant λ, and $\mathrm{cum}_u(L)/\mathrm{cum}_{1^k}(L) \to 1/(\lambda - 1)$.

Proof. $c_{1^k} = (1-x^k)/(1-x)$ gives $F_{1^k} = x^k(1-x)/(1-2x+x^{k+1})$, and $c_u = 1$ gives $F_u = x^{k+1}/(1-2x+x^{k+1})$. ∎ The identity holds exactly for every unbordered pattern of length 3 to 9 up to length 200, and the count ratios at length 300 are 1.618033989, 1.191487884, 1.078095078 for $k = 2, 3, 4$ (EXP-C11-02, EXP-C11-06).

Corollary 12.4 *[PROVEN — written; POTENTIALLY NOVEL (low confidence)].* With $F_1 = F_2 = 1$, $\mathrm{cum}_{11}(L) = F_{L+1} - 1$ and $\mathrm{cum}_{011}(L) = F_{L+2} - L - 1$. If the Fibonacci codeword of n has length $L + 1$, the T_{011} codeword is one bit shorter when $n \le F_{L+2} - L - 1$ and equally long for the last L values of the level; T_{011} is longer than the Fibonacci code only for $n \in \{1, 2, 4\}$. The same holds for every unbordered 3-letter terminator.

Proof idea. Theorem 12.3 with $k = 2$ gives $f_{011}(l) = F_l - 1$; compare the cumulative counts level by level. ∎ For $n \le 2 \cdot 10^6$ the T_{011} codeword is one bit shorter for 1,999,568 values, equally long for 429 and longer for 3; block-wise checks up to $L = 90$ find no exception (EXP-C11-06).

Two predictions of the average gain fail [REFUTED]. For $k \ge 3$, u wins only on part of each level. The guess that the average of $L_{1^k}(n) - L_u(n)$ over $n \le N$ tends to $\log_\lambda(1/(\lambda-1))$ predicts 0.2875 for $k = 3$; the average over $n \le 2^{40}$ is 0.2282 (EXP-C11-06). The corrected guess, convergence to $(2-\lambda)/(\lambda-1)^2 = 0.228155$, fails too: the running average is 0.228181, 0.309709, 0.228548 at $N = 2^{40}, 2^{80}, 2^{120}$ (EXP-C11-07).

Proposition 12.5 *[PROVEN — written; POTENTIALLY NOVEL (low confidence)].* Let $k \ge 2$, λ the k-bonacci constant, u unbordered of length $k + 1$ and $G(N) = \frac{1}{N}\sum_{n \le N}(L_{1^k}(n) - L_u(n))$. Then

$$\liminf_{N \to \infty} G(N) = \frac{2-\lambda}{(\lambda-1)^2}, \qquad \limsup_{N \to \infty} G(N) = \frac{\lambda(2-\lambda)}{\lambda - 1}.$$

Proof idea. Put $r = 1/(\lambda - 1)$. For $k \ge 3$, on the level of T_{1^k} with codeword length L, T_u is one bit shorter exactly on the first $\mathrm{cum}_u(L-1) - \mathrm{cum}_{1^k}(L-1) \approx (r-1)\,\mathrm{cum}_{1^k}(L-1)$ values and

equally long on the rest; G rises along this segment and falls after it, and geometric sums at the two turning points give the constants. For $k = 2$ use Corollary 12.4. ∎ The pair (liminf, limsup) is $(1, 1)$, $(0.228155, 0.352201)$, $(0.084194, 0.150533)$, $(0.036495, 0.069304)$, $(0.016970, 0.033108)$ for $k = 2, \ldots, 6$; exact averages agree to six digits (EXP-C11-08).

12.5 The improved Fibonacci code C3

Fraenkel and Klein (1996) improved C1 by a code C3: delete the final 1 of each C1 codeword, take each length block twice and prefix the copies with 10 and 11 ($101, 111, 1001, 1101, 10001, \ldots$). C3 is uniquely decipherable and complete but *not* prefix-free: its decoder searches for 011 and reads one bit of the next codeword.

Corollary 12.6 *[PROVEN — written; POTENTIALLY NOVEL (low confidence)].* C3 has $2F_{r-2}$ codewords of length $r \geq 3$, so $\mathrm{cum}_{C3}(L) = 2F_L - 2$. C3 is never longer than C1 except at $n = 1$, and T_{011} is never longer than C3 except at $n \in \{2, 4, 8\}$; $\mathrm{cum}_{011}/\mathrm{cum}_{C3} \to \varphi^2/2 \approx 1.309$ and $\mathrm{cum}_{C3}/\mathrm{cum}_{C1} \to 2/\varphi \approx 1.236$. (The count and the comparison with C1 are Fraenkel and Klein's; the comparison with T_{011} is ours.)

Proof. As $F_{L+2} - 2F_L = F_{L-1}$, $\mathrm{cum}_{011}(L) - \mathrm{cum}_{C3}(L) = F_{L-1} - (L - 1)$, negative only for $L = 3, 4, 5$; and $\mathrm{cum}_{C3}(L) - \mathrm{cum}_{C1}(L) = F_{L-2} - 1 \geq 0$ for $L \geq 3$. ∎ Our C3 (its first 12 codewords match the paper's list) confirms this for $n \leq 2 \cdot 10^6$ (EXP-C11-09). The average of $L_{C1} - L_{C3}$ over $n \leq 2^{40}, 2^{80}, 2^{120}$ is 0.462, 0.400, 0.481: C3 recovers less than half of the bit that T_{011} saves.

12.6 Benchmarks

Benchmark result 12.7 *[COMPUTATIONALLY SUPPORTED — algorithmic; improvement MIXED; baselines: Elias γ, δ, ω, Levenshtein, Fibonacci C1, Fraenkel–Klein C3, LEB128, Exp-Golomb(0, 2, 4), and Golomb–Rice and Exp-Golomb codes tuned to each source].* Among untuned codes, a pattern code is best on 5 of 12 synthetic sources and on both posting lists, beating the best classical code there by 0.34–3.06 bits per value; on Zipf 1.1, log-uniform and the bursty posting list it also beats the tuned codes. It loses on light-tailed sources and to tuned codes on geometric and uniform sources, and is 4–10 times slower than the Elias and Exp-Golomb codes.

Expected lengths are exact, from the breakpoints of each length function (no sampling); codes defined on $n \geq 0$ are applied to $n - 1$. Redundancy in bits per value (EXP-C11-03; best entries bold):

source	H	best fixed classical	best pattern code	tuned Golomb	tuned Exp-Golomb
Zipf 1.1	10.90	C3 0.60	T_{101} **0.19**	5.38	1.72
Zipf 1.5	4.62	**Fibonacci 0.10**	T_{011} 0.52	6.07	0.25
Zipf 2	2.36	γ **0.08**	T_{011} 1.38	2.00	**0.08**
Zipf 3	0.98	Levenshtein 0.39	T_{011} 2.23	**0.39**	0.44
geometric 0.5	2.00	Levenshtein 0.19	T_{011} 1.63	**0.00**	0.27
geometric 0.1	4.69	Exp-Golomb(2) 0.30	T_{011} 0.72	**0.04**	0.30

source	H	best fixed classical	best pattern code	tuned Golomb	tuned Exp-Golomb
geometric 0.01	8.08	Exp-Golomb(4) 0.70	T_{011} 0.93	**0.03**	0.28
geometric 0.001	11.41	C3 2.51	T_{0111} 1.74	**0.03**	0.28
uniform 2^8	8	Exp-Golomb(4) 1.75	T_{011} 1.93	**0.50**	1.00
uniform 2^{16}	16	C3 5.71	T_{0111} 3.30	**0.50**	1.00
uniform 2^{32}	32	LEB128 7.50	T_{00111} 4.44	**0.50**	1.00
log-uniform 10^6	13.41	C3 1.36	T_{0111} **0.40**	3.93	2.00

Bits per value on data (EXP-C11-04; H_0 is the zeroth-order entropy; the tuned codes are tuned on the same data, which favours them):

data	H_0	best fixed classical	best pattern code	tuned Golomb	tuned Exp-Golomb
prime gaps $\le 10^7$	4.17	Exp-Golomb(4) 5.72	T_{011} 6.14	**5.27**	5.58
prime gaps, halved	4.17	Exp-Golomb(2) 4.58	T_{011} 5.11	**4.27**	4.58
posting list, geometric gaps	11.36	C3 13.93	T_{0111} 13.15	**11.45**	11.70
bursty posting list	10.71	C3 11.82	T_{101} **11.47**	12.80	11.79

Where pattern codes lose. On light-tailed sources (Zipf ≥ 1.5, geometric ≥ 0.01, prime gaps) γ, Levenshtein, Fibonacci or Exp-Golomb are better; even C3 beats T_{011} on Zipf ≥ 1.5, geometric 0.5 and 0.1 and the prime gaps, where its shorter codewords for $n = 2, 4, 8$ carry much probability. Tuned codes are within 0.04 bits of the entropy on geometric sources and within 0.5–1 bit on uniform ones; no universal code comes close. *Speed:* in our string-based CPython implementations (EXP-C11-05), pattern codes take about 3 µs per value, against 0.4–0.7 µs for the Elias codes and 2.1 µs for the Fibonacci code (machine-dependent; no optimised decoder was attempted).

12.7 Comparison with existing mathematics

Most of the theory of T_p is classical. The Fibonacci code and its order-k generalisation are Apostolico and Fraenkel's and Fraenkel and Klein's. The generating function of Theorem 12.1 comes from Guibas and Odlyzko's theory of string overlaps, and Theorem 12.2 is in substance their comparison of patterns through correlations; every count sequence we used is in the OEIS. The closest prior art for T_{011} is C3, which already improves C1 by decoding with 011 but gives up the prefix property. For prefix-synchronised (fixed-length) codes, unbordered markers $1\cdots10$ are optimal at finite lengths (Gilbert; Guibas and

Odlyzko), the phenomenon of Theorem 12.3 in another setting; the multi-delimiter codes of Anisimov and Zavadskyi are reported to beat Fibonacci codes. We found no statement of Corollaries 12.4 and 12.6 or Proposition 12.5; our confidence in their novelty is low, given C3, the multi-delimiter codes and the generic nature of log-periodic oscillations like Proposition 12.5's.

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- https://en.wikipedia.org/wiki/Fibonacci_coding; OEIS https://oeis.org/A000071 (T_{011}), https://oeis.org/A005314 (T_{101}), https://oeis.org/A008937 (T_{0111}), https://oeis.org/A059633 (T_{1011}), https://oeis.org/A112575 (T_{0101}).

12.8 Applications and limitations

T_{011} is a drop-in prefix-free replacement for the Fibonacci code that saves one bit on almost every integer (Corollary 12.4). For heavy-tailed streams of unknown scale (gaps in posting lists with bursty terms, magnitudes over many orders), a fixed pattern code such as T_{0111} or T_{101} is a strong universal choice; for geometric gaps use Golomb–Rice, for values of known range a tuned Golomb or Exp-Golomb code, for small values γ, Fibonacci or C3. These are empirical findings on 12 synthetic sources and 4 data sets (two simulated), comparing memoryless codes with zeroth-order entropy, with the tuned codes fitted to the evaluated data. No code is claimed optimal, and the 4–10-fold speed penalty of our implementation is unresolved.

12.9 Open questions

1. Can table-driven (byte-wise) decoders make pattern codes as fast as optimised Elias and Fibonacci decoders?
2. Which *sets* of terminators, as in multi-delimiter codes, suit a source?
3. How well do unbordered terminators resynchronise after bit errors?
4. What are the oscillation constants of the gain of T_{011} over C3?

Volume 5 — Mathematical Transformations and Relationships

Structures defined by relations rather than magnitudes: games given by their move relations, operations defined by minimal excluded values, and graphs treated as quantities under local products.

13 Games defined by relations

Volume 5 — Mathematical transformations and relationships. Research cycle C12.

13.1 Motivation

System D asks for structures in which only *relations* matter. In a one-heap game a move from n goes to a smaller position, and who wins is decided by the move relation alone. In Chapter 5 games came from rewriting rules; here the move relation is an arithmetic relation — *divides*, *is coprime to*, *is a submask of*, *differs by a square* — and we ask which relations produce structured values. Nineteen games were solved exactly and compared.

13.2 Definitions

The *Sprague–Grundy value* of a position is

$$g(n) = \operatorname{mex}\{\, g(m) : m \text{ an option of } n \,\},$$

where $\operatorname{mex} A$ is the least natural number not in A. Options are smaller, so g is unique (Lean: `IsGrundy.unique`), and n is a *P-position* (the player to move loses) iff $g(n) = 0$ (Lean: `IsGrundy.zero_pset`). In a *splitting game* a heap n is replaced by two nonempty heaps $a + b = n$ with option value $g(a) \oplus g(b)$ (XOR). By the Sprague–Grundy theorem a sum of games has the XOR of the values, so a formula for g solves all sums of heaps.

$v_2(n)$ is the exponent of 2 in n, $\Omega(n)$ the number of prime factors with multiplicity, $\operatorname{popcount}(n)$ the number of 1-bits, and m in $n = 2^a m$ (m odd) the *odd part*. A sequence is *eventually (arithmetic-)periodic* if $g(n + p) = g(n)$ (resp. $g(n) + s$, $s \neq 0$) for all large n.

game	move from n	relation
G01, G02	$n \to n - d$, $d \mid n$, $d < n$ (G02: also $d = n$)	divisibility
G03	$n \to n - d$, $d \mid n$, $1 < d < n$ (“deleting divisors”)	divisibility
G04	$n \to d$, $d \mid n$, $d < n$	divisibility
G05, G06	$n \to m$, $1 \le m < n$, $\gcd(m, n) = 1$ (G06: > 1)	coprimality
G08–G12	$n \to n - s$, s a square, Fibonacci number, power of 2, 3 or 4, prime, triangular number	additive sets
G13	$n \to$ a proper binary submask of n	bitwise
G16–G18	split $n = a + b$ with $\gcd(a, b) = 1$, $a \mid b$, or $a \neq b$ (Grundy’s game)	splitting
G19, G20	$n \to n \bmod k$ ($1 \le k < n$), $n \to \lfloor n/k \rfloor$ ($2 \le k \le n$)	Euclidean division

13.3 Arithmetic relations give arithmetic values

Theorem 13.1 (CLM-C12-01) *[PROVEN — written; REDISCOVERY].* G01 has $g(n) = v_2(n)$, G02 has $g(n) = v_2(n) + 1$ (for $n \ge 1$), and G04 has $g(n) = \Omega(n)$.

Proof idea. In G01 let $k = v_2(n)$. The options $n - 2^j$ $(j < k)$ have valuation j and realise every smaller value. No option has valuation k: if $v_2(d) < k$ then $v_2(n-d) = v_2(d)$, and if $v_2(d) = k$ then $n - d = 2^k(a-b)$ with a, b odd, so the valuation exceeds k. G02 adds the option 0, which shifts every value by one; for G04 a chain of divisors realises $0, \ldots, \Omega(n) - 1$. ∎

Theorem 13.2 (CLM-C12-02) *[PROVEN — written; REDISCOVERY].* In G05, $g(1) = 0$ and $g(n) = j$ when the smallest prime factor of n is the j-th prime; "subtract k coprime to n" is the same game, since $\gcd(n-m, n) = \gcd(m, n)$. In G06, $g(2k) = k - 1$, $g(m) \le (m-1)/2$ for odd m, and the P-positions $n \ge 1$ are exactly 1 and the primes.

Proof idea. In G05 the primes below the smallest prime factor p_j of n are options with values $1, \ldots, j-1$, and 1 has value 0; an option of value j would be divisible by p_j. In G06 a simultaneous induction proves the even formula and the odd bound: every odd option of $2k$ shares an odd prime q with $2k$, so it is at most $2k - 3$ and its value is at most $k - 2$. ∎ We know no closed form for the odd values of G06; we observed $g(m)/m < 0.24985$ for odd $m \le 5000$.

Theorem 13.3 (CLM-C12-05) *[PROVEN — written; REDISCOVERY].* G13 has $g(n) = \operatorname{popcount}(n)$; G19 has $g(n) = \lfloor \log_2((n+1)/3) \rfloor + 1$ for $n \ge 2$; G20 has $g(n) = \lfloor \log_2 n \rfloor$ for $n \ge 1$.

Proof idea. Proper submasks realise every smaller popcount. The remainders $n \bmod k$ are exactly the r with $0 \le r < n/2$, so $g(n) = \operatorname{mex} g([0, \lceil n/2 \rceil))$ and the value v first appears at $3 \cdot 2^{v-1} - 1$. In G20 the quotients $\lfloor n/2^j \rfloor$ realise every smaller value and every option is at most $n/2$. ∎

13.4 Periodic games and rigid splitting

Theorem 13.4 (CLM-C12-06) *[PROVEN — Lean: `residue_grundy`, `power_of_two_grundy`; general k written; REDISCOVERY].* (a) If no move is divisible by q and $1, \ldots, q-1$ are moves, the subtraction game has $g(n) = n \bmod q$; in particular subtract-a-power-of-2 has $g(n) = n \bmod 3$. (b) For every $k \ge 2$, subtract-a-power-of-k has the Grundy function of the finite game $\{1, k\}$: $n \bmod 2$ for odd k, and for even k the pattern $0, 1, 0, 1, \ldots, 1, 2$ of period $k + 1$.

Proof idea. $k^j \equiv (-1)^j \pmod{k+1}$, so every move changes the residue modulo $k+1$ by ± 1, exactly as the moves 1 and k do; the mex equations are then checked residue by residue. ∎

Theorem 13.5 (CLM-C12-07) *[PROVEN — Lean: `split_parity`; REDISCOVERY].* In every splitting game in which each heap $n \ge 2$ has at least one allowed split, $g(n) = 1$ for even n and $g(n) = 0$ for odd n: every play lasts exactly $n - 1$ moves.

So the coprime and dividing splits (G16, G17) are pure parity games; Grundy's game (G18) escapes only because a heap of size 2 cannot be split.

13.5 Deleting divisors

Theorem 13.6 (CLM-C12-03) *[PROVEN — Lean: `g03_pset`, `g03_P_positions`; REDISCOVERY: Codeforces 1537D].* The P-positions of G03 are exactly 0, the odd numbers, and the powers 2^{2j+1}.

Proof idea. From odd n every option $n-d$ is even and divisible by the odd $d \geq 3$, hence neither odd nor a power of 2. An even n with odd part $m > 1$ moves to the odd number $n-m$. The only power of 2 among the options of 2^a is 2^{a-1}, so the powers of 2 alternate. ∎

The P-positions do not solve *sums* of heaps; the whole Grundy function does.

Theorem 13.7 (CLM-C12-04) *[PROVEN — written + computer-assisted; Skeptic review pending; POTENTIALLY NOVEL].* Let $n = 2^a m$ with m odd. Then $g(n) = 0$ for odd n, and for $a \geq 1$:

odd part m	$g(2^a m)$
$m \geq 17$	a
1	0 for odd a; $a-1$ for even a, except $g(16) = 2$
3	$2, 3, 4, 3, 6$ for $a = 1, \ldots, 5$; $a-2$ for $a \geq 6$
5	$1, 3, 4, 3, 6$ for $a = 1, \ldots, 5$; $a+2$ for $a \geq 6$
$7, 9, 11, 15$	a for $a \leq 2$; $a+1$ for $a \geq 3$
13	a for $a \leq 3$; $a+1$ for $a \geq 4$

Consequently $|g(n) - v_2(n)| \leq 2$ unless n is an odd power of 2, and the record values occur at $n = 5 \cdot 2^a$ for $a \geq 6$.

Proof idea. Call F the tabulated function; we show $g = F$ by strong induction. Up to 10^5 the two agree by exact computation. Odd n have value 0 because the zero set of F is the P-set of Theorem 13.6. For odd part $m \geq 17$ the options $n - 2^j$ realise $0, \ldots, a-1$; an option with $v_2 < a$ again has odd part ≥ 17, and an option $2^a(m-e)$ can only take the value a when $m \in \{3, 13, 15\}$. For odd part $m \leq 15$ (so $a \geq 13$ beyond 10^5) every option value has the form $a + \delta$ with offsets that depend only on m and the parity of a, so the mex equation needs to be checked for one exponent of each parity; the certificate checks all $m \leq 15$ and $13 \leq a \leq 100$. ∎ (EXP-C12-06; the written inductive steps have not yet been reviewed by the Skeptic.)

Compare G01, the same game with the move “subtract 1” allowed, where $g = v_2$ exactly. Forbidding one move perturbs the Grundy function only on the odd powers of 2 and on numbers whose odd part is at most 15; up to 10^5 there are 96 such exceptions.

13.6 Sparse move sets

Theorem 13.8 (CLM-C12-08) *[PROVEN — written, from the Furstenberg–Sárközy theorem; REDISCOVERY].* For subtract-a-square and subtract-a-triangular-number the P-positions form an infinite set of natural density 0. Hence neither Grundy sequence is eventually periodic or eventually arithmetic-periodic.

Proof idea. Two P-positions never differ by a move. So the P-set of subtract-a-square has no two elements differing by a square and has density 0 (Furstenberg 1977, Sárközy 1978); triangular numbers contain the values $m(2m+1)$ of a polynomial vanishing at 0, and the polynomial version (Kamae and Mendès France 1978) applies. A finite P-set would force a tail of $\mathbb{N}$ into finitely many translates of the move set, which has density 0. An infinite eventually periodic zero set has positive density, and an arithmetic-periodic sequence has only finitely many zeros. ∎

The data agree: from $N = 1.25 \cdot 10^4$ to $2 \cdot 10^5$ the P-density falls $0.0538 \to 0.0222$ (squares) and $0.0344 \to 0.0116$ (triangular), while the maxima grow $36 \to 95$ and $50 \to 144$. Primes and

Fibonacci numbers differ.

Conjecture 13.9 (CLM-C12-12) *[COMPUTATIONALLY SUPPORTED; REDISCOVERY: OEIS A014589]*. Subtract-a-prime satisfies $g(n) \le 11$ for all n. (Our check reaches $2 \cdot 10^5$; the OEIS entry records the bound up to 10^9.)

Conjecture 13.10 (CLM-C12-13) *[CONJECTURE]*. Subtract-a-Fibonacci-number has bounded Grundy values (maximum 14 for $n \le 2 \cdot 10^5$; weak support, since it rose from 13 to 14 between $2.5 \cdot 10^4$ and $5 \cdot 10^4$).

13.7 Hypotheses that failed

H-C12-1 [REFUTED] (CLM-C12-10). "In each of the games G01–G06, $g(n)$ depends only on the prime signature of n." Already $n = 2$ and $n = 3$ refute it in G01 ($g = 1$ and 0). The first counterexamples elsewhere are $(2, 3)$ in G02, $(4, 9)$ in G03, $(2, 3)$ in G05 and $(6, 10)$ in G06 (values 2 and 4). Only G04 depends on the signature alone, as $g = \Omega$ predicts.

H-C12-2 [REFUTED] (CLM-C12-11). "Subtraction games whose move sets are the squares, the triangular numbers, the Fibonacci numbers or the primes keep a positive density of P-positions." Squares and triangular numbers have density 0 (Theorem 13.8). For Fibonacci numbers (0.180) and primes (0.155) the question remains open.

H-C12-0 [INCONCLUSIVE] (CLM-C12-09). "Every game is eventually periodic, eventually arithmetic-periodic, or has a closed form in a number-theoretic function." Thirteen of the 19 games conform. Subtract-a-square and subtract-a-triangular-number are provably not (arithmetic-)periodic, but "has no closed form" is not a theorem, so the hypothesis cannot be refuted as stated.

13.8 Experiments

Experiment	Question	Result
EXP-C12-01	census of 19 games	5 eventually periodic, 0 arithmetic-periodic, 14 neither within range; the detector was fixed after spurious end-of-range periods and both runs are kept
EXP-C12-02	closed forms	0 mismatches ($n \le 10^5$, 5000 or $2 \cdot 10^4$; powers of k for $k = 2..20$; 40 random splitting rules)
EXP-C12-03	sparse move sets, $n \le 2 \cdot 10^5$	P-density: squares $0.0538 \to 0.0222$, triangular $0.0344 \to 0.0116$, Fibonacci $0.169 \to 0.180$, primes $0.135 \to 0.155$
EXP-C12-04	OEIS	18 of 19 Grundy sequences catalogued (the G10c hit is spurious); G03 matches nothing
EXP-C12-05	prime signatures	H-C12-1 fails in 5 of 6 games
EXP-C12-06	Grundy function of G03	Theorem 13.7 exact on $[0, 10^5]$; 96 exceptions to v_2, all with

Experiment	Question	Result
		odd part ≤ 15; certificate and 4000 random checks: 0 failures

13.9 Comparison with existing mathematics

Sprague–Grundy theory and the classical subtraction, take-and-break and Grundy's games are treated in Berlekamp, Conway and Guy's *Winning Ways*. Iannucci and Larsson, "Game values of arithmetic functions" (arXiv:2101.07608, 2021), contain G04 (Theorem 1), G02 (Theorem 2), G05 (Theorem 5), the values of G06 shifted by one (Theorem 6), and G19 (Theorem 12, whose proof is left to the reader there; ours supplies it). The win/loss version of G01 is the "Divisor Game" puzzle (LeetCode 1025). The P-positions of G03 are the official solution of Codeforces problem 1537D "Deleting Divisors" (Round #726, June 2021); first believed new (no OEIS match), they were relabelled as a rediscovery when the Historian found the problem. For subtract-a-square, the P-set is a maximal square-difference-free set, of density zero by Furstenberg–Sárközy (Wikipedia, "Square-difference-free set"; Eppstein, FUN 2018). The OEIS entries A014586, A019509, A014588 and A014589 catalogue the sparse games, A002188 Grundy's game, and A279119 is exactly G06.

Theorem 13.7 is the exception: its sequence is not in the OEIS, the game is not among Iannucci and Larsson's, Codeforces 1537D settles only win and loss, and web searches found no analysis of its values. Potentially novel — not a priority claim.

13.10 Applications and limitations

A closed form replaces the dynamic programme over all smaller positions by a direct evaluation and, through XOR, solves sums of games: "Deleting Divisors" on several heaps is solved by Theorem 13.7 but not by Theorem 13.6. Beyond recreational and competitive-programming games we found no application; no benchmark was run and no performance improvement is claimed.

Limitations: the census covers $n \le 10^5$, 5000 or $2 \cdot 10^4$ depending on the game; Theorem 13.8 rests on cited theorems not re-proved here; the written inductive steps of Theorem 13.7 await the Skeptic's review.

13.11 Open questions

1. Are the Grundy values of subtract-a-prime (11) and subtract-a-Fibonacci 14. bounded? Does the Fibonacci P-density converge? (It is at most 1/5 by Pond and Howells, 1965.)
2. For which d_0 does forbidding the move "subtract d_0" in G01 perturb v_2 on only finitely many odd parts, as in Theorem 13.7?
3. Prove $g(m) < m/4$ for odd m in the non-coprime game G06.
4. Formalise Theorem 13.7 in Lean (the certificate is finite).
5. Is Grundy's game (G18) eventually periodic? (A well-known open problem.)

14 Arithmetic from the minimal excludant

Volume 5 — Mathematical transformations and relationships. Research cycle C07.

14.1 Motivation

Most operations are defined by saying what their result *is*. Combinatorial game theory suggests another way: define the result as the least value that is *not* produced by a prescribed set of earlier results — the minimal excludant, or mex. Conway built the nimbers this way:

$$a \oplus b = \operatorname{mex}\{a' \oplus b,\ a \oplus b' : a' < a,\ b' < b\},$$
$$a \otimes b = \operatorname{mex}\{(a' \otimes b) \oplus (a \otimes b') \oplus (a' \otimes b') : a' < a,\ b' < b\},$$

and obtained a field of characteristic 2 on the natural numbers. What else does exclusion produce?

14.2 Definitions

An operation of our family is given by a *spec*: which earlier entries are "options" — $f(a',b)$ for $a' < a$ (L), $f(a,b')$ for $b' < b$ (R), and, for $a' < a, b' < b$, XOR combinations of $x = f(a',b)$, $y = f(a,b')$, $z = f(a',b')$ (D, any of the seven nonempty masks). Then $f(a,b) = \operatorname{mex}$ of the option values. There are $2 \cdot 2 \cdot 2^7 = 512$ specs; nim-sum is L + R and nim-product is D[$x \oplus y \oplus z$].

14.3 Which mex operations are associative?

H-C07-1 [REFUTED]. "Apart from trivial ones, only Conway's two operations are associative."

Theorem 14.1 *[PROVEN — written and computer-assisted; the formulas are classical].* On $[0, 64)^2$ the 512 specs produce 134 different operations. Exactly eleven are associative, and their canonical specs define, for all a, b:

spec	operation
(none)	0
L, R	a, b
D[x], D[y]	a if $b > 0$ (else 0); b if $a > 0$ (else 0)
D[z]	$\min(a,b)$
L + R	$a \oplus b$
D[$x \oplus y \oplus z$]	$a \otimes b$
L + R + D[z]	$a + b$
D[x] + D[y]	$((a-1) \oplus (b-1)) + 1$ on positives, 0 absorbing
D[x] + D[y] + D[z]	$a + b - 1$ on positives, 0 absorbing

Of the other specs, 468 are refuted by an explicit triple, and the remaining 33 define the same 11 operations — an extra option that, evaluated on the formula, already lies in the option set does not change the mex (the argument is the Skeptic's; the first version of this

chapter wrongly said every other spec was refuted). So exactly 44 specs are associative, and **ordinary addition is a mex operation** — the least value not reachable by decreasing either argument or both — and so is min.

Proof idea. For L + R + D[z] the options are $a'+b$, $a+b'$ and $a'+b'$, which together cover exactly $0,\dots,a+b-1$. The other formulas follow the same pattern or Conway's arguments. ∎

14.4 Closure fails

H-C07-2 [REFUTED]. "Commutative mex operations are closed on the segments $[0,2^{2^m})$, like the nimbers." Already for D[$x \oplus y$] we have $f(2,3)=4$, and 30 of the 34 commutative operations fail.

Theorem 14.2 *[PROVEN].* If a spec contains both L and R, 0 is an identity; if it contains neither, 0 is absorbing.

Beyond these facts, most commutative mex operations look chaotic: their values grow roughly quadratically and 88 of their rows and diagonals are not in the OEIS (EXP-C07-02).

14.5 A Jacobsthal row

Wythoff's game, whose Grundy function is also a mex recursion, has *additively periodic* rows: $G(a,b+p)=G(a,b)+p$ for large b. We guessed that this was the rule (H-C07-4). The operation

$$f(a,b)=\operatorname{mex}\{f(a',b)\oplus f(a,b'):a'<a,\ b'<b\}$$

refutes it in an unexpectedly orderly way. Its first rows are $f(0,b)=0$, $f(1,b)=b$, and the second row begins

$$0,\ 2,\ 1,\ 4,\ 3,\ 8,\ 9,\ 10,\ 5,\ 6,\ 7,\ 16,\ 17,\ 18,\ 19,\ 20,\ 11,\ 12,\ \dots$$

Theorem 14.3 (the Jacobsthal row) *[PROVEN — written; POTENTIALLY NOVEL (small)].* Let $J_k=(2^k-(-1)^k)/3$ $(1,1,3,5,11,21,43,\dots)$. For every $k\ge 1$ the second row exchanges the blocks

$$[2^k-J_k,\ 2^k)\quad\text{and}\quad[2^k,\ 2^k+J_k)$$

by translation: $f(2,b)=b+J_k$ on the first and $b-J_k$ on the second. In particular the row is an involution, $|f(2,b)-b|$ is unbounded, and the row is not additively periodic.

Proof idea. The row satisfies $g(b)=\min\{v: v\notin G,\ v\oplus b\notin G\}$ with G the earlier values. Because $J_{k+1}+J_k=2^k$, when the k-th pair of blocks begins the used values are exactly $[0,2^k-J_k)$. Inside the pair, every smaller candidate is excluded because two numbers just below 2^k have a small XOR, while the claimed value survives because adding a number $c>x$ to x always creates a new top bit. ∎ An independent implementation confirms the formula for all $b<2^{15}$ (EXP-C07-03; the Skeptic, computing from the definition: all $b<2^{18}$). Only 222 of the 670 rows tested show an additive period of at most 375 within $b<1500$; rows without a detected period may still be periodic.

14.6 Experiments

Experiment	Question	Result
EXP-C07-01	census of 512 specs	134 operations, 11 associative

Experiment	Question	Result
EXP-C07-02	structure of commutative ones	0 identity/absorbing; 88 sequences not in OEIS
EXP-C07-03	additive periodicity of rows	222/670 rows; Jacobsthal row to 2^{15}

14.7 Comparison with existing mathematics

Nim-sum and nim-product are Conway's (*On Numbers and Games*, 1976); $a+b$ and $\min(a,b)$ as mex values are the rank functions of product orders, folklore in game theory. Additive periodicity is a theorem for Wythoff's game (Dress, Flammenkamp and Pink, 1999). We found no prior description of the Jacobsthal row or of the census.

14.8 Applications and limitations

No application is claimed. The census uses tables up to 64×64 and rows up to $b < 1500$; non-associativity is certified by explicit triples, associativity by the proofs above.

14.9 Open questions

1. Do rows $a \geq 3$ of the same operation have a recursive block structure?
2. Which mex tables are automatic sequences, so that statements about them could be decided mechanically?
3. ~~Are the remaining 33 associative-looking specs associative for all arguments?~~ Yes — answered by the Skeptic's absorption argument (`proofs/paper/C07.md`).

15 Multiplying graphs

Volume 5 — Mathematical transformations and relationships. Research cycle C14.

15.1 Motivation

Volume 1 obtained numbers as what survives transformations of words. Here the roles are reversed: the *objects* are relational structures — graphs — and we ask which ways of combining two graphs into one deserve to be called a multiplication. Only one kind of information is allowed: whether two vertices are equal, adjacent or neither. If graphs are to behave like quantities, the multiplication should be associative, perhaps commutative, have a unit, and admit invariants that turn it into ordinary multiplication of numbers.

15.2 Definitions

For vertices x, y of a graph G let $\mathrm{rel}_G(x,y)$ be E (equal), A (adjacent) or N (distinct and non-adjacent). A **local product** $G * H$ has vertex set $V(G) \times V(H)$; two distinct vertices $(g,h),(g',h')$ are adjacent iff the pair $(\mathrm{rel}_G(g,g'),\mathrm{rel}_H(h,h'))$ lies in a fixed set S of *cells*. There are eight cells besides (E,E), hence $2^8 = 256$ local products. Familiar ones:

product	cells S
Cartesian $\square$	EA, AE
direct $\times$	AA
strong $\boxtimes$	EA, AE, AA
lexicographic $G[H]$	EA, AE, AA, AN
co-normal $\vee$	EA, AE, AA, AN, NA
symmetric difference $\oplus$	EA, AE, AN, NA

The symmetric difference joins (g,h) and (g',h') when exactly one of $g \sim g'$, $h \sim h'$ holds (Harary–Wilcox 1967). The *Seidel matrix* of G is $S(G) = J - I - 2A(G)$; *switching* G with respect to a vertex set Z toggles all adjacencies between Z and its complement, and the switching classes are the *two-graphs*.

15.3 Associativity is a finite fact

The relation of two vertices of $G * H$ is a function φ_S of the relations in the factors. Hence:

Theorem 15.1 *[PROVEN — Lean: `assoc_iso, assoc_of_P3`].* The canonical map $((g,h),k) \mapsto (g,(h,k))$ is an isomorphism $(G*H)*K \to G*(H*K)$ for all graphs iff $\varphi_S(\varphi_S(r_1,r_2),r_3) = \varphi_S(r_1,\varphi_S(r_2,r_3))$ for all relations r_i — and already the path P_3 (taken three times) detects every failure, because it contains pairs in all three relations.

Theorem 15.2 *[PROVEN — Lean kernel computation; REDISCOVERY: Imrich–Izbicki 1975].* Exactly 20 of the 256 local products are associative. Ten of them are commutative: the empty and complete products, Cartesian, direct, strong, co-normal, symmetric difference, and three "complement-duals" $G *' H = \bar{G} * \bar{H}$. Exactly eight have a unit (necessarily K_1): Cartesian, strong, lexicographic and its mirror image, co-normal, symmetric difference and two complement-duals.

Imrich and Izbicki proved the count of twenty in 1975; Nowakowski and Rall (1996) noted that ten depend on both factors and eight of those are commutative — exactly what our computation shows.

A bold hypothesis that survived. We expected that some product might be associative only *up to isomorphism*, through a cleverer identification than the canonical map. It is not so:

Theorem 15.3 *[PROVEN — computer-assisted].* A local product is associative (commutative) up to isomorphism iff it is strictly associative (has a symmetric cell set). For every one of the 236 non-associative and 224 non-commutative products, factors with at most three vertices give products with different degree sequences (EXP-C14-02).

15.4 Graphs become numbers: characters

For constants $c = (c_E, c_A, c_N)$ put $M_c(G) = c_E I + c_A A + c_N (J - I - A)$.

Theorem 15.4 *[PROVEN — written, sympy and Lean; POTENTIALLY NOVEL (small)].* $M_c(G * H) = M_c(G) \otimes M_c(H)$ for all graphs (with $c_A \neq c_N$) for exactly six products:

product	$M_c(G)$
direct	A
strong	$I + A$
co-normal	$J - A$
symmetric difference	$J - 2A = S(G) + I$
two complement-duals	$J - I - A,\ \ I - S(G)$

Theorem 15.5 *[PROVEN — written].* $M_c(G * H) = M_c(G) \otimes I + I \otimes M_c(H)$ holds exactly for the Cartesian product ($M = A$) and its complement-dual. Consequently every commutative associative local product except the empty and complete ones is “linear”: a fixed combination of I, A and J turns it into the Kronecker product or the Kronecker sum.

So each of these products carries a **spectral quantity**: the eigenvalues of M_c multiply (or add) under the product, and $\det M_c$, the traces of powers of M_c, and the characteristic polynomial become homomorphisms into ordinary arithmetic.

Theorem 15.6 *[PROVEN — written; Lean (entry level): `xor_character`; REDISCOVERY — Abiad, Butler and Haemers 2018].* For all graphs,

$$S(G \oplus H) + I = (S(G) + I) \otimes (S(H) + I).$$

Hence the Seidel eigenvalues of $G \oplus H$ are $(\theta + 1)(\eta + 1) - 1$, and $\oplus$ is compatible with switching: it is a multiplication of two-graphs.

The proof is one line — $(-1)^{a \oplus b} = (-1)^a (-1)^b$ — but it changes how one should think about $\oplus$: it only half-sees graphs.

Theorem 15.7 *[PROVEN — written].* (a) $K_{a,b} \oplus K_{c,d} \cong K_{ac+bd,\ ad+bc}$: on complete bipartite graphs the order and the *imbalance* $|a - b|$ simply multiply, like the two components of a split-complex number. (b) $K_2 \oplus X \cong K_2 \oplus Y$ iff X and Y are switching-equivalent.

15.5 A bold hypothesis that failed

For the Cartesian and strong products every connected graph factors uniquely into prime graphs (Sabidussi–Vizing; Dörfler–Imrich, McKenzie). We conjectured (H-C14-3) the same for all six commutative products with a unit, on connected graphs.

Refutation. By Theorem 15.7(b), K_2 cannot tell a graph from its switched versions. The 4-cycle $C_4 = K_2 \oplus K_2$ and the claw $K_{1,3}$ are switching-equivalent, so

$$K_2 \oplus K_2 \oplus K_2 \cong K_2 \oplus K_{1,3},$$

two different factorisations of a connected graph on 8 vertices into connected primes (Theorem 8 of the C14 proofs; an explicit isomorphism is in EXP-C14-06). Among all graphs uniqueness fails already for $C_4 = K_2 \oplus K_2 = K_2 \oplus 2K_1$, and Theorem 15.7(a) produces collisions such as $3K_1 \oplus K_{2,3} \cong K_{1,2} \oplus K_{1,4}$ ($(3,3)(5,1) = (3,1)(5,3)$ in (order, imbalance) coordinates).

The failure points to the right setting:

Conjecture 15.8 *[COMPUTATIONALLY SUPPORTED; POTENTIALLY NOVEL].* Two-graphs form a free commutative monoid under $\oplus$: every switching class factors uniquely into prime switching classes. (No collision among all factorisations up to order 16; EXP-C14-05.)

Conjecture 15.9 *[COMPUTATIONALLY SUPPORTED].* The complement-dual of $\oplus$ (cells EA, AE, AA, NN) has unique factorisation on connected graphs — equivalently $\oplus$ on graphs with connected complement (no collision up to order 15).

15.6 Counting primes

Theorem 15.10 *[PROVEN — written, from the published theorems; REDISCOVERY].* The Cartesian and the strong product make the set of all finite graphs into isomorphic graded monoids: both are the multiplicative monoid of a polynomial semiring $\mathbb{N}[X]$ in their connected primes, and the two prime sets have the same size in every order. Our census reproduces the connected Cartesian prime counts $1, 2, 5, 21, 110, 853, 11111, \ldots$ (OEIS A129582, Ji Li 2007) up to 15 vertices — for the strong product too.

Counting primes over *all* graphs gives sequences we did not find in the OEIS (Theorem 10 of the C14 proofs):

product	primes of order 2, 3, …, 9
Cartesian, strong, co-normal	2, 4, 8, 34, 148, 1044, 12326, 274658
symmetric difference	2, 4, 9, 34, 150, 1044, 12333, 274658
lexicographic	2, 4, 7, 34, 142, 1044, 12310, 274652

The lexicographic product loses uniqueness at order 10:
$2K_1[C_4 + K_1] \cong (2K_2 + K_1)[2K_1] \cong 2C_4 + 2K_1$ — in line with Imrich's theorem that lexicographic factorisations are not unique although the number of factors is.

15.7 Experiments

Experiment	Question	Result
EXP-C14-01	which products are associative, commutative, unital?	20 / 10 / 8; 0 disagreements with graph-level checks

Experiment	Question	Result
EXP-C14-02	is “up to isomorphism” weaker?	no: 460 witnesses with ≤ 3-vertex factors
EXP-C14-03	which products have characters?	6 multiplicative, 2 additive
EXP-C14-04	factorisation census, orders ≤ 15	$\oplus$ collides from order 4; A129582 reproduced
EXP-C14-05	two-graphs under $\oplus$	no collision up to order 16
EXP-C14-06	independent re-implementation	everything reproduced
EXP-C14-07	spectra from factors vs. direct	error $\le 10^{-14}$, orders of magnitude faster

15.8 Comparison with existing mathematics

The twenty associative products (Imrich–Izbicki 1975), the unique factorisation theorems for the Cartesian and strong products, the non-uniqueness for the lexicographic product, the semiring view of disconnected graphs and the prime counts A129582 are all in the literature, and the chapter rediscovers them. So is Theorem 15.6: Abiad, Butler and Haemers (arXiv:1801.01149, 2018, §5) define the same product and state that its Seidel matrix is $(S_1 + I) \otimes (S_2 + I) - I$, using its compatibility with switching. Our four literature searches missed the paper; the Skeptic found it. We found no source for the classification of products by characters (Theorems 15.4–15.5), for Theorem 15.7, or for the factorisation behaviour of two-graphs; these are small results and could well be known to specialists.

15.9 Applications and limitations

- The spectrum of a large product follows from its factors: for $n = m = 40$ (1600 vertices) the factor route agreed with direct diagonalisation to 10^{-14} and was about three orders of magnitude faster (our first measurement, four to five orders, was inflated by competing BLAS threads on a loaded machine — a correction by the Skeptic) — the standard Kronecker gain, newly applicable to $\oplus$.
- Squaring the regular two-graph on 6 points with $\oplus$ gives 36 equiangular lines in $\mathbb{R}^{18}$ at angle $\arccos(1/5)$ — no improvement, since 36 such lines already exist in $\mathbb{R}^{15}$.
- The factorisation censuses are exhaustive only up to 15 (graphs) and 16 (two-graphs) vertices.

15.10 Open questions

1. Are two-graphs a free commutative monoid under $\oplus$ (Conjecture 15.8)?
2. Does the complement-dual of $\oplus$ factor uniquely on connected graphs (Conjecture 15.9)?
3. Which graphs G satisfy $G \oplus X \cong G \oplus X'$ for all switchings X' of X? ($K_{a,a}$ do.)
4. Is there a character theory for directed graphs, or for graphs with loops, where the direct product acquires a unit?

16 The longest Zeckendorf game

Volume 5 — Mathematical transformations and relationships. Research cycle C19.

16.1 Motivation

In Chapter 3 the Fibonacci exchanges turned a heap of n tokens into the Zeckendorf representation of n, and every order of exchanges arrived there. The *number* of exchanges, however, depends on the order. The shortest derivation is easy: $n - Z(n)$ steps, $Z(n)$ the number of Zeckendorf terms. The longest was left open in Chapter 3. As a two-player game this system is the *Zeckendorf game* of Baird-Smith, Epstein, Flynt and Miller, and its longest game has a literature of its own. This chapter reports what the laboratory added to it: an exact identity for the length of every game, the shape of a longest game, a closed form for Fibonacci inputs, and a conjectured recursion for all inputs, a question that a 2025 paper calls open.

16.2 Definitions

We write $F_1 = 1$, $F_2 = 2$, $F_{i+1} = F_i + F_{i-1}$. The game starts with n copies of F_1 and allows

- combines: $F_1 + F_1 \to F_2$ (C_1) and $F_{i-1} + F_i \to F_{i+1}$ (C_i, $i \geq 2$);
- splits: $2F_2 \to F_1 + F_3$ (S_2) and $2F_i \to F_{i-2} + F_{i+1}$ (S_i, $i \geq 3$).

Every game ends at the Zeckendorf representation $n = \sum \delta_i F_i$. Let $L(n)$ be the length of a longest game, MC_i the number of moves C_i in a game, and

$$B(n) = \sum_i (F_{i+2} - i - 2)\,\delta_i,$$

the upper bound of Cusenza et al. (2022), who also proved that a longest game is obtained by splitting whenever possible and otherwise combining the smallest consecutive pair. The *deficit* is $D(n) = B(n) - L(n)$. The *lazy* representation of n writes n as a sum of distinct F_i with no two consecutive indices missing below the largest one.

16.3 What exhaustive search shows

Theorem 16.1 *[PROVEN — computer-assisted; REDISCOVERY].* For every $n \leq 150$ an exhaustive search over all positions, assuming no strategy, gives $L(n) = 0, 0, 1, 2, 3, 5, 6, 8, 10, 11, 13, 15, 17, 20, \ldots$, and the split-first strategy attains it. The bound $B(n)$ is attained exactly at $n = 1, 2, 4, 7, 12, 20, 33, 54, 88, 143$, the numbers $F_k - 1$ (EXP-C19-01).

Proposition 16.2 *[PROVEN].* The positions reachable from n copies of F_1 are exactly the multisets of Fibonacci numbers with sum n. (Undo combines until only copies of F_1 remain, then read the steps backwards.) Their number is the number of partitions of n into Fibonacci parts (OEIS A003107), up to 910,775 positions for $n = 150$.

16.4 Every game has the same “budget”

Theorem 16.3 *[PROVEN — written; arithmetic core in Lean: `length_identity`; REDISCOVERY (implicit in Cusenza et al.)].* Every game G on n satisfies

$$|G| \;=\; B(n) \;-\; \sum_{i \geq 2} (i-1)\,MC_i(G).$$

Proof idea. Give a token F_i the weight $a_i = F_{i+2} - i - 2$. The total weight starts at 0 (since $a_1 = 0$) and ends at $B(n)$, and each split or C_1 raises it by exactly 1 while C_i raises it by i;

for instance a split changes it by $a_{i-2} + a_{i+1} - 2a_i = 1$ by the Fibonacci recurrence. ∎

So a longest game is one that uses the fewest, and the lowest, consecutive combines, and $D(n) = \min_G \sum (i-1)MC_i(G)$. The theorem was checked on 1500 random games as well as on the longest ones (EXP-C19-02).

16.5 The shape of a longest game

The split-first game splits until no level holds two tokens, and only then starts combining. Where does the first phase end?

Theorem 16.4 (the lazy phase) *[PROVEN — written; POTENTIALLY NOVEL (small)].* Split moves, applied in any order until none is possible, always take n copies of F_1 to the **lazy** representation of n, after the same number of moves.

Proof idea. Splits form an abelian chip-firing system, so the end point does not depend on the order, and adding one token and re-stabilising is the same as stabilising from scratch. Adding one F_1 to a lazy representation whose lowest gap is at level p creates a carry: the block $2, 1, 1, \ldots, 1$ below p turns into the alternating block $\ldots, 1, 0, 1$ topped by a token at level p, which is again lazy. ∎

Corollary 16.5 *[PROVEN — a short new proof of Cusenza et al.'s Theorem 1.4].* A game can be played with split moves alone iff $n = F_k - 1$: such a game ends at the lazy representation, and the lazy and Zeckendorf representations coincide exactly for alternating digit strings.

16.6 Fibonacci inputs

For $n = F_k$ the lazy representation is the alternating string …1010 with the two lowest tokens adjacent. The second phase is a single carry running upwards: C_2, C_4, C_6, … (or C_3, C_5, …), each combine landing next to the following token.

Theorem 16.6 *[PROVEN — written, using the strategy theorem of Cusenza et al.; weight sums in Lean; POTENTIALLY NOVEL].*

$$L(F_k) = F_{k+2} - k - 2 - \left\lfloor \frac{(k-1)^2}{4} \right\rfloor : \quad 0,\ 1,\ 2,\ 5,\ 10,\ 20,\ 37,\ 67,\ 117,\ 201,\ \ldots$$

The deficit $D(F_k)$ is the quarter-square $\lfloor (k-1)^2/4 \rfloor$, the total weight $1 + 3 + 5 + \cdots$ or $2 + 4 + 6 + \cdots$ of the carry. ∎ Checked for every $k \le 34$ (EXP-C19-03). This sequence is not in the OEIS.

16.7 Four failed guesses

Before the recursion below was found, four simpler descriptions of $D(n)$ were proposed and refuted (EXP-C19-02, EXP-C19-03):

- *H-C19-1 [REFUTED]*: D is a sum over Zeckendorf terms with corrections for consecutive pairs only. The two-term values are given exactly by such a rule (Conjecture 16.8), but already $n = 12 = F_5 + F_3 + F_1$ has $D = 0$ where the rule predicts 1.
- *H-C19-2 [REFUTED]*: the effect of the largest term depends only on the two largest terms.
- *H-C19-3 [REFUTED]*: D is additive over the runs of the lazy representation — the corresponding linear system is inconsistent already for $n < 160$ (rank 46 against 47).

- *H-C19-4 [REFUTED]*: a longest game on $n+1$ is a longest game on n followed by the longest continuation after adding one token ($n=3$ is a counterexample). Longest games avoid the Zeckendorf form until the very end.

16.8 A recursion for all n

The failures pointed to carries that travel through alternating blocks. The right bookkeeping turned out to be the *saving* that a smaller part m grants to a top term F_{j+1}:

$$\rho = D(F_{j+1}) + D(m) - D(F_{j+1}+m).$$

Among 17,711 cases it depends only on m and on the parity of j.

Conjecture 16.7 *[COMPUTATIONALLY SUPPORTED; POTENTIALLY NOVEL]*. Let $h(k) = \lfloor k^2/4 \rfloor$. For $p \in \{0,1\}$ define $\rho_p(0) = 0$ and, for $m = F_{i+1} + t$ with Zeckendorf top index $i+1$,

$$\rho_p(m) = r_p(i) + [\, i \equiv p \,] \cdot \max(0,\ \rho_p(t) - h(i-2)),$$

with $r_p(0) = 1$, $r_p(i) = h(i) + 2$ if $i \equiv p \pmod 2$ $(i \geq 1)$, and $r_p(i) = h(i+1) + 1$ otherwise. Then

$$D(F_{j+1} + m) = h(j) + D(m) - \rho_{j \bmod 2}(m), \qquad L(n) = B(n) - D(n).$$

The recursion was read off all $n < 30000$. It was then tested on 6,272 numbers it had not seen, including every n in $[30000, 36000)$ and random n up to $2 \cdot 10^7$, without a single mismatch (EXP-C19-04). It is not proved.

Conjecture 16.8 *[COMPUTATIONALLY SUPPORTED]* (two terms). For index gaps ≥ 2, $D(F_{j+1} + F_{i+1}) = h(j) + h(i) - R(j,i)$ with $R(j,i) = \lfloor (i+1)^2/4 \rfloor + 1$ when $j - i$ is odd, $\lfloor i^2/4 \rfloor + 2$ when $j - i$ is even $(i \geq 1)$, and $R(j,0) = 1$ (all 276 pairs with $j \leq 24$).

16.9 Experiments

Experiment	Question	Result
EXP-C19-01	exact longest games	$n \leq 150$; strategy optimal; equality iff $F_k - 1$
EXP-C19-02	identity; lazy phase; insertion	0 violations; lazy for all $n < 5000$; H-C19-4 refuted
EXP-C19-03	structure of D	Fibonacci law to $k = 34$; three additive models refuted
EXP-C19-04	recursion out of sample	6,272 numbers, 0 mismatches

16.10 Comparison with existing mathematics

The game and its shortest length are due to Baird-Smith, Epstein, Flynt and Miller (2020). Li et al. (2020) bounded the longest game by $3n - 3Z(n) - lZ(n) + 1$. Cusenza et al. (*Fibonacci Quarterly* 60, 2022) proved that the split-first strategy is longest, that $L(n) \leq B(n)$, and that equality holds iff $n = F_k - 1$. Cheigh et al. (2025) proved that every length between the shortest and longest game occurs, and stated that a convenient closed form for non-Fibonacci inputs is open. Theorem 16.3 is implicit in Cusenza et al.'s argument, and Corollary 16.5 reproves their Theorem 1.4. We found no earlier statement of the lazy phase, of the formula for $L(F_k)$, or of a recursion for general n. The sequences $L(n)$ and $L(F_k)$ are not in the OEIS.

16.11 Applications and limitations

If Conjecture 16.7 holds, $L(n)$ can be computed with $O(\log^2 n)$ arithmetic operations instead of playing $\approx 2.618\,n$ moves. At $n = 10^7$ the recursion takes $9 \cdot 10^{-5}$ s against 8.3 s for playing the game. That speed-up rests on an unproved recursion. All data beyond $n = 150$ rely on the published optimality of the split-first strategy.

16.12 Open questions

1. Prove Conjecture 16.7, for instance by following the carries of the second phase from the lazy representation.
2. Prove that $D(n) \le \lfloor j^2/4 \rfloor$, where j is the level of the largest term, and that L is strictly increasing (both hold for $n < 30000$).
3. Do the lazy phase and the Fibonacci-input formula extend to the generalised Zeckendorf games built on other linear recurrences?

Volume 6 — New Conjectures and Verified Theorems

This volume is the laboratory’s ledger: a catalogue of every claim that was proved, supported or conjectured, generated directly from the database so that it cannot drift from the evidence.

Catalogue of results

Volume 6 — New conjectures and verified theorems. Generated from the claim database (`python -m lab catalogue`); every entry links to its record in `db/claims/`.

Proof kinds: **Lean** — machine-checked; **computer-assisted** — a finite certificate checked by a program; **written** — a complete written proof in `proofs/paper/`. Novelty labels are those of the Historian's audit; *rediscovery* means the result is in the literature.

16.13 C01 — Quantity as the complete invariant of rewriting

Theorems.

- **CLM-C01-01** (Lean), rediscovery: Symbol counts are the complete invariant of the commutation congruence.
- **CLM-C01-02** (Lean), rediscovery: One symbol, one relation a^m = a^(m+p) gives exactly m+p classes (index m, period p).
- **CLM-C01-03** (written), rediscovery: Several one-symbol relations have index min m_i and period gcd(n_i - m_i).
- **CLM-C01-04** (written), rediscovery: Multiplication emerges as composition of substitutions; with two symbols it is non-commutative.
- **CLM-C01-06** (written), rediscovery: Border-containment theorem for one length-preserving relation.
- **CLM-C01-07** (computer-assisted), rediscovery: Common-border formula proved for 1632 specific relations by verified finite completions.
- **CLM-C01-09** (written), rediscovery: Common-border formula when u and v share no border (c = 1) - consequence of published results.

Computationally supported.

- **CLM-C01-08**, potentially novel: Common-border conjecture: one-relation class counts depend only on common borders.
- **CLM-C01-10**, potentially novel: Common-border formula with exactly one common border (derivation sketch from Gray-Steinberg 2022).
- **CLM-C01-11**, rediscovery: Converse of border containment: u -> v (|u| = |v|) is confluent iff Borders(u) is contained in Borders(v).

16.14 C02 — Exchange systems: positional numeration as normal forms

Theorems.

- **CLM-C02-01** (written), rediscovery: Integer bases emerge as normal forms of the exchange rule b[k] -> [k+1].
- **CLM-C02-02** (written), rediscovery: Abelian property of carry rules (order of exchanges is irrelevant).
- **CLM-C02-03** (written), rediscovery: Termination dichotomy for carry rules away from sum b = m.

- **CLM-C02-04** (written), potentially novel: Carry rules with sum b = m never terminate from all unary starts (uniform glider).
- **CLM-C02-05** (computer-assisted), potentially novel: Termination classification for all 189 small carry rules (computer-assisted).
- **CLM-C02-07** (Lean), rediscovery: Uniqueness of representations when the reduced carry polynomial is primitive and irreducible.
- **CLM-C02-08** (written), rediscovery: Exchange representations respect the radix order.
- **CLM-C02-09** (written), rediscovery: The Fibonacci exchange system terminates at the Zeckendorf representation; shortest derivation n - z(n).
- **CLM-C02-10** (computer-assisted), rediscovery: Longest Fibonacci exchange derivations (longest Zeckendorf games) for n <= 40.
- **CLM-C02-11** (written), potentially novel: Termination classification of carry rules: terminating from every unary start iff sum b < m.

16.15 C03 — Termination of one-rule string rewriting 0^p 1^q -> 1^r 0^s

Theorems.

- **CLM-C03-01** (Lean), rediscovery: A one-rule system that decreases the count of some letter terminates.
- **CLM-C03-02** (written), rediscovery: Conserving rules terminate (s = p or r = q).
- **CLM-C03-03** (Lean), rediscovery: Self-embedding derivations prove non-termination; the doubling region r >= 2q, s >= 2p always loops.
- **CLM-C03-04** (Lean), rediscovery: 011 -> 11100 is non-terminating although r < 2q.
- **CLM-C03-05** (Lean), rediscovery: Lifting lemma - loops transfer along letter morphisms (explains divisibility).
- **CLM-C03-06** (computer-assisted), rediscovery: Classification of 601 of the 625 rules with p, q, r, s <= 5 (computer-assisted).
- **CLM-C03-10** (written), rediscovery: Divisibility characterization of termination for 0^p 1^q -> 1^r 0^s (H-C03-4).

Computationally supported.

- **CLM-C03-11**, rediscovery: Loop lengths in the base family grow like k^q.

16.16 C04 — Derivation lengths, random descent and rewriting games

Theorems.

- **CLM-C04-01** (computer-assisted), rediscovery: Termination census of one-rule systems with |l|, |r| <= 3: non-terminating iff l is a factor of r.
- **CLM-C04-03** (written), rediscovery: The rules aab -> ababa and aab -> babaa (and their six symmetric images) terminate.
- **CLM-C04-04** (Lean), rediscovery: ab -> ba: every derivation has inv(w) steps, D(n) = floor(n^2/4), P-positions = even inversion count.
- **CLM-C04-05** (Lean), rediscovery: One-rule systems with an unbordered left-hand side have random descent (Lean-verified).
- **CLM-C04-06** (written), rediscovery: Every one-rule system has a constant-step potential; for terminating ones confluence is equivalent to random descent.

- **CLM-C04-08** (written), rediscovery: Rewriting games of random-descent rules are parity games (Grundy value = D(w) mod 2).
- **CLM-C04-09** (written), potentially novel: The rule aab -> ba has derivational complexity D(n) = n - 1 - min{k : 2^k + k >= n} (non-rational).
- **CLM-C04-10** (written), potentially novel: Identity: OEIS A247983 (truncation index of the series for log 2) equals D(n+2) for aab -> ba.
- **CLM-C04-13** (written), rediscovery: Rewriting games with lhs a^k are take-and-break games: aa -> b is Dawson's Kayles, aa -> ab is Officers.
- **CLM-C04-15** (computer-assisted), rediscovery: The rewriting game of aa -> b on a^n has Grundy values periodic with period 34 from n = 53.

Computationally supported.

- **CLM-C04-12**, rediscovery: Complexity classes and closed forms of the 162 terminating rules with |l|, |r| <= 3.

Conjectures.

- **CLM-C04-16**, rediscovery: Open: are the rewriting games of aa -> ab (Officers), aaa -> b (0.007) and aaa -> ab (0.06) ultimately periodic?.

16.17 C05 — Spectra of single-identity magma varieties

Theorems.

- **CLM-C05-04** (Lean), rediscovery: Law 167, x = (y*x)*(x*y): every finite model has order 0 or 1 mod 4.
- **CLM-C05-05** (Lean), rediscovery: Identities x = y*s(x, y) make every left translation of a finite model bijective.
- **CLM-C05-07** (Lean), potentially novel: New spectrum memberships for open ETP laws: 677 has models of orders 5, 7, 9; 1286, 1313 and 1516 have models of order 9.
- **CLM-C05-08** (computer-assisted), potentially novel: New spectrum gaps for open ETP laws (certified UNSAT): 501 has no models of orders 6, 7; 677 none of order 6; 883 none of 6; 1076 none of 8; 1083 none of 5, 6.
- **CLM-C05-10** (computer-assisted), not assessed: Linear models over finite fields: the orders q <= 128 at which the 15 open laws have a model x*y = ax + by over GF(q).

Computationally supported.

- **CLM-C05-02**, rediscovery: Spectrum census of the 115 nontrivial two-variable identities t = x with 2-4 operations (n <= 6, restricted ones to n <= 8).

16.18 C06 — Successor-anchored magmas: spectra of composition laws

Theorems.

- **CLM-C06-02** (Lean), rediscovery: Anchored associativity forces x*y = x + y + 1, for every N.
- **CLM-C06-03** (Lean), rediscovery: Anchored right-distributivity forces x*y = x + 1, for every N.

- **CLM-C06-05** (written), rediscovery: Anchored right-permutable magmas are x*y = x + c(y); there are N^(N-1) of them.
- **CLM-C06-06** (Lean), potentially novel: An ‘odd’ law: (xy)z = x((yz)z) has an anchored model exactly for odd N, and it is unique.
- **CLM-C06-07** (written), potentially novel: A ‘golden’ law: anchored models of (xy)z = x(y(zz)) correspond to roots of a^2 + a = 1 mod N.
- **CLM-C06-08** (Lean), rediscovery: Anchored left-permutable magmas are commutative monoids in disguise (counted by OEIS A058156).
- **CLM-C06-12** (written), rediscovery: Anchored spectra are closed under coprime products.

Computationally supported.

- **CLM-C06-01**, rediscovery: Left-distributivity with a successor anchor: the Laver tables (rediscovery).
- **CLM-C06-10**, potentially novel: Laws with finite anchored spectra {1, 2} and {1, 3}.
- **CLM-C06-11**, potentially novel: Mirror twins of the golden and odd laws.

16.19 C07 — Arithmetic from the minimal excludant: a census of mex-defined operations

Theorems.

- **CLM-C07-01** (computer-assisted), potentially novel: The associative mex operations are 0, projections, min, +, XOR, nim-product and four relatives.
- **CLM-C07-03** (written), not assessed: In a mex operation with both L and R options 0 is an identity; with neither, 0 is absorbing.
- **CLM-C07-05** (written), potentially novel: The Jacobsthal row: row 2 of f(a,b) = mex{f(a’,b) XOR f(a,b’)} swaps the blocks [2^k - J_k, 2^k) and [2^k, 2^k + J_k).

Computationally supported.

- **CLM-C07-08**, not assessed: Additive periodicity is the exception among mex operations.

16.20 C08 — Circle multiplications on recurrence numeration systems

Theorems.

- **CLM-C08-01** (Lean), rediscovery: Every shifted Knuth product a o_t b = sum F(k+l+t) over Zeckendorf indices is associative (t >= 0).

Computationally supported.

- **CLM-C08-02**, potentially novel: Circle products over non-Pisot base sequences are never associative.
- **CLM-C08-03**, rediscovery: Recurrences with non-increasing coefficients and canonical initial values give associative circle products for large shifts.
- **CLM-C08-04**, potentially novel: Non-monotone associative-shift sets require a negative conjugate root.

- **CLM-C08-08**, rediscovery: Associativity thresholds after the targeted search - certified lower bounds, no failure up to s = 16.

16.21 C09 — Arithmetic of Knuth’s circle semigroup

Theorems.

- **CLM-C09-01** (written), rediscovery: Knuth’s circle semigroup embeds in Z[phi]; Fibonacci numbers are its units-images.
- **CLM-C09-03** (written), potentially novel: Window obstruction: at least a proportion 1/phi^2 of all integers are circle-irreducible.
- **CLM-C09-06** (written), potentially novel: Factorisation lengths are bounded by the conjugate coordinate.
- **CLM-C09-07** (written), rediscovery: The norm N(n) = n*^2 + n n* - n^2 is multiplicative for the circle product.
- **CLM-C09-08** (computer-assisted), potentially novel: Factorisation statistics of the circle semigroup up to 2e5.

Conjectures.

- **CLM-C09-05**, potentially novel: The density of circle irreducibles is exactly 1/phi^2.

16.22 C10 — The cost of counting across representations

Theorems.

- **CLM-C10-01** (written), rediscovery: Average cost of counting in classical representations (b/(b-1), e-1, 1).
- **CLM-C10-02** (written), rediscovery: Zeckendorf counting: T_m = 2 G_m - ceil(m/2) - 2, average cost 2.
- **CLM-C10-04** (written), rediscovery: Exchange systems fire 1/(m - sum b) times per increment on average.
- **CLM-C10-05** (written), rediscovery: Single-offset exchange rules cost exactly firings + 1 per increment (average 1 + 1/(m - q)).
- **CLM-C10-07** (written), rediscovery: Skew binary counting has worst-case cost 2 and average 3/2.

Computationally supported.

- **CLM-C10-06**, potentially novel: Multi-offset exchange rules have rule-specific average counting costs.

16.23 C11 — Pattern-terminated universal integer codes

Theorems.

- **CLM-C11-01** (written), rediscovery: Pattern-terminated codes T_p are complete prefix codes with counts x^k / (x^k + (1-2x) c_p(x)); T_11 is the Fibonacci code.
- **CLM-C11-02** (written), rediscovery: The all-ones terminator 1^k (and 0^k) uniquely maximises the growth rate among length-k patterns.

- **CLM-C11-03** (written), rediscovery: Unbordered terminators: F_u = x/(1-x) F_(1^k), so codeword counts are partial sums of the all-ones counts.
- **CLM-C11-04** (written), potentially novel: T_011 is never longer than the Fibonacci code for n >= 5 and one bit shorter on all but L values per level.
- **CLM-C11-07** (written), potentially novel: The average gain oscillates between (2-lambda)/(lambda-1)^2 and lambda(2-lambda)/(lambda-1).
- **CLM-C11-09** (written), potentially novel: T_011 is never longer than Fraenkel-Klein's C3 except at n = 2, 4, 8; C3 is never longer than C1 except at n = 1.

Computationally supported.

- **CLM-C11-08**, not assessed: Fixed pattern-terminated codes beat classical universal codes on heavy-tailed and wide-range sources (mixed result).

16.24 C12 — Relational one-heap games

Theorems.

- **CLM-C12-01** (written), rediscovery: Divisibility games: subtract a proper divisor = v2, subtract a divisor = v2 + 1, move to a divisor = Omega.
- **CLM-C12-02** (written), rediscovery: Coprimality games: index of the smallest prime factor; the non-coprime game has g(2k) = k - 1.
- **CLM-C12-03** (Lean), rediscovery: P-positions of the deleting-divisors game (subtract d | n with 1 < d < n): 0, odd numbers, 2^(2j+1).
- **CLM-C12-04** (computer-assisted), potentially novel: Grundy function of the deleting-divisors game: v2(n) except on odd parts <= 15.
- **CLM-C12-05** (written), rediscovery: Bitwise and quotient games: popcount, floor(log2((n+1)/3)) + 1 and floor(log2 n).
- **CLM-C12-06** (written), rediscovery: Residue lemma; subtract-a-power-of-k has the Grundy function of the subtraction game {1, k}.
- **CLM-C12-07** (Lean), rediscovery: Splitting games in which every heap can be split have g(n) = [n even].
- **CLM-C12-08** (written), rediscovery: Subtract-a-square and subtract-a-triangular-number have infinite density-zero P-sets and are not periodic.

Computationally supported.

- **CLM-C12-12**, rediscovery: Subtract-a-prime has Grundy values at most 11.

Conjectures.

- **CLM-C12-13**, not assessed: Subtract-a-Fibonacci-number has bounded Grundy values.

16.25 C13 — Anchored spectra II: which sets of integers are spectra?

Theorems.

- **CLM-C13-01** (written), rediscovery: Affine anchored spectra are root sets of polynomial systems - divisor-closed, CRT-closed, finite ones are divisor sets.
- **CLM-C13-02** (Lean), potentially novel: Right-normed laws (xy)z = x V(y,z) have only translation models.

- **CLM-C13-03** (Lean), potentially novel: The W-family: anchored models of (xy)z = x(y W(z)) are the affine tables with a*c_W(a) = 1.
- **CLM-C13-04** (Lean), potentially novel: Exact finite spectra {1,3} and {1,2} for x(yz) = ((zx)x)y and x(yz) = ((xy)z)x.
- **CLM-C13-05** (Lean), not assessed: The law x = x(x(xx)) has anchored models of every size except 2.
- **CLM-C13-06** (computer-assisted), not assessed: Offset congruence closure is a sound prover of finite anchored spectra; it settles 16 of the 17 laws of CLM-C06-10.
- **CLM-C13-11** (written), potentially novel: Single laws whose affine spectra are classical arithmetic sets.

Computationally supported.

- **CLM-C13-10**, not assessed: Spectra of one-variable laws are finite or cofinite.
- **CLM-C13-12**, not assessed: Non-affine models typically exist for all sizes beyond a small threshold; no Laver-like sparse spectrum found.

16.26 C14 — Graphs as quantities: the 256 local graph products

Theorems.

- **CLM-C14-01** (Lean), not assessed: Associativity of a local graph product is a finite identity on relations, detected by P3 x P3 x P3 (Lean).
- **CLM-C14-02** (Lean), rediscovery: Exactly 20 of the 256 local graph products are associative, 10 commutative, 8 with unit K1 (Lean decide).
- **CLM-C14-03** (computer-assisted), not assessed: For local graph products, associativity and commutativity up to isomorphism are no weaker than the strict notions.
- **CLM-C14-04** (written), potentially novel: Exactly six local products are Kronecker products in disguise (multiplicative characters).
- **CLM-C14-05** (written), potentially novel: Additive characters: only the Cartesian product and its complement-dual; every non-trivial commutative associative product has a character.
- **CLM-C14-06** (written), rediscovery: The symmetric-difference product is the Kronecker product of Seidel matrices; it acts on two-graphs.
- **CLM-C14-07** (written), potentially novel: Complete bipartite graphs multiply as (order, imbalance); K2 absorbs Seidel switching.
- **CLM-C14-09** (written), rediscovery: The Cartesian and strong product monoids of all finite graphs are isomorphic graded monoids; equal prime counts (A129582).
- **CLM-C14-10** (computer-assisted), potentially novel: Numbers of prime graphs (all graphs) for the unital associative products, orders <= 15.
- **CLM-C14-11** (written), rediscovery: The lexicographic product has non-unique factorisation already at order 10 (different multisets of primes).

Computationally supported.

- **CLM-C14-12**, not assessed: Code 141 (EA AE AA NN) has unique factorisation on connected graphs (equivalently, symmetric difference on co-connected graphs).
- **CLM-C14-13**, potentially novel: H-C14-4: two-graphs form a free commutative monoid under the symmetric-difference (Kronecker) product.

16.27 C15 — Magma cellular automata: light cones, streaming width and fast-forwarding

Theorems.

- **CLM-C15-01** (Lean), rediscovery: Light cones of semigroups - the popcount word, Pascal exponents, bounded streaming width.
- **CLM-C15-02** (written), rediscovery: Affine magmas - binomial light cone and an O(L log L log t) fast-forward.
- **CLM-C15-03** (written), not assessed: Commutative semigroups - fast-forward by exponent arithmetic.
- **CLM-C15-04** (written), potentially novel: The fuse magma: linear streaming width, yet an O(L) fast-forward for every t.
- **CLM-C15-05** (computer-assisted), rediscovery: Surjective iff permutive for magmas of order at most 3; false at order 4.
- **CLM-C15-06** (written), rediscovery: The two-element magma automata are the elementary rules 17m.

Computationally supported.

- **CLM-C15-07**, potentially novel: Census of order-3 magma automata by streaming width.
- **CLM-C15-13**, rediscovery: Affine fast-forward by operator powers.
- **CLM-C15-14**, not assessed: Commutative-semigroup fast-forward by exponent arithmetic.
- **CLM-C15-15**, potentially novel: Fuse fast-forward by burn times.
- **CLM-C15-16**, rediscovery: Cycle detection as a generic fast-forward.
- **CLM-C15-17**, not assessed: Earlier structures of the lab as automata.

Conjectures.

- **CLM-C15-18**, not assessed: Bounded streaming width implies a fast-forward.

16.28 C16 — Fibonacci dynamics in invented finite algebras

Theorems.

- **CLM-C16-01** (Lean), rediscovery: The Fibonacci pair map of a magma is a permutation iff every left translation is a bijection.
- **CLM-C16-03** (written), rediscovery: Full-period magmas are de Bruijn cycles: there are (n!)^n / n^2 of them.
- **CLM-C16-04** (written), not assessed: A parity condition for full periods.
- **CLM-C16-05** (written), potentially novel: Full-period quasigroups are cycles in which every pair occurs once at distance 1 and once at distance 2.
- **CLM-C16-07** (computer-assisted), potentially novel: Full-period quasigroups: none of order 2, 3, 4; 3360 of order 5; examples of every order 6-16.
- **CLM-C16-08** (written), not assessed: Direct products of full-period magmas of coprime orders are full-period.
- **CLM-C16-09** (written), potentially novel: No isotope of a group of even order has a full-period Fibonacci pair map.
- **CLM-C16-11** (Lean), not assessed: Fixed points of the pair map are idempotents.

- **CLM-C16-12** (written), rediscovery: Affine magmas: periods are orders of companion matrices; the anchored magma x + y + 1 is the Fibonacci sequence mod N.
- **CLM-C16-14** (written), rediscovery: Fibonacci sequences of nimbers: maximal period pi(2^(2^k) - 1).
- **CLM-C16-15** (written), potentially novel: Laver tables: every Fibonacci orbit ends in the fixed point or one of the N - 1 two-cycles through the top element.
- **CLM-C16-18** (computer-assisted), not assessed: Maximal Fibonacci periods of small groups.

Computationally supported.

- **CLM-C16-10**, not assessed: Full-period quasigroups exist for every order n >= 5.
- **CLM-C16-13**, not assessed: For affine magmas the maximal period equals the order of the 3x3 affine matrix.
- **CLM-C16-16**, not assessed: Fibonacci orbits in Laver tables reach the top element within 5 steps (n <= 10).
- **CLM-C16-17**, rediscovery: Magma Fibonacci generators are not competitive pseudo-random generators.
- **CLM-C16-19**, not assessed: Random sampling finds full-period quasigroups where exact search fails.

16.29 C17 — Deciding laws of finite magmas: generator, anchor and randomised tests against the exhaustive check

Theorems.

- **CLM-C17-01** (written), rediscovery: Light’s test: generators suffice for associativity.
- **CLM-C17-02** (written), not assessed: Associativity of an anchored magma is decided by one O(n^2) identity.
- **CLM-C17-03** (written), rediscovery: The Rajagopalan-Schulman randomised associativity test is one-sided.

Computationally supported.

- **CLM-C17-04**, rediscovery: Benchmark - generator and anchor tests versus the exhaustive check.
- **CLM-C17-05**, not assessed: The anchor instance of left-distributivity singles out the Laver tables (every N <= 4096).

16.30 C18 — Invented operations as mixing primitives: quasigroup string transformations versus standard hashes

Theorems.

- **CLM-C18-01** (Lean), rediscovery: e-transformations are bijections of strings of fixed length.
- **CLM-C18-02** (Lean), not assessed: Differences propagate forward only; forward-only tail hashes fail the strict avalanche criterion.

- **CLM-C18-03** (Lean), not assessed: The tail-information bound: an e-transformation tail carries one byte of state.
- **CLM-C18-04** (written), rediscovery: T-function quasigroups (affine mod 256, XOR, (x + 1) XOR y) cannot mix high input bits into low output bits.
- **CLM-C18-05** (written), rediscovery: GF(2)-affine quasigroups give GF(2)-affine hashes with key-independent differences.
- **CLM-C18-06** (written), not assessed: The anchored quasigroup x + c(y): additive differences persist, and the one-pass hash is anagram-blind.
- **CLM-C18-07** (written), not assessed: The 8-pass cascade has a bijective 64-bit state update.

Computationally supported.

- **CLM-C18-08**, not assessed: H-C18-1: QG-LS with 2 passes and a reverse pass has BLAKE2b-level avalanche.
- **CLM-C18-13**, not assessed: Quasigroup hashes against non-cryptographic baselines - no significant improvement.

16.31 C19 — The longest Zeckendorf game

Theorems.

- **CLM-C19-01** (written), rediscovery: Every Zeckendorf game has length B(n) - sum_{i>=2} (i-1) MC_i.
- **CLM-C19-02** (written), potentially novel: The split phase of the longest game ends at the lazy Fibonacci representation.
- **CLM-C19-03** (written), potentially novel: Closed form of the longest game on a Fibonacci number: L(F_k) = F_{k+2} - k - 2 - floor((k-1)^2/4).
- **CLM-C19-04** (written), rediscovery: A short proof that only n = F_k - 1 can be played with split moves alone (Cusenza et al., Thm 1.4).
- **CLM-C19-07** (computer-assisted), rediscovery: Exact longest games for n <= 150 and independent confirmation of the split-first strategy.
- **CLM-C19-08** (written), not assessed: Every multiset of Fibonacci numbers with sum n is a reachable game position.

Computationally supported.

- **CLM-C19-05**, potentially novel: A two-sequence recursion for the longest Zeckendorf game on every n.
- **CLM-C19-06**, potentially novel: The deficit of a two-summand number depends only on the lower level and the parity of the gap.
- **CLM-C19-09**, not assessed: D(n) <= floor(j^2/4) for top level j, and L is strictly increasing.
- **CLM-C19-14**, potentially novel: Computing the longest game in O(log^2 n) operations instead of simulating Theta(n) moves.

16.32 C20 — Open problems revisited: the last one-rule systems of C04, and a literature check for C03

Theorems.

- **CLM-C20-01** (written), rediscovery: aab -> ababa terminates, with derivational complexity D(n) = 2^(n-2) - 1.

- **CLM-C20-02** (written), potentially novel: aab -> babaa terminates; exact per-word derivation length and D(n) = max_b b(2^floor((n-b)/2) - 1).
- **CLM-C20-03** (computer-assisted), not assessed: The C04 extended termination census is complete - 480 of the 868 rules with |l| <= 3, |r| <= 5 terminate.
- **CLM-C20-04** (written), rediscovery: C03 Conjecture 4.6 coincides with the Zantema-Geser characterisation.

Volume 7 — Computational Experiments and Algorithms

Algorithms that the research produced or needed, measured against baselines.

17 Magma automata and fast-forwarding

Volume 7 — Computational experiments and algorithms. Research cycle C15.

17.1 Motivation

A binary operation can be read as a law of motion. Put symbols in a row and let every cell look at its right neighbour:

$$c'_i = c_i * c_{i+1}.$$

Every magma of the earlier chapters — the invented operations of Volume 2, the Laver tables, the nimbers — becomes a one-dimensional cellular automaton. Two questions follow. How *complicated* is the state after t steps as a function of the initial row? And can it be computed without simulating all t steps ("fast-forwarding")?

17.2 Definitions

Cell i at time t is $F_t(c_i, \dots, c_{i+t})$, where $F_0(x_0) = x_0$ and

$$F_t(x_0, \dots, x_t) = F_{t-1}(x_0, \dots, x_{t-1}) * F_{t-1}(x_1, \dots, x_t).$$

A cell x_k is *essential* if F_t depends on it. To measure complexity we read the arguments of F_t from left to right and count how many different residual functions of the unread arguments can occur after a prefix of fixed length; the maximum over prefix lengths is the *forward width* (the width of the reduced ordered decision diagram). Reading from the right gives the backward width, and the smaller of the two is the **streaming width** $\lambda(t)$ — the memory a one-pass reader needs, up to a logarithm. The automaton is *permutive* if all rows or all columns of the table are permutations, *surjective* if every bi-infinite row has a preimage.

17.3 Two letters: the elementary rules

Theorem 17.1 *[PROVEN; REDISCOVERY].* The 16 automata of two-element magmas are exactly the elementary cellular automata that ignore the left neighbour — the Wolfram rules $0, 17, 34, \dots, 255$. All have streaming width at most 3; XOR (rule 102) depends on $2^{\mathrm{popcount}(t)}$ cells (Gould's sequence).

17.4 Structure makes the light cone simple

Theorem 17.2 *[PROVEN — Lean: `ca_comm_binomial`; REDISCOVERY].* If $*$ is associative, $F_t(x_0, \dots, x_t) = \prod_{j < 2^t} x_{s(j)}$ with $s(j)$ the number of ones in the binary expansion of j. If it is also commutative,

$$F_t(x_0, \dots, x_t) = \prod_{k=0}^{t} x_k^{\binom{t}{k}},$$

Pascal's triangle in the exponents, and the streaming width is at most $|S|$.

Theorem 17.3 *[PROVEN — written; REDISCOVERY: linear cellular automata].* For an affine operation $x * y = \varphi(x) + \psi(y) + c$ on an abelian group (φ, ψ commuting endomorphisms), $F_t = \sum_k \binom{t}{k} \varphi^{t-k} \psi^k(x_k) + \mathrm{const}$, the streaming width is at most $|A|$, and a cyclic row at time t is $P^t r + \mathrm{const}$ with $P = \varphi + \psi S$ (S the shift), computable with $O(\log t)$ polynomial multiplications modulo $S^L - 1$.

For commutative semigroups the exponents $\binom{t}{k}$ can be grouped by residue mod L and reduced with the index and period of each element, which gives a second exact fast-forward (Theorem 3 of `proofs/paper/C15.md`).

17.5 Surjectivity

Theorem 17.4 *[PROVEN — computer-assisted; REDISCOVERY: Kari, Salo and Törmä].* For magmas of order at most 3 the automaton is surjective if and only if it is permutive; there are $1, 6, 420$ surjective tables of orders $1, 2, 3$, namely $2(n!)^n$ minus the number of Latin squares. For order 4 the rectangular band $(b, c) * (b', c') = (b, c')$ is surjective — even reversible, an identity on one coordinate and a shift on the other — without being permutive.

The order-3 statement is the prime-alphabet case of a 2013 theorem; our exhaustive pair-graph computation is an independent confirmation, and the order-4 band shows why the hypothesis “prime” cannot simply be dropped.

17.6 A census of order 3

There are 19683 tables of order 3, 1734 up to isomorphism and mirror image.

Observation 17.5 *[COMPUTATIONALLY SUPPORTED, $t \le 9$].* The streaming width is bounded for 808 classes, grows linearly for 104, roughly geometrically for 300 (up to $\lambda(9) = 649$) and irregularly for 522. All 18 semigroup and all 8 affine classes are bounded; 43 classes are surjective (exactly the permutive ones), 7 are reversible, 34 are nilpotent.

Two bold hypotheses fail. H-C15-1 predicted that widths are either bounded or geometric. H-C15-2 predicted that bounded width means “semigroup, affine, or a light cone of at most two cells”. In fact 685 bounded-width classes have none of these structures — for example the reversible automaton with table $[[0, 0, 2], [2, 2, 0], [1, 1, 1]]$, whose light cone uses every cell but whose width stays 3 — and linear growth exists, as the next result proves.

17.7 The burning fuse

Let 0 mean *burnt*, 1 *burning* and 2 *intact*, with $0 * y = 0$; $1 * y = 1$ if $y = 2$, else 0; $2 * y = 2$ if $y \neq 0$, and $2 * 0 = 1$. An intact cell starts burning when its right neighbour has burnt, and burns out one step later.

Theorem 17.6 *[PROVEN — written; POTENTIALLY NOVEL (small)].* Let τ_i be the time at which cell i burns out. Then $\tau_i = \tau_{i+1} + 2$ for an intact cell, $\tau_i = 0$ for a burnt one, and $\tau_i \in \{1, \tau_{i+1}\}$ for a burning one, according as its right neighbour is not or is intact. Consequently

1. the row at any time t can be computed in $O(L)$ operations, whatever t;
2. the streaming width grows linearly: with $m = \lfloor t/2 \rfloor$,
$$\left\lfloor \tfrac{m+1}{2} \right\rfloor \le \lambda(t) \le 3t + 3,$$
and in fact $\lambda(t) = t + 2$ for all $t \le 12$.

Proof idea. A run of intact cells burns from the right, two steps per cell, so the light cone has to compare t with twice the length of a run — a counter, which no bounded memory can hold. The prefixes $2\,(12)^b\,2^{m-1-2b}$ and suffixes $2^j\,02^{\cdots}$ produce output 2 exactly when $j \ge b + 1$ (for even t), so they generate pairwise distinct residuals. ∎

The fuse refutes the “dichotomy” H-C15-1, and it also refutes H-C15-4 — the brief’s hypothesis that an automaton can be fast-forwarded exactly when its light-cone complexity is bounded: its width is unbounded, yet the future is computable in $O(L)$. The reverse direction is open (Conjecture CLM-C15-18).

17.8 Experiments

Experiment	Question	Result
EXP-C15-01	order 2	rules $17m$; widths ≤ 3
EXP-C15-02	order-3 census	808 / 104 / 300 / 522 width classes; surjective = permutive
EXP-C15-03	exact fast-forwards	99/99 agree with simulation
EXP-C15-04	cycle detection	1590 of 1734 rows ($L = 32$) repeat within 5000 steps
EXP-C15-05	case studies	fuse $\lambda(t) = t + 2$ to $t = 12$; Laver tables; order-4 band

17.9 Benchmark

Direct numpy simulation is the baseline; rows of length $L = 1024$; times are single runs on a shared machine and only their order of magnitude is meaningful. (Our benchmark timed the fast methods as the best of three runs and the simulation once; the Skeptic’s re-run with one protocol for both gave, for example, ×202–314 and ×1459–2262 for the affine method at $t = 10^4$ and 10^5, and ×1363 instead of ×4000 for the fuse method.)

method	$t = 10^2$	$t = 10^3$	$t = 10^4$	$t = 10^5$	$t = 10^9$
affine (5 magmas)	×3–4	×22–29	×138–215	×1800–2500	1.5–3 ms
fuse burn times	×2.5	×27	×239	×4000 (re-run ×1363)	0.6 ms
commutative semigroups (3)	×1–3	×0.9–2.1	×6–9	×62–100	21–42 ms

With the same protocol on both sides, the affine method loses below $t \approx 20$ and the fuse method below $t \approx 20$–70. | cycle detection | exact whenever the row repeats; 1590/1734 classes within 5000 steps at $L = 32$; fails for the affine quasigroups mod 3 | | | | |

A third hypothesis fails. H-C15-6 predicted more than ×100 for every exact method at $t = 10^5$ and losses for small t. The commutative-semigroup method stays below ×100 — its final product costs $O(L^2)$ and it is even slower than simulation near $t = 10^3$. The first version said the affine and fuse methods “never lose”; timed with the same protocol on both sides (our benchmark had used the best of three runs for the fast method and a single run for the simulation) they lose for t below about 20–70, so the small-t clause of H-C15-6 is partly true, and the refutation rests on the ×100 clause. And H-C15-5 (“long cycles only for geometric width”) fails because the affine quasigroups mod 3 have bounded width but long cycles.

17.10 Comparison with existing mathematics

Automata defined by a binary operation were studied algebraically by Pedersen (1992) and by Moore with Drisko and with Pnin (1995–1997), who showed that rules given by nilpotent or solvable groups can be predicted in logarithmic parallel time. Linear automata and their fast-forward go back to Martin, Odlyzko and Wolfram (1984); the surjectivity theorem for prime alphabets is due to Kari, Salo and Törmä (2013), building on Hedlund (1969). We found no prior use of the streaming width of t-step rules, no census of order-3 magma automata and no analysis of the fuse rule.

17.11 Applications and limitations

Exact fast-forwarding gives the state of affine and fuse automata after 10^9 steps in milliseconds. The census judges growth on $t \le 9$ only, and the “geometric” label is empirical; timings come from a loaded shared machine.

17.12 Open questions

1. What do the 685 bounded-width automata without known structure have in common?
2. Does bounded streaming width always allow a fast-forward polynomial in L and $\log t$?
3. Is the width of the most complex order-3 automaton truly exponential?
4. How large are the light cones of Laver tables in general?

18 Fibonacci dynamics in invented algebras

Volume 7 — Computational experiments and algorithms. Research cycle C16.

18.1 Motivation

The Fibonacci numbers need two things: a pair of starting values and a rule that turns the last two values into the next one. Any invented operation provides such a rule:

$$x_{k+1} = x_k \star x_{k-1}.$$

Modulo m with ordinary addition this gives the Pisano periods. What happens in a finite magma with no laws at all — and what do the laws studied earlier in this book (anchoring, left-distributivity, the odd and golden laws, nim arithmetic) do to these sequences?

18.2 Definitions

On a magma M with n elements the state is the pair (x_{k-1}, x_k) and the *pair map* is

$$P(a,b) = (b,\ b \star a)$$

on the n^2 states. Periods are the cycle lengths of P; a table is *full-period* if P is a single cycle through all n^2 pairs. Write $L_b(a) = b \star a$ for the left translations. Every function $f : M \times M \to M$ is the rule of exactly one magma, so the pair maps of n-element magmas are exactly the *feedback shift registers of order 2* over an n-letter alphabet — a translation that turned several of our findings into rediscoveries.

18.3 How long can a magma's Fibonacci sequence cycle?

Theorem 18.1 *[PROVEN — Lean: `pairMap_injective_iff`, `pairMap_bijective_iff`; REDISCOVERY: Golomb's nonsingularity criterion].* P is a permutation iff every left translation L_b is a bijection; exactly $(n!)^n$ tables have this property. The mirror recurrence $x_{k+1} = x_{k-1} \star x_k$ is the forward recurrence of the opposite magma.

A bold hypothesis fails. For Fibonacci numbers modulo m the pair $(0,0)$ is fixed, so the period is at most $m^2 - 1$. H-C16-1 guessed the same bound for every magma, attained only by quasigroups. Already for $n = 2$ the table $x \star y = 1 - y$ produces $0, 0, 1, 1, 0, 0, 1, 1, \ldots$ — period 4.

Theorem 18.2 *[PROVEN — written; REDISCOVERY: van Aardenne-Ehrenfest and de Bruijn, 1951].* A table is full-period iff the symbols along its cycle form a de Bruijn cycle of order 2 (every ordered pair once as consecutive symbols), and this correspondence is a bijection. Hence there are exactly

$$\frac{(n!)^n}{n^2}$$

full-period tables of order n — $1, 24, 20736$ for $n = 2, 3, 4$, as the exhaustive census confirms — and the maximal period is n^2 for every n.

Theorem 18.3 (parity) *[PROVEN — written].* If every L_b is a bijection, $\operatorname{sgn} P = (-1)^{n(n-1)/2} \prod_b \operatorname{sgn} L_b$, because P is a coordinate swap after fibrewise left translations. A full period therefore requires

$$\prod_b \operatorname{sgn} L_b = (-1)^{(n-1)(n+2)/2}.$$

18.4 Quasigroups

None of the 24 full-period tables of order 3, nor any of the 20736 of order 4, is a quasigroup, and the largest quasigroup period was $n^2 - 1$ for $n = 2, 3, 4$.

H-C16-5 [REFUTED]. "No quasigroup is full-period."

Theorem 18.4 *[PROVEN — written; POTENTIALLY NOVEL].* A full-period table is a quasigroup iff every ordered pair also occurs exactly once at *distance 2* (x_i, x_{i+2}) along its cycle. Equivalently, the windows (x_{i-1}, x_i, x_{i+1}) run once through the n^2 triples $(a, b, b \star a)$ of the Latin square: the cycle is a universal cycle of the quasigroup.

Theorem 18.5 *[PROVEN — computer-assisted; POTENTIALLY NOVEL].*

order n	**2**	**3**	**4**	**5**	**6 … 16**
full-period quasigroups	0	0	0	3360 (28 classes)	examples verified

The counts for $n \le 5$ come from all $2 + 12 + 576 + 161280$ Latin squares and, independently, from a depth-first enumeration of cycles with distinct pairs at distances 1 and 2. An example of order 5 is the cycle

$$0, 0, 1, 0, 2, 2, 0, 3, 3, 1, 2, 1, 4, 0, 4, 4, 2, 3, 2, 4, 1, 1, 3, 4, 3.$$

Theorem 18.6 *[PROVEN — written].* The direct product of full-period quasigroups of coprime orders is full-period (cycles of coprime lengths combine into one), giving orders such as $30, 35, 40, 42, 45, 56, 63, 72$.

Random sampling revealed a striking pattern: isotopes of the cyclic group $\mathbb{Z}_n$ were full-period for every odd n from 5 to 15 and *never* for even n. The parity condition explains it.

Theorem 18.7 *[PROVEN — written; POTENTIALLY NOVEL].* No isotope $\gamma(\alpha(x)\beta(y))$ of a group of even order $n = 2m$ is full-period: its rows have sign product $(-1)^m$, while Theorem 18.3 demands $(-1)^{m+1}$.

Proof idea. The sign product equals that of the left regular representation, $\prod_g \operatorname{sgn} \lambda_g = (-1)^{\#\{g\,:\, n/o(g) \text{ odd}\}}$, and by Burnside's normal 2-complement theorem exactly m elements (or none, with m even) have $n/o(g)$ odd. ∎

Conjecture 18.8 *[COMPUTATIONALLY SUPPORTED].* Full-period quasigroups exist for every $n \ge 5$. Random Latin squares are full-period with probability of the order of $1/n^2$ — roughly what a random permutation of the forced parity would give, though not close to it (exactly $0.52/n^2$ at $n = 5$). Exhaustive search and a SAT encoding, by contrast, stall at $n = 9$. Examples are known for every $n \le 26$ (the Skeptic's sampler covers 17–26).

18.5 Laws shape the dynamics

Theorem 18.9 *[PROVEN — Lean: `pairMap_fixed_iff`].* The fixed points of P are the pairs (a, a) with $a \star a = a$; an idempotent magma has period at most $n^2 - n$, and no idempotent quasigroup is full-period.

Theorem 18.10 (affine magmas) *[PROVEN — written, Lean for the anchored case; REDISCOVERY: periods of linear recurrences].* For $x \star y = \alpha x + \beta y + \gamma$ on $\mathbb{Z}/N$ with a fixed point, the maximal period is the order of the companion matrix $\left(\begin{smallmatrix} 0 & 1 \\ \beta & \alpha \end{smallmatrix}\right)$. So the anchored

associative magma $x + y + 1$ of Chapter 7 is the Fibonacci sequence in disguise ($y_k = x_k + 1$), with maximal period the Pisano period $\pi(N)$. The odd law's $x + 1 + y/2$ and the golden law's $x + 1 + ay$ have the orders of $\left(\begin{smallmatrix} 0 & 1 \\ 1/2 & 1 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 0 & 1 \\ a & 1 \end{smallmatrix}\right)$. Their period sequences, $6, 24, 48, 18, 5, \ldots$ and $4, 120, 342, \ldots$, are not in the OEIS. For general affine magmas, the maximal period matched the order of the 3×3 affine matrix in 300 of 300 random cases (a conjecture).

Theorem 18.11 *[PROVEN — written; REDISCOVERY].* In the nimber fields $[0, 2^{2^k})$ of Chapter 14, nim-multiplication gives maximal period $\pi(2^{2^k} - 1)$: 40 on $[0, 16)$ and 360 on $[0, 256)$. (Logarithms turn the recurrence into Fibonacci numbers modulo $2^{2^k} - 1$.)

Theorem 18.12 (Laver tables) *[PROVEN — written, using Laver's monotonicity; POTENTIALLY NOVEL (small)].* For $n \geq 1$ the pair map of the Laver table A_n has exactly one fixed point and $2^n - 1$ two-cycles, all through the top element. Every Fibonacci sequence climbs strictly ($x \star y > x$ below the top) until it reaches the top, then alternates.

So the bold guess that Laver periods are powers of 2 was true, for the dullest possible reason. Observed but unexplained: the climb never takes more than five steps ($n \leq 10$).

For small groups the maximal periods are S3: 8, D4: 6, Q8: 6, A4: 16, S4: 18; the dihedral value agrees with Knox's period 6 for generating pairs.

18.6 Experiments

Experiment	Question	Result
EXP-C16-01	all magmas, orders 2–4	max period n^2; de Bruijn counts; no full-period quasigroup
EXP-C16-02	quasigroups	0, 0, 0, 3360; examples to 16; parity obstruction confirmed
EXP-C16-03	structured families	Laver: period ≤ 2; affine = matrix orders; nimbers 40, 360
EXP-C16-04	PRNG test	no improvement over PCG64
EXP-C16-05	search methods	sampling beats DFS and SAT from $n = 9$

generator (2^20 outputs)	result
PCG64 (baseline)	passes every test
LCG $1664525x + 1013904223$ (baseline)	lowest bit of period 2; low byte too regular
anchored Fibonacci $x + y + 1$ on $\mathbb{Z}/2^{32}$	low byte non-uniform; lowest bit of period 3
affine $x + \beta y + 1$ on $\mathbb{Z}/2^{32}$	low byte non-uniform; lowest bit of period 3
de Bruijn byte magma (period 65536)	too regular in every frequency test

18.7 Comparison with existing mathematics

Feedback shift registers (Golomb, 1967), de Bruijn counts (van Aardenne-Ehrenfest and de Bruijn, 1951), periods of linear recurrences (Ward, 1933; Wall, 1960) and Fibonacci lengths of groups (Knox, 1992; Campbell and coauthors) cover the general theory. Tuscan squares (Etzion, Golomb and Taylor, 1989) impose similar distance-2 conditions on arrays

rather than single cycles. We found no description of full-period quasigroups, their counts, the even-order isotope obstruction, or Fibonacci sequences in Laver tables.

18.8 Applications and limitations

The test against standard generators was negative: order-2 recurrences on n symbols cannot have period longer than n^2, and affine versions modulo 2^{32} inherit the known low-bit weaknesses of lagged Fibonacci generators. The test battery is a sanity check, not TestU01. The existence of full-period quasigroups beyond order 16 is unproved.

18.9 Open questions

1. Do full-period quasigroups exist for every $n \geq 5$? A construction for even n must avoid isotopes of groups (Theorem 18.7).
2. How many are there of order 6, and is the count close to $L(6)/36$?
3. Why do Fibonacci sequences in Laver tables reach the top within five steps?

19 Deciding laws of finite operations

Volume 7 — Computational experiments and algorithms. Research cycle C17.

19.1 Motivation

Every model found by the laboratory's SAT searches had to be checked: is this table really associative, left-distributive, …? The obvious check examines all n^3 triples. This chapter measures how much the structure found in earlier chapters can save, against honest baselines.

19.2 Definitions

A finite magma is a table T with $T[a,b] = a \star b$ on n elements. The cost of a test is its number of *probes* (table look-ups) — an exact, machine-independent measure. Wall-clock times were also recorded, on a shared machine, and are indicative only.

19.3 Three tests

Theorem 19.1 (Light's test) *[PROVEN — written; REDISCOVERY: Light, Clifford–Preston].* If S generates the magma, then it is associative iff $(x \star s) \star y = x \star (s \star y)$ for all x, y and all $s \in S$.

Proof idea. The set of "good middles" s is closed under $\star$: if a and b are good, four applications of the definition show that $a \star b$ is good. A submagma containing the generators is everything. ∎

Corollary 19.2 *[PROVEN].* For an anchored magma ($x \star 0 = x + 1$), associativity holds iff $x \star (y + 1) = (x \star y) + 1$ for all x, y — a check of $2n^2$ probes. (Chapter 7: this identity alone forces $x \star y = x + y + 1$.)

Theorem 19.3 *[PROVEN (one-sidedness); detection bound cited — REDISCOVERY: Rajagopalan–Schulman 2000].* Comparing $(u \star v) \star w$ with $u \star (v \star w)$ for random subsets u, v, w, computed in the magma algebra over GF(2), never rejects an associative table and rejects a non-associative one with probability at least $1/8$ per round, at $O(n^2)$ probes per round.

19.4 The benchmark

Inputs: cyclic groups, anchored tables $x + y + 1$, left-zero bands ($x \star y = x$), random magmas, and single-entry corruptions of the first two — the hardest case for sampling, since one wrong entry spoils only about $4n$ of the n^3 triples.

family ($n = 512$)	exhaustive / Light probes	exhaustive time	Light + generators	anchored test
cyclic group	383	1.7 s	6 ms	—
anchored $x + y + 1$	614	1.3 s	4 ms	3 ms
left-zero band	**1.7**	1.4 s	0.94 s	—
one corrupted entry	0.15–1.1	0.16 s*	4–5 ms	3 ms

family ($n = 512$)	exhaustive / Light probes	exhaustive time	Light + generators	anchored test
random	$4 \cdot 10^{-6}$ (4 probes against $\approx 9 \cdot 10^{5}$)	0.19 s*	4–5 ms	—

* Our vectorised implementation evaluates a whole slab of 64 rows; the probe column counts a sequential check that stops at the first violating triple.

The first three rows are *certification*: the input is associative and every exact test must do its full work. There Light's test wins by two to three orders of magnitude whenever a small generating set exists. The last two rows are *refutation*, and there the exhaustive check wins: a random magma fails at the first triple it looks at. The first version of this table showed gains of 48–77 on these rows; the Skeptic traced them to an accounting error (the exhaustive check was charged a whole slab as soon as the slab contained a violation), and the numbers above are re-measured.

All exact tests and the randomised test (40 rounds) returned the correct verdict on all 90 instances; random-triple sampling with $n^2/16$ samples missed 27 of the 30 near-associative inputs.

H-C17-1 [REFUTED]. "Light's test always wins by a factor of at least $n/10$." Left-zero bands need every element as a generator, and the gain collapses to 1.7×; on non-associative inputs there is no gain at all.

19.5 The anchor instance of a law

For anchored magmas, the instance $z = 0$ of a three-variable law costs only $O(n^2)$ probes to check. For associativity it decides the law (Corollary 19.2). Does it always?

H-C17-2 [REFUTED]. Among the 382 laws of Chapter 7, the anchor instance decides the law for 339 (all anchored models with $N \le 5$) and is insufficient for 43 — e.g. $x(yz) = y(y(xz))$ has 11 anchored models of its instance at $N = 4$, two of which violate the law, and $x(yz) = x(zy)$ fails already at $N = 3$. (The first version reported 333 / 18 and 31 undecided laws; a classification bug, found by the Skeptic, had counted laws with a violation at one size as undecided because the enumeration was capped at another.)

Observation 19.4 *[COMPUTATIONALLY SUPPORTED; uniqueness PROVEN].* For every $N \le 4096$, the single identity $x \star (y + 1) = (x \star y) \star (x + 1)$ (for all x, y modulo N) has an anchored solution exactly when N is a power of 2, and then it is the Laver table. So left-distributivity of an anchored table is decided by an $O(N^2)$ check, at least for these sizes.

The first version checked only $N \le 10$, by SAT. The Skeptic then noticed why the search space is so small: the instances with $y \le N - 2$ already *force* the table. Row $N - 1$ must be the identity, and going down, every entry $x \star (y + 1) = v \star (x + 1)$ with $v = x \star y > x$ refers to a row that is already determined — this is Laver's recursion. So for every N there is at most one candidate, and the question is only whether it satisfies the remaining wrap-around equations. Building the candidates for every $N \le 4096$ (EXP-C17-03) shows they do exactly at the powers of 2; why they never do otherwise is open.

This experiment also corrected the laboratory's own citation of Laver's theorem in Chapter 7: the classical recursion is imposed only for $q < N$; with the wrap-around instance there are no solutions at all for most N.

19.6 Comparison with existing mathematics

Light’s test and the randomised test are classical; the anchored criteria are small corollaries of Chapter 7. The benchmark is ours.

19.7 Applications and limitations

Certifying a table found by a solver: compute a small generating set and run Light’s test — hundreds of times faster than the exhaustive check in our runs at $n = 512$ — or, for anchored tables, a single identity. Limitations: Python implementations on a shared machine; no comparison with optimised C code or with model checkers.

19.8 Open questions

1. Prove Observation 19.4 for all N: show that the forced table fails the wrap-around equations whenever N is not a power of 2.
2. Which other laws admit a Light-type reduction to generators?

Volume 8 — Optimization and Practical Applications

Applications are claimed only with a baseline, a benchmark and an honest statement of limitations. Several candidate applications failed that test; they are reported here as well.

20 Quasigroups as mixers: an honest negative

Volume 8 — Optimization and practical applications. Research cycle C18.

20.1 Motivation

The laboratory invented and studied many finite operations: Latin squares, the nimber field of Chapter 14, the anchored structures of Chapter 7. A *quasigroup* — an operation whose every row and column is a permutation — is the natural candidate for a mixing primitive: Markovski, Gligoroski and Bakeva's *e-transformation*

$$b_0 = l \star a_0, \qquad b_i = b_{i-1} \star a_i$$

turns a byte string into another one, and iterating it is the core of published quasigroup hash designs. Can our operations serve as practical (non-cryptographic) hash functions, measured honestly against standard baselines? *Nothing in this chapter is a security evaluation.*

20.2 Definitions

We use seven quasigroups of order 256: a random Latin square (built row by row from random matchings), a random isotope of $\mathbb{Z}/256$, the nimber inversion quasigroup $(x \oplus y)^{-1}$ (the nim-product itself is not a quasigroup — its row 0 is zero), an affine operation $77x + 141y + 5$, XOR, and the anchored quasigroups $x + c(y)$ and $(x+1) \oplus y$ of Chapter 7. A *tail hash* QG(T, p, rev) applies p e-transformations to key ‖ length ‖ padding, optionally one more to the reversed string, and outputs the last 8 bytes. Baselines: FNV-1a-64, MX64 (multiply-xorshift with the splitmix64 finaliser), BLAKE2b-64 and CRC32.

20.3 What can be proved in advance

Theorem 20.1 *[PROVEN — Lean: `eT_injective`, `eT_length`; REDISCOVERY: Markovski et al. 1999].* Every e-transformation is a bijection of strings of a given length; the *d-transformation* $a_i = b_{i-1} \setminus b_i$ inverts it.

Theorem 20.2 *[PROVEN — Lean: `eT_ne_of_ne`].* A difference at position i changes exactly the outputs at positions $\geq i$. So a tail hash without a reverse pass cannot let a late input byte influence earlier tail bytes: its maximal avalanche bias is $1/2$.

Theorem 20.3 *[PROVEN — written; REDISCOVERY: Klimov–Shamir T-functions].* If the operation is a T-function — built from $+$, $\times$, $\oplus$ modulo 2^8, like the affine and the anchored $(x+1) \oplus y$ quasigroups — then no hash built from it can move information from high input bits to low output bits.

Theorem 20.4 *[PROVEN — written].* For XOR the whole hash is affine over GF(2): output differences do not depend on the key (as for CRC32). For the anchored $x + c(y)$, differences between runs over the same inputs stay constant, so its one-pass hash cannot see the order of the message bytes.

20.4 The avalanche test: a misleading success

On 2000 random 16-byte keys, flipping each of the 128 input bits:

hash	mean bias	max bias	bit independence (max corr.)
BLAKE2b-64 (reference)	0.0089	0.045	0.11
MX64	0.0090	0.045	0.11
QG random Latin square, 2 passes + reverse	0.0089	0.041	0.11
QG nimber inversion, 2 passes + reverse	0.0090	0.042	0.11
QG $x + c(y)$, 2 passes + reverse	0.0091	0.043	**1.00**
QG random Latin square, forward only	0.090	**0.5**	1.00
FNV-1a-64	0.098	**0.5**	1.00
QG affine / $(x+1) \oplus y$	0.39 / 0.34	0.5	1.00
QG XOR, CRC32	0.5	0.5	1.00

The bold hypothesis H-C18-1 — a random Latin square with two passes and a reverse pass reaches BLAKE2b's avalanche — **holds**. The anchored $x + c(y)$ has ideal per-bit flip rates but perfectly correlated bit changes, exactly as Theorem 20.4 predicts.

20.5 The distribution test: a collapse, and its theorem

On 65,536 decimal strings, near-duplicate keys and word-like strings, every reverse-pass quasigroup hash failed badly: up to 2.7 times the expected collisions in 2^{16} buckets and **32,624 identical 64-bit hash values** among the near-duplicates. The nimber hash, conjectured to be competitive with FNV-1a (H-C18-3), is **refuted** (chi-square $z = 1084$ on decimal strings). The explanation is not a bug:

Theorem 20.5 (tail-information bound) *[PROVEN — Lean core: `eT_append, eT_tail_factors`]*. The last 8 outputs of an e-transformation are a bijective function of **one state byte** and the last 7 inputs. Consequently a reverse-pass tail hash gives at most 256 different values to all keys sharing their first 7 bytes, and a p-pass forward hash at most 256^p to keys sharing their last 7 bytes.

Proof idea. b_j is determined by b_{j-1} and a_j, and conversely $a_j = b_{j-1} \setminus b_j$; the rest of the message enters only through b_{n-8}. ∎

The bound is attained exactly on data (256 values in every one of 100 test families), and it predicts the observed number of collisions exactly: $16 \times (2295 - 256) = 32,624$. A byte-level e-transformation is a pipeline with an 8-bit state, however good its avalanche looks — which is why published quasigroup hashes used enormous quasigroups.

20.6 A repair that works — and does not pay

Theorem 20.5 suggests keeping a full 64-bit state: run eight e-transformations in parallel and output the final state of each (a *cascade*). Its state update is a bijection of the 8-byte state (**Theorem 20.6** *[PROVEN — written]*), so the bound no longer applies.

hash	max bias	BIC	flagged key sets	MB/s (indicative)
cascade, random Latin square	0.048	0.11	none	13
cascade, $\mathbb{Z}/256$ isotope	0.046	0.11	none	10
cascade, nimber inversion	**0.085**	0.15	one marginal	10

hash	max bias	BIC	flagged key sets	MB/s (indicative)
cascade, $x + c(y)$	**0.116**	0.18	none	9
MX64	0.045	0.11	none	165
BLAKE2b-64	0.045	0.11	none	41

The cascades over random-like quasigroups pass every test we ran. The bold hypothesis that all three random-like cascades would pass (H-C18-8) is **refuted** by the nimber cascade, whose structure ($x \star x = 0$, commutativity) leaves a measurable bias.

Result 20.7 *[COMPUTATIONALLY SUPPORTED — improvement: NO IMPROVEMENT].* No quasigroup construction improves significantly on the best baseline in any measured dimension (some are numerically slightly better on a single statistic, within noise): the good cascades only match MX64 and BLAKE2b in quality and are several times slower in the same run. Throughput depends on machine load — the tail-output variants reached 11–52 MB/s in one run and 20–98 MB/s in the Skeptic's re-run, against 640 and 428 MB/s for numpy-vectorised FNV-1a and MX64 — but the ordering is stable. Against FNV-1a alone the good cascades mix better, at a large speed cost. (FNV-1a's $z = -9.8$ on sequential keys means *more* uniform than random; the first version of this chapter read it as a defect.)

20.7 Comparison with existing mathematics

The e-transformation and its bijectivity are Markovski, Gligoroski and Bakeva's (1999); T-functions are Klimov and Shamir's (2002). Quasigroup hashes submitted to the SHA-3 competition (Edon-R, NaSHA) used huge quasigroups in wide-pipe designs and were nevertheless attacked. We found no explicit statement of the byte-level tail-information bound, which is elementary.

20.8 Applications and limitations

The outcome is negative and is reported as such. Limitations: Python and numpy implementations on a shared machine (timings indicative); a small subset of standard hash tests; one random Latin square, not uniformly sampled; 16-byte keys for the avalanche test. None of this is a security evaluation.

20.9 Open questions

1. Do cascades over larger quasigroups (order 2^{16}), in optimised code, become competitive?
2. Which properties of a quasigroup (commutativity, relations such as $x \star x = 0$, its isotopy class) leave a bias in cascades?
3. How does the avalanche bias of a cascade over a random Latin square decay with its depth?

Volume 9 — Failed Theories and Mathematical Counterexamples

The laboratory’s rule is that failures are never deleted.

Failed theories and counterexamples

Volume 9 — Failed theories and mathematical counterexamples. Generated from the claim database.

Every bold hypothesis the laboratory proposed and refuted is kept here with the counterexample that refuted it. Failures are part of the record: several of them (the Fibonacci pattern of Chapter 2, the non-rational complexity of Chapter 5, the odd and golden spectra of Chapter 7) led directly to the positive results of the book.

20.10 C01 — Quantity as the complete invariant of rewriting

- **CLM-C01-05** (refuted). *Claimed:* For every relation u <-> v over {a,b} with u != v and equal symbol counts, the generated congruence coincides with counting (Parikh) equivalence on all words of length >= |u|. *Counterexample:* u = aab, v = aba: the word baa is fixed by every rewrite, so its class {baa} is smaller than its counting class {aab, aba, baa}. All 33 relations of length 3-4 fail (EXP-C01-04).

20.11 C02 — Exchange systems: positional numeration as normal forms

- **CLM-C02-06** (refuted). *Claimed:* For every carry rule with sum b < m, each integer has exactly one stable configuration of that value. *Counterexample:* 4[k] -> 2[k+1] (base 2 with digits 0..3): the stable configurations ‘2’ and ‘10’ both have value 2. Also 4[k] -> [k+2] on all levels, 4[k] -> 2[k+2], 4[k] -> 2[k+3] (EXP-C02-04).

20.12 C03 — Termination of one-rule string rewriting 0^p 1^q -> 1^r 0^s

- **CLM-C03-07** (refuted). *Claimed:* The rule 0^p 1^q -> 1^r 0^s terminates iff s <= p or r <= q. *Counterexample:* 0011 -> 111000 (both counts grow) is terminating - a published theorem (A. Geser, ‘A Solution to Zantema’s Problem’, Passau MIP-9314, 1993; Zantema-Geser 2000; automated RFC match-bound proofs). Our own evidence agrees (all 3747 words of length <= 12 containing 0011 have finite derivation trees) but we have no independent proof; the refutation rests on the cited proof.
- **CLM-C03-08** (refuted). *Claimed:* The rule 0^p 1^q -> 1^r 0^s terminates iff r < 2q or s < 2p. *Counterexample:* 011 -> 11100 (r = 3 < 4 = 2q) loops: 001111 ->4 111 001111 00100 (Lean-verified, not_terminating_0_11). 40 counterexamples with parameters <= 5 (35 loops of EXP-C03-02 plus the 5 mirror loops of EXP-C03-03; count corrected after the skeptic review).
- **CLM-C03-09** (refuted). *Claimed:* The rule is non-terminating iff r > q, s > p and (r >= 2q or s >= 2p). *Counterexample:* 0011 -> 11100000 (p = q = 2, r = 3, s = 5): H-C03-3 predicts non-termination (r > q, s > p, s >= 2p), but the rule terminates by Zantema-Geser Theorem 1.1, case 4 (q < r < 2q and p does not divide s). The refutation rests on the published proof.

20.13 C04 — Derivation lengths, random descent and rewriting games

- **CLM-C04-02** (refuted). *Claimed:* A one-rule system l -> r with |l| <= 3 is non-terminating iff l is a factor of r. *Counterexample:* ab -> bbaa is non-terminating although ab is not a factor of bbaa: aab -> abbaa -> bbaabaa = bb.aab.aa. In total 18 such rules with |r| <= 5 (EXP-C04-02).
- **CLM-C04-07** (refuted). *Claimed:* A terminating one-rule system has random descent iff its left-hand side is unbordered. *Counterexample:* aa -> a has the bordered left-hand side aa but random descent (every derivation from w has sum over a-runs of (length - 1) steps and ends in the same word); 18 bordered rules with |l|, |r| <= 3 have random descent (EXP-C04-05). The ‘if’ direction is CLM-C04-05.
- **CLM-C04-11** (refuted). *Claimed:* For every terminating one-rule system over {a,b} with |l|, |r| <= 3, the derivational complexity D(n) satisfies a linear recurrence. *Counterexample:* aab -> ba (and its 3 symmetric images): D(n) = n - 1 - kappa(n) is not given by any linear recurrence (CLM-C04-09).
- **CLM-C04-14** (refuted). *Claimed:* Every rewriting game of a terminating one-rule system with |l|, |r| <= 3 has Sprague-Grundy values <= 3. *Counterexample:* aa -> b: the word a^14 has Grundy value 4 (computed from its own derivation graph); Dawson’s Kayles reaches 9, Officers 91 for n <= 3000 (EXP-C04-08).

20.14 C05 — Spectra of single-identity magma varieties

- **CLM-C05-01** (refuted). *Claimed:* Every identity t = x has spectrum {1} or the set of all positive integers. *Counterexample:* ETP law 167, x = (y*x)*(x*y): Lean-certified models of orders 4 and 5 (T4_law167, T5_law167) and no model of order 2, 3, 6 or 7 (law167_card_mod_four: every finite model has order 0 or 1 mod 4). The census EXP-C05-01 finds 61 of 115 nontrivial two-variable identities with spectra on n <= 6 that are neither {1} nor everything.
- **CLM-C05-03** (refuted). *Claimed:* If x occurs in t but is neither its leftmost nor its rightmost leaf, every finite model of t = x is a quasigroup. *Counterexample:* The 4-element model T4 of law 167, x = (y*x)*(x*y), has row 1 = (0, 2, 2, 2) (Lean: T4_law167, T4_not_latin). In the census, 18 of the 115 identities already have non-quasigroup models of order <= 3.
- **CLM-C05-06** (refuted). *Claimed:* A nontrivial identity t = x has models of all orders 1..6 iff it has an affine model x*y = a x + b y + c over Z_n for every n <= 6. *Counterexample:* x = (y*x)*(x*(y*y)) has models of every order 1..6 (EXP-C05-01); of order 2 it is satisfied by NAND (Lean: nand2_law), yet no affine operation over Z_2 satisfies it (Lean: no_affine_model_mod2); its affine models exist only for n in {1, 5}. Three further identities behave alike (EXP-C05-05).
- **CLM-C05-09** (refuted). *Claimed:* The finite models of x = y(x((yx)y)) (ETP 677) and of x = y(((yx)x)y) (ETP 1313) have exactly the orders 1 and the odd integers n >= 5. *Counterexample:* Linear models of order 16 (found by the Skeptic, reviews/skeptic_C05_C08_C12.md, Appendix B) - over GF(16) = GF(2)[t]/(t^4 + t + 1), x*y = (1 + t^3)x + t^3 y satisfies law 677 and x*y = t^3 x + (1 + t^3)y satisfies law 1313 (each has 8 such pairs (a, b)). Re-found by EXP-C05-07 and Lean-certified (E677_order16_law, E1313_order16_law, kernel decide on the 16 x 16 tables).

20.15 C06 — Successor-anchored magmas: spectra of composition laws

- **CLM-C06-04** (refuted). *Claimed:* Among the laws x(yz) = t and (xy)z = t with t a term of 3-4 leaves over exactly {x, y, z}, every law other than left-distributivity has anchored spectrum {1} or {1, ..., 6}. *Counterexample:* 21 other laws have restricted spectra (EXP-C06-03): (xy)z = x((yz)z) - odd N; (xy)z = x(y(zz)) - N = 1, 5 (and 11, 19, ... by Theorem 6 of proofs/paper/C06.md); x(yz) = ((zx)x)y - N = 1, 3; x(yz) = ((xx)y)z - N = 1, 2.
- **CLM-C06-09** (refuted). *Claimed:* For every law of EXP-C06-03 other than left-distributivity, the anchored spectrum for N <= 6 equals the set of N for which some affine table x*y = x + a*y + 1 satisfies the law. *Counterexample:* x(yz) = y(x(yz)), x(yz) = y(x(xz)), x(yz) = y(y(xz)) and x(yz) = (yx)(yz) have anchored models for every N <= 6 but no affine model for N > 1; x(yz) = y((yx)z) has a non-affine model at N = 2 (EXP-C06-04).

20.16 C07 — Arithmetic from the minimal excludant: a census of mex-defined operations

- **CLM-C07-02** (refuted). *Claimed:* Apart from trivial operations (0, projections), the only associative mex operations of the family are Conway's nim-sum and nim-product. *Counterexample:* L+R+D[z] gives ordinary addition a + b and D[z] gives min(a, b) (Theorem 1); also the zero-absorbing shifts of XOR and +.
- **CLM-C07-04** (refuted). *Claimed:* Every commutative mex operation of the family is closed on [0, 2), [0, 4) and [0, 16), like the nimbers. *Counterexample:* D[x^y]: f(2,3) = 4, so [0,4) is not closed; 30 of the 34 commutative operations fail (EXP-C07-01).
- **CLM-C07-06** (refuted). *Claimed:* For every operation of the family and every a, the row b -> f(a,b) satisfies f(a, b + p) = f(a, b) + s for all large b, for some p >= 1 and s, as for Wythoff's game. *Counterexample:* Row 2 of D[x^y] (Theorem 3): |f(2,b) - b| = J_k on the k-th block pair, which rules out every (p, s). Numerically only 222 of 670 rows (a = 1..5, b < 1500) have a period <= 375.
- **CLM-C07-07** (refuted). *Claimed:* For every commutative mex operation of the family, 0 is absorbing or an identity, and the diagonal f(n,n) is a sequence in the OEIS. *Counterexample:* The first part holds (Theorem 2); the second fails - e.g. the diagonal 0, 2, 1, 5, 7, 4, 3, 16, 13, 10, 15, 6, ... of L+R+D[x^y^z] and 0, 1, 3, 6, 8, 11, 10, 19, ... of D[z]+D[x^y^z] have no OEIS match (88 unmatched sequences, EXP-C07-02, OEIS state of 2026-10-18).

20.17 C08 — Circle multiplications on recurrence numeration systems

- **CLM-C08-05** (refuted). *Claimed:* Smallest shift s0 with o_s associative for all tested s0 <= s <= max tested: Fibonacci 2, Pell (1,2,5) 1, G(k+1)=2G(k)+2G(k-1) (1,2) 5, tribonacci 3, 4-bonacci 7, G(k+1)=G(k)+G(k-2) 10, G(k+1)=3G(k)+3G(k-1) (1,5) 10; G(k+1)=G(k)+G(k-3) none up to 20 (full table in results/EXP-C08-02.json and EXP-C08-03.json). *Counterexample:* Certified violations (EXP-C08-06) - tribonacci s = 3: (10016 o 10016) o 1 = 3414681926 != 3414681924 = 10016 o (10016 o 1) (the example of Ambroz-Frougny-Masakova-Pelantova 2003); 4-bonacci s = 7: triple (623729, 581581, 1); G(k+1) = 2G(k) + 2G(k-1) (1,2) s = 5: triple (29957, 280628, 1); G(k+1) = G(k) + G(k-2) s = 10: triple (78, 248, 1) (found by the Historian). The

random tests behind the measurement (triples up to 10^6) missed these rare failures. The Skeptic's review adds G(k+1) = 3G(k) + 3G(k-1) (1,5) at s = 10: (173938 o 173938) o 1 != 173938 o (173938 o 1) - the fifth wrong value. Thresholds after the targeted search in CLM-C08-08.

- **CLM-C08-06** (refuted). *Claimed:* Among digit-convolution products a o_s b = sum d_i d_j G_(i+j+s), only Fibonacci with s = 2 is associative. *Counterexample:* Fibonacci with s = 3 (t = 1) is associative by Theorem 1 (CLM-C08-01); geometric bases are associative for every s; many Pisot sequences are associative at larger shifts (EXP-C08-02).
- **CLM-C08-07** (refuted). *Claimed:* For the circle product o_s: Fibonacci (1,2) fails for s <= 1 and is associative for s >= 2 (Lean, CLM-C08-01); Pell (1,2,5) fails at s = 0, no failure found for 1 <= s <= 5 (associative for s >= 1 by Messaoudi 2000); G(k+1) = 2G(k) + 2G(k-1) (1,2) fails for s <= 5, no failure found for 6 <= s <= 8; tribonacci (1,2,4) fails for s <= 3, no failure found for 4 <= s <= 9; 4-bonacci (1,2,4,8) fails for s <= 7, no failure found for 8 <= s <= 10; G(k+1) = G(k) + G(k-2) (1,2,3) fails for s <= 10, no failure found for s = 11, 12; G(k+1) = 3G(k) + 3G(k-1) (1,5) fails for s <= 9, no failure found for 10 <= s <= 12. Every failure is certified by an explicit triple; every "no failure" is computational support only. *Counterexample:* Two of its 'no failure found' ranges were wrong (Skeptic review 2026-10-18, reviews/skeptic_C05_C08_C12.md, P2) - G(k+1) = 2G(k) + 2G(k-1) (1,2) fails at s = 6: (314965 o 314965) o 1 != 314965 o (314965 o 1); G(k+1) = 3G(k) + 3G(k-1) (1,5) fails at s = 10 and s = 11: (173938 o 173938) o 1 != 173938 o (173938 o 1). Both re-verified with the lab's code and re-found by EXP-C08-07 (e.g. (5653, 42197, 1) at s = 6; (3192, 173938, 1) at s = 10; (12100, 12100, 1) at s = 11). The certified lower bounds of this claim stand; the thresholds it implied for these two sequences do not. Superseded by CLM-C08-08.

20.18 C09 — Arithmetic of Knuth's circle semigroup

- **CLM-C09-02** (refuted). *Claimed:* Every n >= 1 has a unique multiset of circle-irreducible factors. *Counterexample:* 8 = 1 o 1 o 1 = 2 o 2 with 1, 2 irreducible (every product is >= 3); the factorisations even differ in length.
- **CLM-C09-04** (refuted). *Claimed:* The proportion of circle-irreducible numbers up to x tends to 0 as x tends to infinity. *Counterexample:* By CLM-C09-03 the irreducibles contain a set of natural density 1/phi^2 = 0.3819…; observed density is 0.539 at 1e6.

20.19 C10 — The cost of counting across representations

- **CLM-C10-03** (refuted). *Claimed:* The average number of digit changes per increment in the Zeckendorf system is smaller than in binary. *Counterexample:* Both averages tend to exactly 2 (Theorem 2 and the binary case of Theorem 1); at n < 1e6 the Zeckendorf average is 1.999992 vs binary 1.999988.

20.20 C11 — Pattern-terminated universal integer codes

- **CLM-C11-05** (refuted). *Claimed:* For every k, the all-ones terminator 1^k gives the shortest codeword among all length-k terminators for every n beyond a small

threshold. *Counterexample:* k = 4, n = 1000: terminators 1000, 1001 and 1010 give codewords 1 bit shorter than 1111. Rivals beat 1^k at 762, 972 and 1127 sampled n >= 1000 (all n <= 2048 and quarter-powers of 2 up to 2^64) for k = 4, 5, 6, up to n = 311743, 2^45 and about 1.3e19 respectively (EXP-C11-02). By Theorem 2 (CLM-C11-02), 1^k does win eventually.

- **CLM-C11-06** (refuted). *Claimed:* (a) The average over n <= N of L_(1^k)(n) - L_u(n), u unbordered of length k + 1, tends to log_lambda(1/(lambda - 1)). (b) Corrected version: this average converges to (2 - lambda)/(lambda - 1)^2. *Counterexample:* k = 3 (111 vs 0111): the average over n <= 2^40 is 0.2282 against the prediction 0.2875 of (a) (EXP-C11-06); the running average is 0.228181, 0.309709 and 0.228548 at N = 2^40, 2^80 and 2^120, so it does not converge (EXP-C11-07). Both statements hold only for k = 2 (value 1).

20.21 C12 — Relational one-heap games

- **CLM-C12-09** (inconclusive). *Claimed:* Every game of the catalogue is eventually periodic, eventually arithmetic-periodic, or has a Grundy function given by a closed form in a number-theoretic function. *Status:* The “closed form” clause is not precisely defined, so the hypothesis cannot be refuted as stated; recorded as inconclusive rather than refuted.
- **CLM-C12-10** (refuted). *Claimed:* For each of the games G01-G06, g(n) depends only on the prime signature (sorted exponent vector) of n. *Counterexample:* G01: n = 2 and n = 3 have signature (1) but g = 1 and 0. Further first counterexamples: G02 (2, 3: g = 2, 1), G03 (4, 9: 1, 0), G05 (2, 3: 1, 2), G06 (6, 10: 2, 4). Only G04 (g = Omega) depends on the signature alone.
- **CLM-C12-11** (refuted). *Claimed:* Subtraction games whose move sets are the squares, the triangular numbers, the Fibonacci numbers or the primes have P-positions of positive natural density. *Counterexample:* Squares and triangular numbers: the P-set has density 0 (CLM-C12-08); measured P-density 0.0538 -> 0.0222 and 0.0344 -> 0.0116 as N grows from 1.25e4 to 2e5. For Fibonacci numbers (0.180) and primes (0.155) the question remains open.

20.22 C13 — Anchored spectra II: which sets of integers are spectra?

- **CLM-C13-07** (refuted). *Claimed:* If a set of laws has an anchored model of size N, it has one of every size d dividing N. *Counterexample:* x = x(x(xx)) has models of every size except 2 (CLM-C13-05, Lean). Violations are common - 24 of 253 one-variable laws (N <= 12), 81 of 300 random 2-variable laws (N <= 9), 29 of 150 random 3-variable laws and 17 of 80 random pairs of laws (N <= 8); e.g. ((xy)x)y = (y(xx))x has models at N = 6 and 8 but none at N = 2, 3, 4, 5.
- **CLM-C13-08** (refuted). *Claimed:* For every law with at most 4 leaves per side (3 variables) or 5 leaves per side (2 variables) whose affine spectrum is infinite and not all N, every anchored model is affine. *Counterexample:* 198 of the 199 laws tested have non-affine models (EXP-C13-02), e.g. x = x(x(xx)) (affine spectrum A034017, non-affine models for N = 3..8) and (xx)x = (yy)y (non-affine models for N = 2..8).
- **CLM-C13-09** (refuted). *Claimed:* Every law t1(x) = t2(x) in one variable has anchored spectrum {1} or contains all sufficiently large N. *Counterexample:* x = (xx)x has spectrum exactly {1, 2}: at x = 0 the anchor gives 0 = 2, and x*y = x + 1 is a model of size 2; likewise x = ((xx)x)x has spectrum {1, 3} and x = (((xx)x)x)x has {1, 2, 4} (proofs/paper/C13.md, remark after Theorem 5).

20.23 C14 — Graphs as quantities: the 256 local graph products

- **CLM-C14-08** (refuted). *Claimed:* Each of the six commutative associative local products with unit K1 (Cartesian, strong, symmetric difference, and their complement-duals 221, co-normal, 141) has unique prime factorisation on connected graphs, or, for the complement-duals, on co-connected graphs. *Counterexample:* Symmetric difference: K2 xor K2 xor K2 is isomorphic to K2 xor K_{1,3}; K2 and K_{1,3} are connected and prime, so this connected graph of order 8 has the two factorisations {K2,K2,K2} and {K2,K_{1,3}} (proofs/paper/C14.md Theorem 8; explicit isomorphism in EXP-C14-06). In all graphs already C4 = K2 xor K2 = K2 xor 2K1. The other five products show no collision up to order 15 in the stated classes, consistent with Sabidussi-Vizing and Doerfler-Imrich/McKenzie.

20.24 C15 — Magma cellular automata: light cones, streaming width and fast-forwarding

- **CLM-C15-08** (refuted). *Claimed:* For every magma automaton of order <= 3 the streaming width lambda(t) is either bounded or grows geometrically; there is no intermediate growth. *Counterexample:* The fuse magma [[0,0,0],[0,0,1],[1,2,2]] has floor((floor(t/2)+1)/2) <= lambda(t) <= 3t + 3 (Theorem 4; lambda(t) = t + 2 for t <= 12); 104 order-3 classes show linear growth on t <= 9 (EXP-C15-02).
- **CLM-C15-09** (refuted). *Claimed:* An order-3 magma automaton has bounded streaming width iff the magma is a semigroup, is affine over Z_3, or its light cone depends on at most two cells. *Counterexample:* The ‘if’ direction holds for all 18 semigroup and 8 affine classes, but 685 bounded-width classes lie outside the three classes, e.g. the reversible automaton [[0,0,2],[2,2,0],[1,1,1]] whose light cone uses all t + 1 cells and whose width stays 3 (EXP-C15-02).
- **CLM-C15-10** (refuted). *Claimed:* A magma automaton admits a fast-forward (rows at time t computable in time polynomial in L and log t) if and only if its streaming width is bounded. *Counterexample:* The fuse magma has unbounded (linear) streaming width but its row at any time t is computable in O(L) operations (Theorem 4). The converse direction is open (CLM-C15-18).
- **CLM-C15-11** (refuted). *Claimed:* For at least half of the order-3 classes a random cyclic row of length 32 repeats within 5000 steps, and only classes with geometric streaming width have rows that do not. *Counterexample:* The first part holds (1590/1734 classes at L = 32), the second fails - the bounded-width affine quasigroups x + y, 2x + y, 2x + 2y, 2x + 2y + 1 (mod 3, up to relabelling) do not repeat within 5000 steps at L = 32 and 64 (EXP-C15-04). The cap matters (Skeptic) - at L = 32 they do repeat, with periods 6,560 or 19,680 (still long, so the refutation stands); at L = 64 none repeats within 200,000 steps.
- **CLM-C15-12** (refuted). *Claimed:* Each of the affine, commutative-semigroup and fuse fast-forwards is exact, beats direct numpy simulation by more than 100x at L = 1024, t = 10^5, and loses for t below a few hundred steps. *Counterexample:* Exactness holds (99/99), but the commutative-semigroup method reaches only x62 (mult mod 3) and x77 (min) at L = 1024, t = 10^5 (the Skeptic’s re-run - 79-97x). The clause ‘loses for small t’ is partly true - timed with the same protocol on both sides, the affine method loses for t below about 20 and the fuse method below about 20-70 (Skeptic review, Appendix C). The first version said the affine and fuse

methods never lose; EXP-C15-03 had timed the fast method as the best of 3 runs and the simulation as a single run.

20.25 C16 — Fibonacci dynamics in invented finite algebras

- **CLM-C16-02** (refuted). *Claimed:* As for Pisano periods (where (0, 0) is fixed), the maximal period of the pair map on an n-element magma is at most n^2 - 1, and the maximum is attained only by quasigroups. *Counterexample:* x*y = 1 - y on {0,1} has the single cycle (0,0) -> (0,1) -> (1,1) -> (1,0) of length 4 = n^2 and is not a quasigroup; every n has full-period tables (Theorem 2); for n <= 4 none of them is a quasigroup (EXP-C16-01).
- **CLM-C16-06** (refuted). *Claimed:* For every n, no quasigroup of order n has a full-period pair map; the maximal period of a quasigroup of order n is n^2 - 1. *Counterexample:* True for n = 2, 3, 4, false for n = 5 - exactly 3360 of the 161280 Latin squares of order 5 are full-period (e.g. the cycle 0,0,1,0,2,2,0,3,3,1,2,1,4,0,4,4,2,3,2,4,1,1,3,4,3); examples exist for every order 6-16 (EXP-C16-02).

20.26 C17 — Deciding laws of finite magmas: generator, anchor and randomised tests against the exhaustive check

- **CLM-C17-06** (refuted). *Claimed:* Light's test with a greedy generating set uses at most a fraction 10/n of the exhaustive check's probes on every input family. *Counterexample:* Left-zero band x*y = x (associative; n generators needed) - exhaustive/Light probe ratio 1.71 at n = 512 (EXP-C17-01). The first version also cited one-entry corruptions of cyclic groups (ratio 47.9 at n = 512); that number was an artefact of the probe accounting (Skeptic review, P1). With correct accounting an early-exit exhaustive check is about as cheap as Light's test on corruptions and far cheaper on random magmas, so the hypothesis fails on non-associative inputs as well.
- **CLM-C17-07** (refuted). *Claimed:* For every law of the C06 family, every anchored magma satisfying the instance z = 0 of the law satisfies the law (N <= 5). *Counterexample:* 43 of the 382 laws (first violating anchored model at N = 3 for 33 laws, at N = 4 for 10), e.g. x(yz) = x(zy) (54 violating models among 82 at N = 3), x(yz) = y(y(xz)) (two violating models among 11 at N = 4), x(yz) = x((xy)z), (xy)z = (zy)x, (xy)z = (x(zz))y (EXP-C17-02). For the other 339 laws the instance decides the law for N <= 5, among them associativity, and (N <= 10) left-distributivity. The first version reported 18 insufficient and 31 undecided laws because of a classification bug - a law whose enumeration was capped at one N was counted as undecided even when a violation had been found at another N; the Skeptic's independent census gives the same 339 / 43 / 0 as the corrected experiment.

20.27 C18 — Invented operations as mixing primitives: quasigroup string transformations versus standard hashes

- **CLM-C18-09** (inconclusive). *Claimed:* Every structured quasigroup tested (affine, XOR, x + c(y), (x + 1) XOR y) gives a larger maximal avalanche bias than the random

Latin square, in the reverse-pass (p2+rev) construction. (As first worded - ‘in the same construction’ - the hypothesis is false for the forward-only construction, where the random Latin square and x + c(y) both have bias exactly 1/2 by CLM-C18-02; the experiment compared every structured variant with QG-LS p2+rev.) *Status:* Proved (bias 1/2) for the T-function and GF(2)-affine quasigroups (CLM-C18-04, CLM-C18-05). For x + c(y) with a reverse pass the difference (0.043 versus 0.041) is within sampling noise; its defect shows in bit independence instead (CLM-C18-06), which the hypothesis did not measure.

- **CLM-C18-10** (refuted). *Claimed:* QG-NIMINV p2+rev (x*y = (x XOR y)^-1 in the nimber field of order 256) has avalanche at least as good as FNV-1a-64, |chi-square z| <= 3 and a bucket-collision ratio within 2% on all four key sets. *Counterexample:* Avalanche is better than FNV-1a’s (max bias 0.0415 vs 0.5), but on the decimal strings 0..65535 the 1024-bucket z-score is 1084 and the bucket-collision ratio 2.62, and near-duplicate keys give 32,624 full 64-bit collisions - the tail-information bound (CLM-C18-03). (FNV-1a itself has z = -9.8 there - more uniform than random, bucket-collision ratio 0.995 - and 0 full collisions; the first wording wrongly attributed |z| <= 3 on all four key sets to FNV-1a.)
- **CLM-C18-11** (refuted). *Claimed:* The cascades QC-LS, QC-ISO and QC-NIMINV (depth 8, 8 zero bytes of finalisation) have avalanche within 0.01 of BLAKE2b-64, |z| <= 3, collision ratios within 2% and no full collisions on all four key sets. *Counterexample:* QC-NIMINV d8 - maximal avalanche bias 0.0845 against 0.045 for BLAKE2b (about 7 standard deviations of a cell; the Skeptic’s replication with new keys - 0.0735, against a null maximum of 0.048; worst cell key byte 15, bit 0 -> top bit of s1). QC-LS d8 and QC-ISO d8 pass every test (EXP-C18-05). (A z = -3.21 on synthetic words was also cited at first; it is 1 of 48 z-scores and in the benign direction - more uniform than random - so the refutation rests on the avalanche bias alone.)
- **CLM-C18-12** (refuted). *Claimed:* The cascade QC-ARP d8 has a maximal bit-independence correlation of at least 0.5. *Counterexample:* QC-ARP d8 has maximal correlation 0.176 (noise level about 0.11) - the cascade removes the perfect correlation of QG-ARP p2+rev, although 5 avalanche cells still exceed bias 0.1 (EXP-C18-05).

20.28 C19 — The longest Zeckendorf game

- **CLM-C19-10** (refuted). *Claimed:* D(n) = sum_t floor(j_t^2/4) - sum_t R(j_t, j_{t+1}) over consecutive Zeckendorf levels j_1 > j_2 > … of n. *Counterexample:* n = 12 = F_5 + F_3 + F_1 (levels 4, 2, 0): D(12) = 0 but the model gives 1; 23,680 of the n < 30000 violate it (EXP-C19-03).
- **CLM-C19-11** (refuted). *Claimed:* D(n) - D(n - F_top) is a function of the two highest Zeckendorf levels of n. *Counterexample:* Levels (4, 2) occur with D(n) - D(n - F_5) in {0, 1} (n = 12 and n = 11); 87 such ambiguous pairs below 30000 (EXP-C19-03). The dependence is through the parity chain of lower levels, captured by rho in CLM-C19-05.
- **CLM-C19-12** (refuted). *Claimed:* There is a function phi(start, length) with D(n) = sum of phi over the maximal runs of ones of the lazy representation of n. *Counterexample:* For n < 160 the linear system D(n) = sum phi(run) has coefficient rank 46 and augmented rank 47 (exact rational elimination), so it has no solution (EXP-C19-03).
- **CLM-C19-13** (refuted). *Claimed:* L(n + 1) = L(n) + lambda(n), where lambda(n) is the longest game from the Zeckendorf representation of n plus one extra F_1.

Counterexample: n = 3: L(4) = 3, L(3) = 2, lambda(3) = 0 (Zeckendorf(3) + F_1 is already Zeckendorf); fails for 288 of the first 300 n (EXP-C19-02). Longest games avoid early normalisation, so they do not pass through Zeckendorf(n).

Volume 10 — Open Problems and Future Mathematical Research

What remains open, stated precisely enough to be attacked.

21 Closing an open problem, and what remains open

Volume 10 — Open problems and future mathematical research. Research cycle C20.

21.1 Motivation

Chapter 5 classified 860 of the 868 one-rule systems $l \to r$ over $\{a, b\}$ with $|l| \le 3$, $|r| \le 5$. Eight rules — the images of $aab \to ababa$ and $aab \to babaa$ under reversal and the exchange of letters — resisted every certificate the laboratory tried: no loop, no interpretation. The last research cycle of the batch went back to them.

21.2 Two tools from earlier chapters

Random descent. The left side aab has no border, so by Theorem 5.4 all maximal derivations from a word have the same length; one terminating strategy settles termination for all of them.

Runs. Writing $w = a^{x_0} b\, a^{x_1} b \cdots b\, a^{x_m}$, a redex is a run $x_k \ge 2$ followed by a b. For $aab \to ababa$ a step turns (x_k, x_{k+1}) into $(x_k - 1, 1, x_{k+1} + 1)$: one surplus a moves one b to the right and a new run of length 1 appears behind it. For $aab \to babaa$ a *pair* moves: $(x_k - 2, 1, x_{k+1} + 2)$.

21.3 The results

Theorem 21.1 *[PROVEN — written; termination REDISCOVERY (barren rules), complexity POTENTIALLY NOVEL].* $aab \to ababa$ terminates, and its derivational complexity is exactly

$$D(n) = 2^{n-2} - 1 \qquad (n \ge 2),$$

attained by $a^{n-1} b$.

Proof idea. Always rewrite the rightmost redex. A surplus a then travels to the right — each step brings it one b closer to the end — until it fills an empty run or reaches the last run: a finite *journey*. Journeys never create new surplus, so there are at most (number of a's) -1 of them. Each step inserts one b, and a journey has at most as many steps as there are b's, so every journey at most doubles the number of b's. For $a^{n-1} b$ every journey has exactly that many steps: $1 + 2 + \cdots + 2^{n-3}$. ∎

Theorem 21.2 *[PROVEN — written; POTENTIALLY NOVEL].* $aab \to babaa$ terminates. Every maximal derivation from w has length

$$D(w) = \sum_{k=m-1}^{0} d_k \,(2^{\lfloor x_k/2 \rfloor} - 1),$$

where d_k is the number of b's to the right of run k when its turn comes (the original $m - k$ plus one for every step already made), and

$$D(n) = \max_{1 \le b \le n} b\,(2^{\lfloor (n-b)/2 \rfloor} - 1) = \Theta(2^{n/2}),$$

explicitly $D(n) = 2^{n/2} - 2$ for even $n \ge 2$ and $3 \cdot 2^{(n-3)/2} - 3$ for odd $n \ge 5$ (the Skeptic's simplification).

Proof idea. A pair of a's, once released, always travels all the way to the last run, so a run releases its pairs one after another, each journey twice as long as the previous one. ∎

Corollary 21.3 *[PROVEN].* All 868 rules of the extended census are classified: 480 terminate and 388 do not.

All three statements were checked on every word of length ≤ 12 for all eight rules; the rightmost and leftmost strategies agree on every word, as random descent predicts (EXP-C20-01). The independent Skeptic re-checked the written proofs step by step and extended the checks to all words of length ≤ 18 and ≤ 21.

What the literature already knew. Both rules are *right barren* in McNaughton's sense, and barren one-rule systems are known to terminate (McNaughton; Geser, ICASE Report 2002-33). So the termination half is a rediscovery. Geser's report proves linear derivation lengths for its non-barren rules; for barren rules, as these examples show, the lengths can be exponential — and here they are known exactly. (The report's introduction announces linear lengths for *all* terminating rules with one pair of overlaps; $aab \to ababa$ has one pair of overlaps and exponential length, so only the body's narrower statement is correct — a point the Skeptic made.)

21.4 A literature question, settled by the Historian

Chapter 4 conjectured a divisibility criterion for when $0^p 1^q \to 1^r 0^s$ does not terminate. Zantema and Geser published a complete characterisation of these rules (tech. report 1994, journal 2000). The Director's attempts to obtain the full text failed (HTTP 403, a bot challenge, HTTP 500); the Historian then found the scanned report on an older archive host. Their Theorem 1.1 is equivalent to our conjecture (it characterises termination, ours non-termination, with the same cases) — now Theorem 4.6. The 24 rules we had left open all terminate.

21.5 The open problems of batch 1

Volume 10's catalogue (next chapter) lists every unproven claim of the database. The most interesting ones, in the laboratory's judgement:

1. **Common borders** (Chapter 2): the class-count formula with two or more common borders — and, in the one-border case, the injectivity lemma that would finish the proof.
2. **Finite spectra** (Chapters 7–8): of the 17 laws whose anchored models seemed to exist only for $N \in \{1, 2\}$ or $\{1, 3\}$, 16 are now proved (Chapter 8); $x(yz) = y((yx)z)$ remains open (spectrum $\{1, 2\}$ supported up to $N = 12$).
3. **The anchor instance of left-distributivity** (Chapter 19): the instance $x \star (y + 1) = (x \star y) \star (x + 1)$ forces at most one table for every N, and for $N \le 4096$ a model exists exactly at the powers of 2. Prove that it never exists otherwise.
4. **Rows of mex operations** (Chapter 14): the structure beyond the Jacobsthal row.
5. **Octal games** (Chapter 5): the periodicity of 0.06 and Officers (famous and hard).
6. **Circle-product thresholds** (Chapter 9): the exact associativity thresholds, e.g. whether tribonacci is associative at shift 4 or 5. The first version of the book got five of them wrong, and the laboratory's first correction two more; only lower bounds are certified.
7. **Spectra of ETP laws 677 and 1313** (Chapter 6): which even orders below 16 occur, and which odd ones — the conjectured answer of the first version was refuted by models of order 16.
8. **The longest Zeckendorf game** (Chapter 16): prove the recursion that predicts $L(n)$ for every n.

Open problems

Volume 10 — Open problems and future research. Generated from the claim database and the cycle records.

21.6 Unproven claims

Each of these is supported by the evidence cited in its record but has no proof. A proof or a counterexample would settle it.

- **CLM-C01-08** (computationally supported): For all words u != v of equal length k over a q-letter alphabet, the number of congruence classes of length-n words modulo u = v has generating function c(x) / (x^k + (1 - q x) c(x)), where c(x) = 1 + sum over words b that are borders of both u and v of x^(k - |b|). *Evidence:* EXP-C01-05, EXP-C01-06, EXP-C01-07.
- **CLM-C01-10** (computationally supported): If u != v of length k share exactly one common border z (neither border set containing the other), the class counts have generating function c/(x^k + (1 - q x) c) with c = 1 + x^(k - |z|). *Evidence:* EXP-C01-06, EXP-C01-07, EXP-C01-08.
- **CLM-C01-11** (computationally supported): For words u != v of equal length, the one-rule system {u -> v} is confluent if and only if every border of u is a border of v. *Evidence:* EXP-C01-08.
- **CLM-C03-11** (computationally supported): For 0 1^q -> 1^(q+1) 0^k the rightmost strategy from 0 1^(q(q+1)) reaches a self-embedding after 16, 32, 64 steps (k = 2) and 260, 1556, 9332 steps (k = 6) for q = 4, 5, 6. *Evidence:* EXP-C03-04.
- **CLM-C04-12** (computationally supported): The 162 terminating rules (55 symmetry classes) have derivational complexity linear (144 rules, of which 4 are n - Theta(log n)), quadratic (14: floor(n^2/4), floor(n(n-1)/6), floor(n^2/8)) or exponential (4: 2^(n-1) - 1); for 54 of the 55 classes D(n) = floor((a n^2 + b n + c)/d) or 2^(n-1) - 1 on all computed terms. *Evidence:* EXP-C04-03.
- **CLM-C04-16** (conjecture): The Grundy sequences of the rewriting games of aa -> ab, aaa -> b and aaa -> ab on the words a^n are ultimately periodic. *Evidence:* EXP-C04-08, EXP-C04-09.
- **CLM-C05-02** (computationally supported): Of the 115 identities t = x (x occurring in t, neither leftmost nor rightmost), 52 have models of every order 1..6, 2 have none of orders 2..6, and 61 have restricted spectra; the 63 non-full ones fall into 15 patterns on n = 1..8 listed in research/cycles/C05_magma_spectra/README.md (e.g. 10111011 for 16 identities including the Dupont law, 10110010 = Mendelsohn pattern for law 66 and three equivalents, 1010001? for law 907). *Evidence:* EXP-C05-01, EXP-C05-02.
- **CLM-C06-01** (computationally supported): An anchored left-distributive magma (x*0 = x+1 mod N, x(yz) = (xy)(xz)) of size N exists exactly for N = 1, 2, 4, 8 among N <= 10 and is then unique (the Laver table A_n); Laver's recursion yields left-distributive tables exactly at the powers of 2 up to N = 64 (verified to N = 256 exhaustively). *Evidence:* EXP-C06-01, EXP-C06-02.
- **CLM-C06-10** (computationally supported): The seven laws x(yz) = ((zx)x)y, x(yz) = ((zx)y)x, x(yz) = ((zy)x)x, (xy)z = y((zx)y), (xy)z = y((zy)x), (xy)z = z((yx)z), (xy)z = z((yz)x) have anchored models exactly for N in {1, 3}; the ten laws x(yz) = y((yx)z), x(yz) = ((xx)y)z, x(yz) = ((xx)z)y, x(yz) = ((xy)x)z, x(yz) = ((xy)y)z, x(yz) = ((xy)z)x,

x(yz) = ((xy)z)y, x(yz) = ((xz)x)y, x(yz) = ((xz)y)x, x(yz) = ((xz)y)y have anchored models exactly for N in {1, 2}. *Evidence:* EXP-C06-03, EXP-C06-04.

- **CLM-C06-11** (computationally supported): (xy)z = x(z(yy)) has the same anchored model counts as the golden law (xy)z = x(y(zz)), and (xy)z = x((zy)y) the same as the odd law (xy)z = x((yz)z), for N <= 9; their affine spectra coincide for N <= 40. *Evidence:* EXP-C06-04.
- **CLM-C07-08** (computationally supported): Among the 134 distinct mex operations, 22 have additively periodic rows a = 1..5 (period <= 375 within b < 1500), 40 have none, and the rest are mixed. *Evidence:* EXP-C07-03.
- **CLM-C08-02** (computationally supported): If the minimal linear recurrence of an increasing base sequence G (G_0 = 1) has a dominant root alpha > 1 and another characteristic root beta with |beta| >= 1 (counted with multiplicity: a repeated dominant root counts as such a beta), then the circle product o_s is non-associative for every s >= 0. *Evidence:* EXP-C08-02, EXP-C08-07.
- **CLM-C08-03** (computationally supported): For base sequences G_k = a_1 G_(k-1) + ... + a_d G_(k-d) with a_1 >= a_2 >= ... >= a_d >= 1 and canonical initial values (the conditions of Grabner-Petho-Tichy-Woeginger 1994), the circle product o_s is associative once the shift s is large enough. (The earlier title, “Pisot base sequences ...”, overstated the published theorem.) *Evidence:* EXP-C08-02, EXP-C08-03, EXP-C08-06, EXP-C08-07.
- **CLM-C08-04** (computationally supported): For quadratic Pisot base sequences G(k+1) = pG(k) + qG(k-1) with non-canonical initial values (for canonical ones the associative shifts are upward closed whatever the sign of beta, by Messaoudi’s Thm 2 as derived by the Historian): if the conjugate root beta is positive, the set of shifts s for which o_s is associative is upward-closed; sets that are not upward-closed (e.g. {2} and s >= 4 for G = 1,3,4,7,11,...) occur only when beta < 0. *Evidence:* EXP-C08-07, EXP-C08-06, EXP-C08-04, EXP-C08-03.
- **CLM-C08-08** (computationally supported): For the circle product o_s: Fibonacci (1,2) fails for s <= 1, associative for s >= 2 (Lean, CLM-C08-01); Pell (1,2,5) fails at s = 0 (associative for s >= 1, Messaoudi 2000); G(k+1) = 2G(k) + 2G(k-1) (1,2) fails for s <= 6; G(k+1) = 3G(k) + 3G(k-1) (1,5) fails for s <= 11; tribonacci (1,2,4) fails for s <= 3; 4-bonacci (1,2,4,8) fails for s <= 7; G(k+1) = G(k) + G(k-2) (1,2,3) fails for s <= 10; G(k+1) = G(k) + G(k-3) (1,2,3,4) fails for every s <= 16 (the Skeptic - every s <= 24). Every failure is certified by an explicit triple. Above these bounds no failure was found up to s = 16 by the targeted search of EXP-C08-07 and by the Skeptic’s independent extreme-conjugate search - computational support only. *Evidence:* EXP-C08-07, EXP-C08-06.
- **CLM-C09-05** (conjecture): Almost every n whose conjugate lies in the product zone is a circle product, so the circle irreducibles have natural density 1/phi^2. *Evidence:* EXP-C09-04, EXP-C09-01.
- **CLM-C10-06** (computationally supported): For carry rules with two or more nonzero b_j, the average digit-change cost per increment converges to a constant exceeding 1 + 1/(m - B) (measured excess: 0.500 for 3->(1,1), 0.333 for 4->(1,1), 0.643 for 3->(0,1,1), 0.714 for 3->(1,0,1), 0.831 for 4->(1,1,1)), equal to an expectation under the invariant measure of the digit process. *Evidence:* EXP-C10-03.
- **CLM-C11-08** (computationally supported): Among fixed (untuned) universal codes, a pattern-terminated code (T_101, T_0111 or T_00111) has the smallest expected length on 5 of 12 synthetic sources (Zipf 1.1, geometric 0.001, uniform 2^16, uniform 2^32, log-uniform on [1, 1e6]) and on both posting-list data sets, beating the best classical universal code (Elias, Levenshtein, Fibonacci, Fraenkel-Klein C3, LEB128, fixed Exp-Golomb) by 0.34-3.06 bits per value; on Zipf 1.1, log-uniform and

the bursty posting list it also beats Golomb and Exp-Golomb codes tuned to the source. It loses on light-tailed sources (Zipf >= 1.5, geometric >= 0.01, prime gaps) and to tuned Golomb/Exp-Golomb codes on geometric and uniform sources, and the reference implementation is 4-10 times slower than the Elias and Exp-Golomb codes. *Evidence:* EXP-C11-03, EXP-C11-04, EXP-C11-05, EXP-C11-09.

- **CLM-C12-12** (computationally supported): The Grundy values of the subtraction game whose move set is the set of primes satisfy g(n) <= 11 for all n. *Evidence:* EXP-C12-03, EXP-C12-01.
- **CLM-C12-13** (conjecture): The Grundy values of the subtraction game with move set {1, 2, 3, 5, 8, ...} (Fibonacci numbers) are bounded; the observed maximum is 14 for n <= 2e5. *Evidence:* EXP-C12-03, EXP-C12-01.
- **CLM-C13-10** (computationally supported): Every one-variable law t1(x) = t2(x) has an anchored spectrum that is either finite or contains all sufficiently large N. *Evidence:* EXP-C13-07.
- **CLM-C13-12** (computationally supported): Among the 198 non-rigid laws of EXP-C13-02, 190 have anchored models at every N = 6, 7, 8 (the others have UNKNOWN instances or small exceptions); no law other than left-distributivity was found whose non-affine models occur only at sparse sizes such as powers of two. *Evidence:* EXP-C13-02, EXP-C13-05, EXP-C13-06.
- **CLM-C14-12** (computationally supported): Every connected graph has exactly one factorisation into connected primes with respect to the product (g,h) ~ (g',h') iff (EA, AE, AA or NN); equivalently every graph with connected complement factors uniquely under the symmetric-difference product into primes with connected complement. *Evidence:* EXP-C14-04.
- **CLM-C14-13** (computationally supported): Every switching class of finite graphs has a unique factorisation into prime switching classes with respect to the product induced by the symmetric difference (equivalently, symmetric +-1 matrices with unit diagonal up to signed permutation similarity, under the Kronecker product). *Evidence:* EXP-C14-05.
- **CLM-C15-07** (computationally supported): Of the 1734 order-3 magma automata (up to isomorphism and mirror image), on t <= 9 the streaming width is bounded for 808 classes, grows linearly for 104, geometrically for 300 (largest lambda(9) = 649) and irregularly for 522; 43 are surjective, 7 injective and 34 nilpotent by t = 9. *Evidence:* EXP-C15-02.
- **CLM-C15-13** (computationally supported): Computing the row at time t of an affine magma automaton by binary exponentiation of P = a + bS modulo S^L - 1 (FFT products) is exact and, at L = 1024, faster than direct numpy simulation by about one order of magnitude at t = 10^3, two orders at 10^4 (138-314x in two runs) and three orders at 10^5 (1,459-2,500x); t = 10^9 takes milliseconds for L <= 4096. With the same timing protocol on both sides it loses for t below about 20. *Evidence:* EXP-C15-03.
- **CLM-C15-14** (computationally supported): The exponent-arithmetic fast-forward (Theorem 3) is exact; at L = 1024 it is about two orders of magnitude faster than direct simulation at t = 10^5 (62-100x; the Skeptic's re-run 79-97x) but no faster around t = 10^3 (0.5-1.0x) for L >= 1024, and equal for t < 2L (where it falls back to simulation). *Evidence:* EXP-C15-03.
- **CLM-C15-15** (computationally supported): The burn-time fast-forward of the fuse magma (Theorem 4) is exact and takes O(L) time for every t - about three orders of magnitude faster than direct simulation at t = 10^5 (L = 1024; 4,021x in the first run, 1,363x on the Skeptic's re-run), with t = 10^9 in about a millisecond; because

the O(L) method is a Python loop it loses to simulation for t below about 20-70. *Evidence:* EXP-C15-03.

- **CLM-C15-16** (computationally supported): Storing visited rows answers a query t in mu + lambda simulated steps; for random rows of length 32 this is at most 5000 steps for 1590 of 1734 order-3 classes (median transient 5, median period 2), but it fails for the affine quasigroups and for 128 of 300 geometric-width classes. *Evidence:* EXP-C15-04.
- **CLM-C15-17** (computationally supported): The Laver tables A_2 and A_3 give automata whose light cones depend on at most 3 resp. 4 cells (t <= 8 resp. 5) with streaming (forward) widths 4 and 8 (backward widths 5 and 13-14); GF(4) multiplication and XOR on [0,4) have streaming width 4; the anchored golden- and odd-law models of C06 are permutive (hence surjective) affine automata. *Evidence:* EXP-C15-05.
- **CLM-C15-18** (conjecture): Every magma automaton with bounded streaming width admits an algorithm computing a cyclic row of length L at time t in time polynomial in L and log t. *Evidence:* EXP-C15-02, EXP-C15-03.
- **CLM-C16-10** (computationally supported): For every n >= 5 there is a quasigroup of order n whose Fibonacci pair map is a single cycle through all n^2 pairs. *Evidence:* EXP-C16-02, EXP-C16-05.
- **CLM-C16-13** (computationally supported): For every affine magma x*y = alpha x + beta y + gamma on Z/N with beta a unit, the maximal period of the pair map equals the order of [[0,1,0],[beta,alpha,gamma],[0,0,1]] mod N. *Evidence:* EXP-C16-03.
- **CLM-C16-16** (computationally supported): The longest transient of the Laver pair map (steps until the state contains the top element) is 1, 3, 3 for n = 1, 2, 3 and exactly 5 for every n = 4, ..., 10 (the Skeptic - every n = 4, ..., 13). *Evidence:* EXP-C16-03.
- **CLM-C16-17** (computationally supported): On a simple battery (byte chi-square, serial pairs, lowest-bit period, lag-1 correlation; 2^20 outputs) the anchored Fibonacci and affine magma generators on Z/2^32 fail low-byte uniformity and have lowest-bit period 3, and the de Bruijn byte magma (maximal possible period 65536) is too regular; PCG64 passes every test. *Evidence:* EXP-C16-04.
- **CLM-C16-19** (computationally supported): With fixed budgets, lexicographic DFS (3e6 nodes) and a SAT encoding (3e5 conflicts) find full-period quasigroups only up to n = 8, while seeded sampling of random Latin squares finds them for every n = 5..16 at rates close to 1/n^2 (H-C16-11). *Evidence:* EXP-C16-05, EXP-C16-02.
- **CLM-C17-04** (computationally supported): Certification (associative inputs, n = 32..512) - Light's test with a greedy generating set uses 23-383 times (cyclic groups) and 38-614 times (anchored tables) fewer table probes than the exhaustive n^3 check (generating-set cost included), the anchored test 2n^2 probes; on left-zero bands the gain is only 1.7x. Refutation (one-entry corruptions, random magmas) - an exhaustive check that stops at the first violating triple is about as cheap as Light's test or much cheaper (4 probes on random magmas against about 9e5; 0.1-1.1x Light's probes on corruptions), so the gains concern certification only. The randomised test (40 rounds) made no error; random-triple sampling with n^2/16 samples missed 27 of 30 near-associative inputs. *Evidence:* EXP-C17-01.
- **CLM-C17-05** (computationally supported): For every N, the instances y <= N - 2 of x*(y+1) = (x*y)*(x+1) force at most one anchored table (Laver's recursion; written proof in proofs/paper/C17.md). At N = 2^n it is the Laver table A_n, which satisfies the full instance including the wrap-around y = N - 1. For every N <= 4096 (EXP-C17-03, and independently the Skeptic's C program) the full instance has a model

exactly at the powers of 2, so for these N left-distributivity of an anchored table is decided by an O(N^2) check. *Evidence:* EXP-C17-03, EXP-C17-02.

- **CLM-C18-08** (computationally supported): QG-LS p2+rev has mean and maximal avalanche bias within 0.01 of BLAKE2b-64 on 2000 random 16-byte keys. *Evidence:* EXP-C18-02.
- **CLM-C18-13** (computationally supported): Among 13 tail-output quasigroup hashes and 4 cascades, measured against FNV-1a-64, MX64 (multiply-xorshift with the splitmix64 finaliser), BLAKE2b-64 and CRC32 on avalanche, bit independence, chi-square, collisions and throughput, no quasigroup construction improves significantly on the best baseline in any measured dimension (some have numerically smaller maximal bias or bit-independence values, within noise). *Evidence:* EXP-C18-02, EXP-C18-03, EXP-C18-04, EXP-C18-05.
- **CLM-C19-05** (computationally supported): Let h(k) = floor(k^2/4) and define rho_p (p = 0, 1) by rho_p(0) = 0 and, for m = F_{i+1} + t with Zeckendorf top level i (0 <= t < F_i): rho_p(m) = r_p(i) + [i = p mod 2] * max(0, rho_p(t) - h(i-2)), where r_p(0) = 1, r_p(i) = h(i) + 2 for i >= 1, i = p mod 2, and r_p(i) = h(i+1) + 1 otherwise. Then D(0) = 0 and D(F_{j+1} + m) = h(j) + D(m) - rho_{j mod 2}(m) for 0 <= m < F_j, and L(n) = B(n) - D(n). *Evidence:* EXP-C19-03, EXP-C19-04.
- **CLM-C19-06** (computationally supported): For levels j >= i + 2, D(F_{j+1} + F_{i+1}) = floor(j^2/4) + floor(i^2/4) - R(j,i) with R(j,0) = 1, R(j,i) = floor((i+1)^2/4) + 1 if j - i is odd and R(j,i) = floor(i^2/4) + 2 if j - i is even (i >= 1). *Evidence:* EXP-C19-03.
- **CLM-C19-09** (computationally supported): For 1 <= n < 30000, the deficit satisfies D(n) <= floor(j^2/4) where j is the top Zeckendorf level of n (equality at the Fibonacci numbers), and L(n+1) > L(n). *Evidence:* EXP-C19-03.
- **CLM-C19-14** (computationally supported): The recursion of CLM-C19-05 evaluates L(n) with O(log^2 n) arithmetic operations; the published method (playing the split-first game) needs L(n) ~ phi^2 n moves. On the tested range both agree. *Evidence:* EXP-C19-04.

21.7 Directions proposed by the research cycles

C01 — Quantity as the complete invariant of rewriting.

- C01: prove the injectivity lemma that completes the one-common-border case (CLM-C01-10), then attack two or more common borders
- C01: full Lean formalisation of the border-containment theorem
- C01: Lean formalisation of the min/gcd theorem (Bezout over N)
- C01: common-border formula for pairs of relations?

C02 — Exchange systems: positional numeration as normal forms.

- C02: explicit bound n0 for critical rules; Lean formalisation of Theorem 10 (Perron-Frobenius argument)
- C02: longest Zeckendorf game - closed form or asymptotics
- C02: which carry rules have regular representation languages?

C03 — Termination of one-rule string rewriting 0^p 1^q -> 1^r 0^s.

- C03: retrieve the Zantema-Geser theorem statement and compare with H-C03-4
- C03: implement RFC match-bounds to prove termination of the 24 open rules (start with 0011 -> 111000 and 0011 -> 11100000)

- C03: prove that every base rule 0 1^q -> 1^r 0^k (q < r < 2q, k >= 2) loops from 0 1^(q(q+1)); explain k^q loop lengths
- C03: Lean formalisation of the conservation theorem (s = p)

C04 — Derivation lengths, random descent and rewriting games.

- C04: decide termination of aab -> ababa and aab -> babaa
- C04: games of aba -> u and bab -> u (alternating runs)
- C04: complexity census for |l| = 4 - new non-rational classes? cubic growth?
- C04: OEIS submissions (0.06 Grundy values; formula for A247983) - human decision

C05 — Spectra of single-identity magma varieties.

- C05: decide law 1286 at orders 8, 10 and laws 677/1313 at 8, 10, 12, 14 with stronger symmetry breaking; certify
- C05: which odd orders are in the spectra of 677 and 1313? (677 has no linear model at 17, 23, 29; 1313 none at 11, 29; 15 is not a prime power)
- C05: extend the census to order-5 laws and three-variable identities
- C05: Lean-formalise orbit-counting arguments for other laws with a finite-order pair map

C06 — Successor-anchored magmas: spectra of composition laws.

- C06: prove the finite spectra {1,2} and {1,3} (CLM-C06-10) by showing every anchored model of those laws is affine
- C06: which subsets of N arise as anchored spectra? They are closed under coprime products (CLM-C06-12), so never the primes; are they always closed under divisors? (bold)
- C06: larger terms - are there non-LD laws with power-of-two (Laver-like, non-affine) spectra?
- C06: Lean formalisation of the golden-law correspondence (Theorem 6)

C07 — Arithmetic from the minimal excludant: a census of mex-defined operations.

- C07: rows a >= 3 of D[x^y] - an exploratory look suggested row 3 is injective and rows 4-6 are not; is there a recursive block structure?
- C07: prove (or refute) that the 33 remaining specs equal their associative formulas for all a, b
- C07: submit the Jacobsthal row and selected census sequences to the OEIS (human decision)
- C07: which mex operations are automatic sequences (2-automatic tables)? A Walnut-style decision procedure would turn observations into proofs

C08 — Circle multiplications on recurrence numeration systems.

- C08: complete the Lean proof of the shifted Knuth theorem
- C08: prove non-Pisot never-associativity (CLM-C08-02, dominant root > 1)
- C08: compute greedy windows exactly (Rauzy fractal boundaries) to turn the threshold searches of EXP-C08-07 into proofs
- C08: compute greedy windows to prove the sign-of-beta conjecture (CLM-C08-04)
- C09: arithmetic of the circle semigroup - irreducibles and factorisation

C09 — Arithmetic of Knuth’s circle semigroup.

- C09: prove the density-1/phi^2 conjecture
- C09: characterise circle irreducibles via prime factorisation in Z[phi] plus a unit-window condition
- C09: arithmetic of other associative circle products (Pell, tribonacci)

C10 — The cost of counting across representations.

- C10: exact multi-offset counting constants via the invariant measure of the digit process

C11 — Pattern-terminated universal integer codes.

- C11: table-driven (byte-wise) decoders for T_p and a fair speed comparison with optimised Elias and Fibonacci decoders
- C11: multi-pattern terminators (sets of delimiters, as in multi-delimiter codes) and the best set per source
- C11: synchronisation and robustness after bit errors for unbordered terminators (the classical selling point of Fibonacci codes)
- C11: exact oscillation constants for the average gain of T_011 over C3 (as Proposition 5 does for 1^k)

C12 — Relational one-heap games.

- C12: boundedness of the Grundy values of subtract-a-prime (11) and subtract-a-Fibonacci (14); convergence of the Fibonacci P-density (<= 1/5)
- C12: forbidden-move perturbations - for which d0 does forbidding ‘subtract d0’ in G01 give finitely many exceptional odd parts as in Theorem 8?
- C12: prove g(m) < m/4 for odd m in the non-coprime game G06 (max ratio 0.24985 up to 5000)
- C12: formalise Theorem 8 in Lean (finite certificate plus the shift-invariance lemma)

C13 — Anchored spectra II: which sets of integers are spectra?.

- C13: prove the last case of CLM-C06-10, x(yz) = y((yx)z) (spectrum {1,2}, with a non-affine model at N = 2)
- C13: which arithmetic sets are TRUE spectra of rigid laws? Find a rigid law realising ‘-1 is a square mod N’ (A008784)
- C13: prove that spectra of one-variable laws are finite or cofinite (CLM-C13-10)
- C13: search larger laws for a non-affine sparse (Laver-like) spectrum other than left-distributivity
- C13: submit the tribonacci-law and plastic-number spectra to the OEIS (human decision)

C14 — Graphs as quantities: the 256 local graph products.

- C14: prove or refute that two-graphs are a free commutative monoid under the symmetric-difference product (twin reducts; analogue of McKenzie’s theorem for Kronecker products of +-1 matrices)
- C14: unique factorisation of product 141 on connected graphs (no collision to order 15)
- C14: classify the graphs G that absorb switching (G xor X = G xor X^Z for all X); K_{a,a} do
- C14: characters for directed graphs and for products of graphs with loops

C15 — Magma cellular automata: light cones, streaming width and fast-forwarding.

- C15: characterise bounded streaming width - what do the 685 unexplained bounded-width order-3 classes have in common?
- C15: prove or refute CLM-C15-18 (bounded width => polylog-time fast-forward); find fast-forwards for non-commutative semigroups via the popcount word
- C15: are the ‘geometric’ widths really exponential? Lower-bound proofs for one chaotic order-3 magma
- C15: light cones of Laver tables - is the dependence always bounded by the order’s 2-adic size?

C16 — Fibonacci dynamics in invented finite algebras.

- C16: prove that full-period quasigroups exist for every n >= 5 (CLM-C16-10) - a recursive or algebraic construction for even n is needed, since isotopes of groups of even order are excluded
- C16: count full-period quasigroups of order 6 (a partial DFS found 1136 normalised cycles within 2e7 nodes without finishing) and compare with L(6)/36
- C16: prove CLM-C16-13 (affine maximal period = order of the 3x3 matrix) and explain the Laver transient bound 5
- C16: which laws force short periods? (idempotency gives n fixed points; left-distributivity with an anchor forces period <= 2)

C17 — Deciding laws of finite magmas: generator, anchor and randomised tests against the exhaustive check.

- C17: prove CLM-C17-05 for all N (anchored + wrap-around LD instance => Laver table, N a power of 2)
- C17: generator-based tests for other laws (left-distributivity, mediality) - which laws admit a Light-type reduction?
- C17: benchmark against optimised C implementations and against SAT-based model checkers

C18 — Invented operations as mixing primitives: quasigroup string transformations versus standard hashes.

- C18: wide-state quasigroup cascades with larger quasigroups (e.g. order 2^16) and an optimised C implementation, run through a full SMHasher-style suite before any claim of usefulness
- C18: explain the residual bias of the nimber-inversion cascade (x*x = 0 and commutativity) - which quasigroup properties hurt cascades?
- C18: a probabilistic model of the cascade over a random Latin square - how does the avalanche bias decay with the depth?

C19 — The longest Zeckendorf game.

- C19: prove the recursion CLM-C19-05 by analysing the normalisation phase from the lazy representation (carries through alternating blocks)
- C19: prove D(n) <= floor(j^2/4) and the strict monotonicity of L
- C19: generalise the lazy-phase theorem and the Fibonacci-input formula to the generalised Zeckendorf games of Baird-Smith et al. (other linear recurrences)
- C19: submit L(n) and L(F_k) to the OEIS (human decision)

C20 — Open problems revisited: the last one-rule systems of C04, and a literature check for C03.

- C20: derivational complexity of all terminating barren one-rule systems with |l| <= 4 - is it always n - O(1), polynomial or 2^Theta(n)?
- C20: formalise Theorems 1-2 of proofs/paper/C20.md in Lean (run encoding + journeys)